



Generating SMPS Bode Plots in LTspice

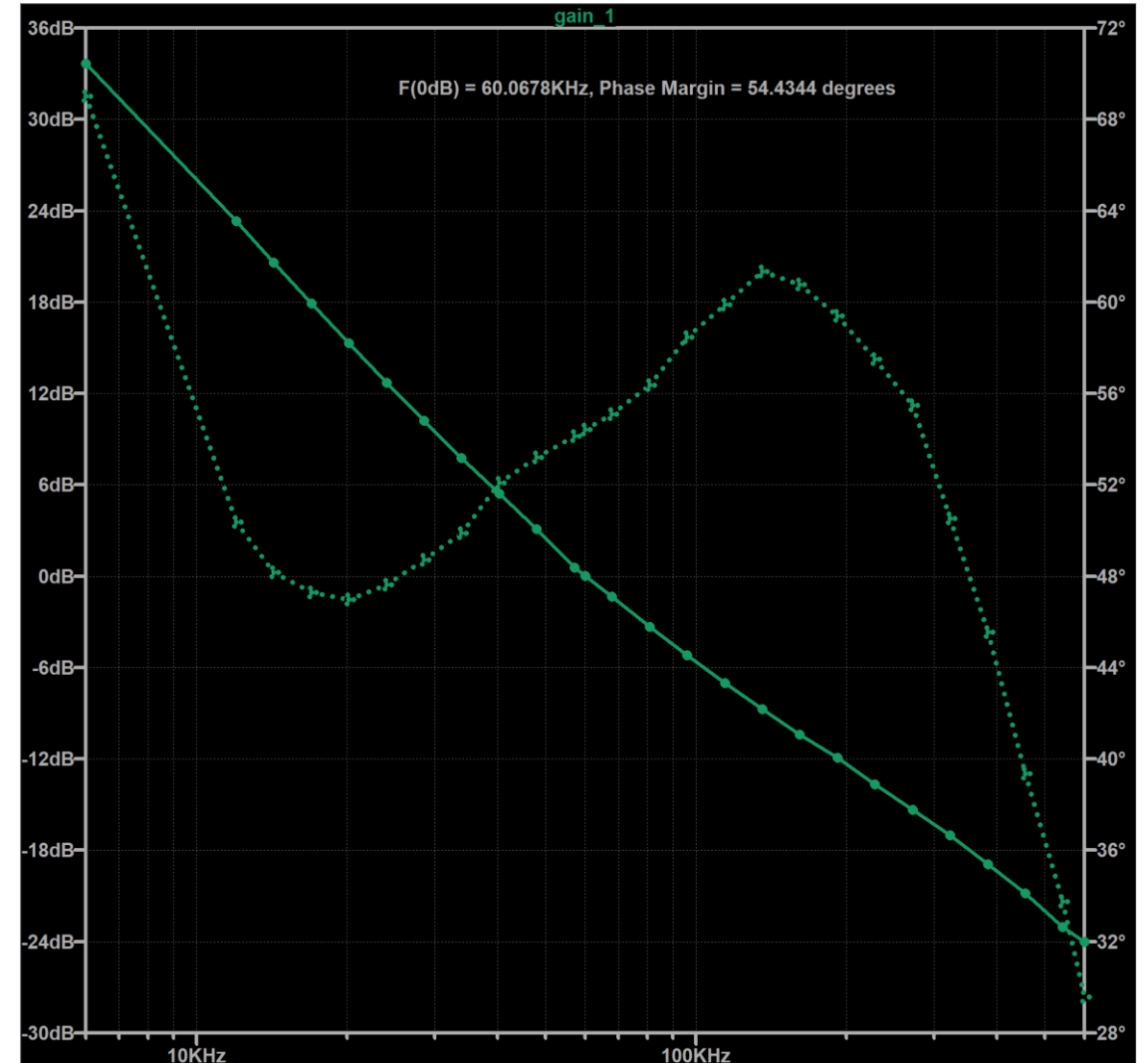
Anne Mahaffey

Principal Engineer, LTspice Applications

analog.com

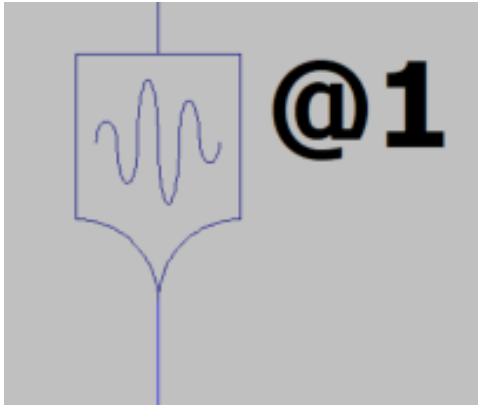
Agenda

- Why use LTspice?
- Overview of FRA components
- FRA Settings
- Step-by-Step Procedure
- Configuration Wizard
- Considerations when correlating to LTpowerCAD and bench
- Configurations that require FRA Probe



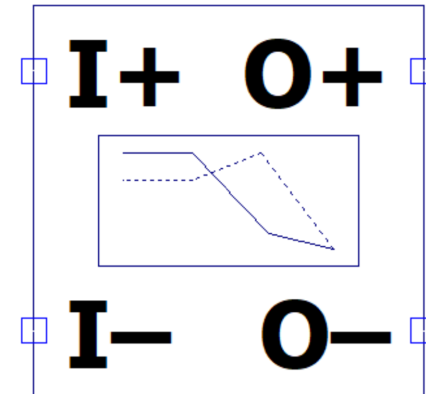
FRA Components in LTspice

FRA Device



FRA Probe

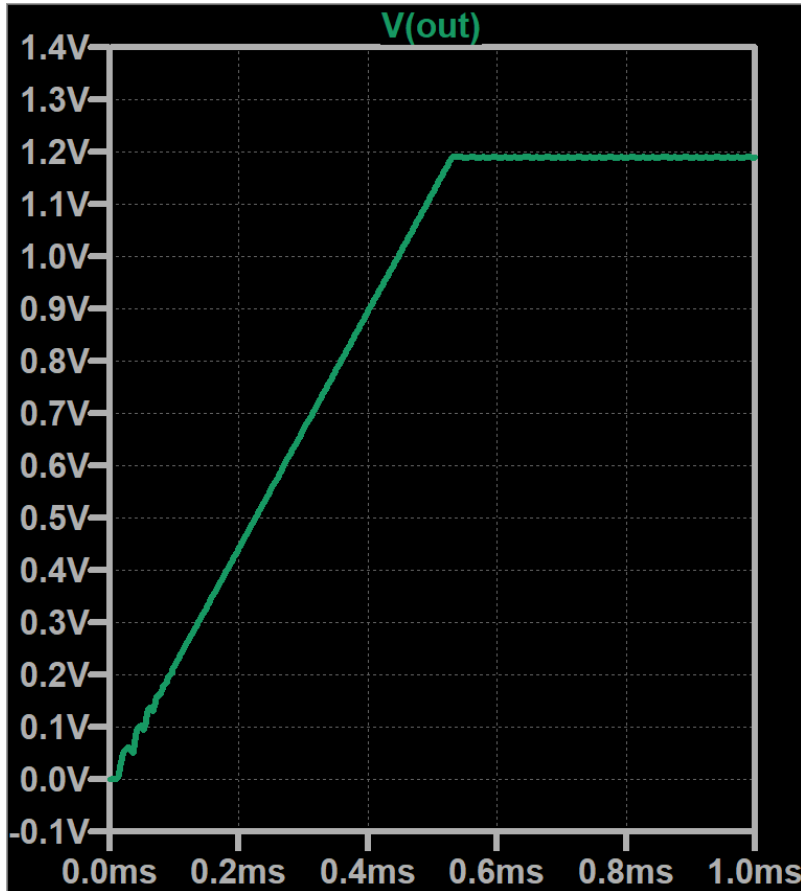
&1



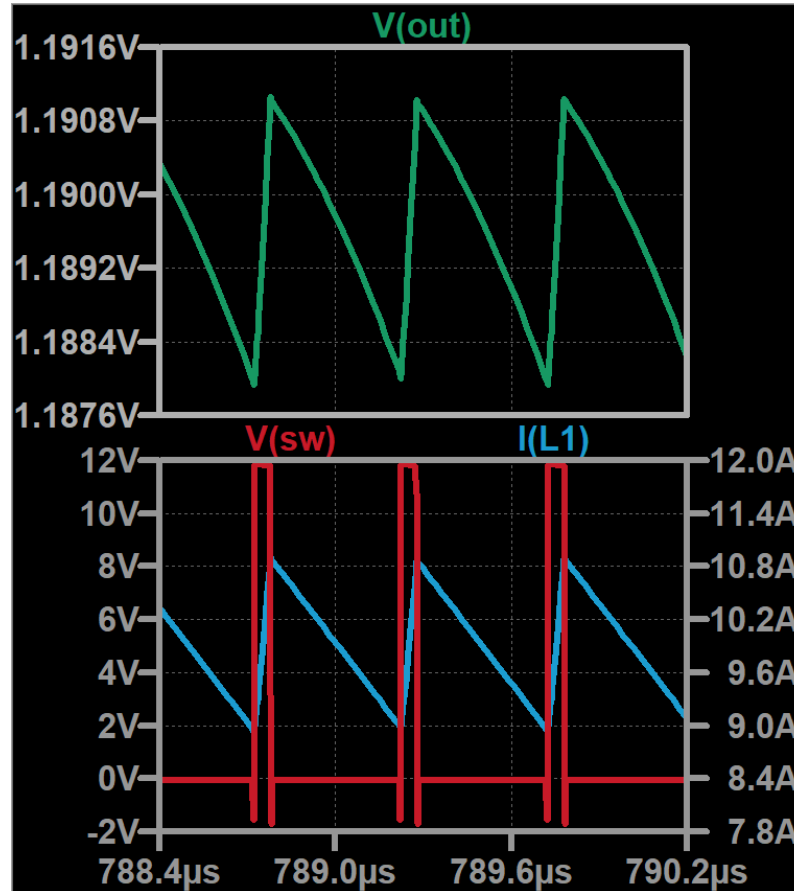
Why LTspice?

Free download @ www.analog.com/LTpowerCAD

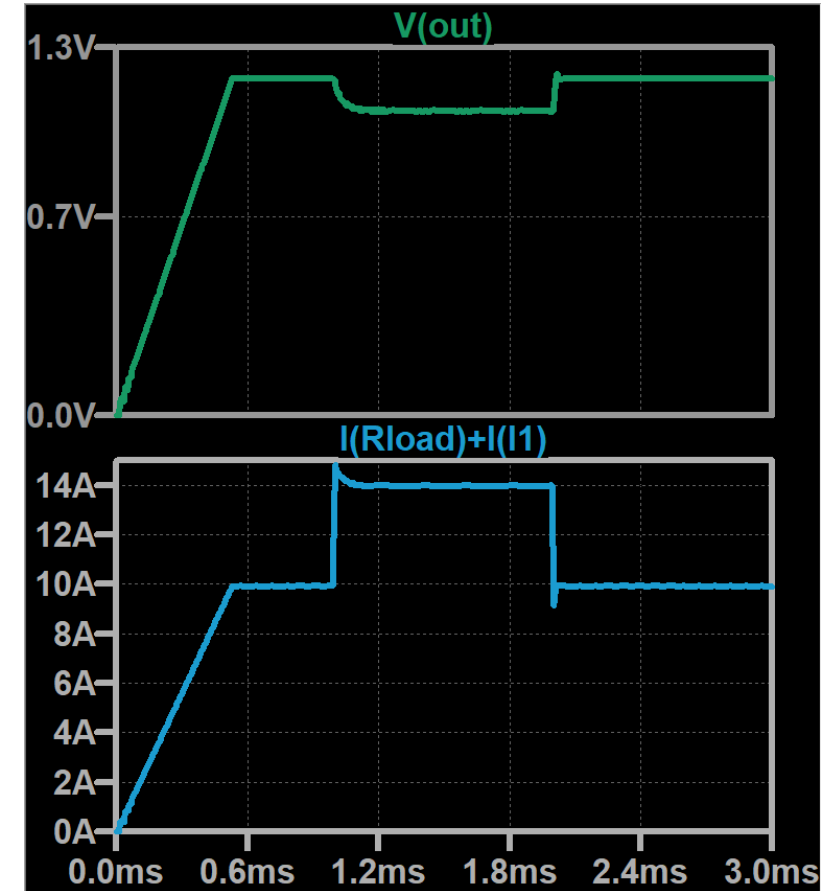
Why use LTspice?



Startup



Steady-state

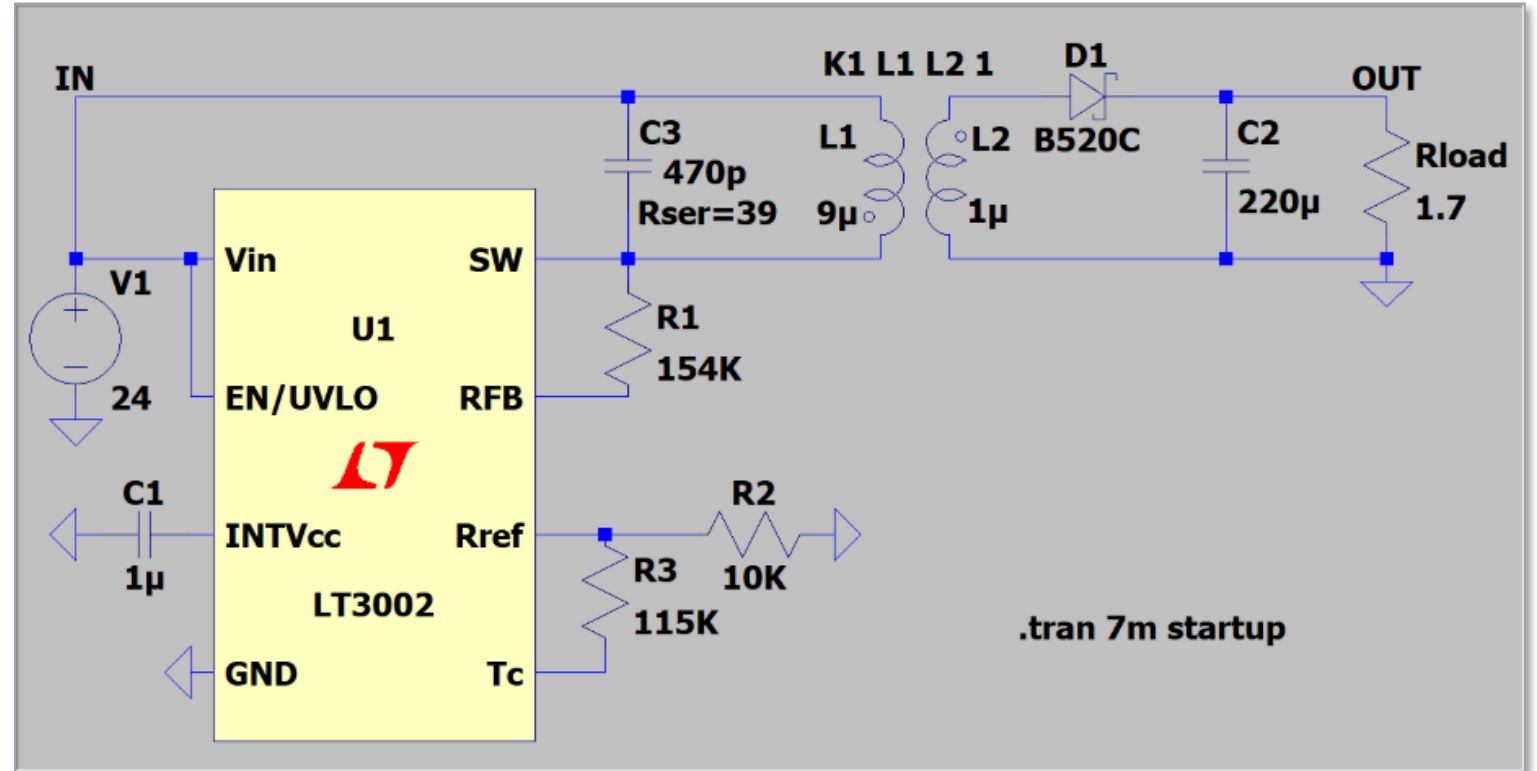


Failure Mode - OCP

Why use LTspice?

To analyze more complex topologies not in LTpowerCAD

- Inverting power supplies
- Isolated supplies
- LDOs

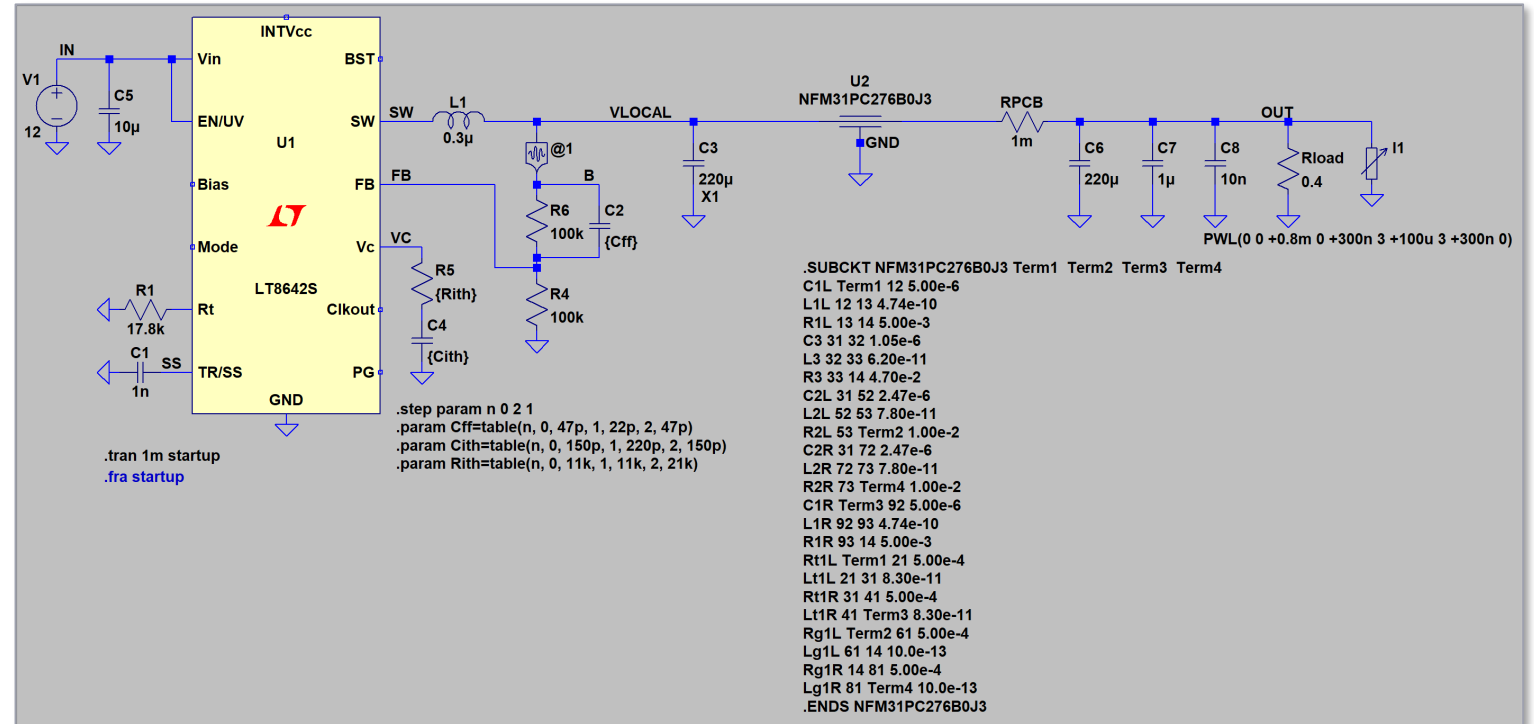


Flyback Converter

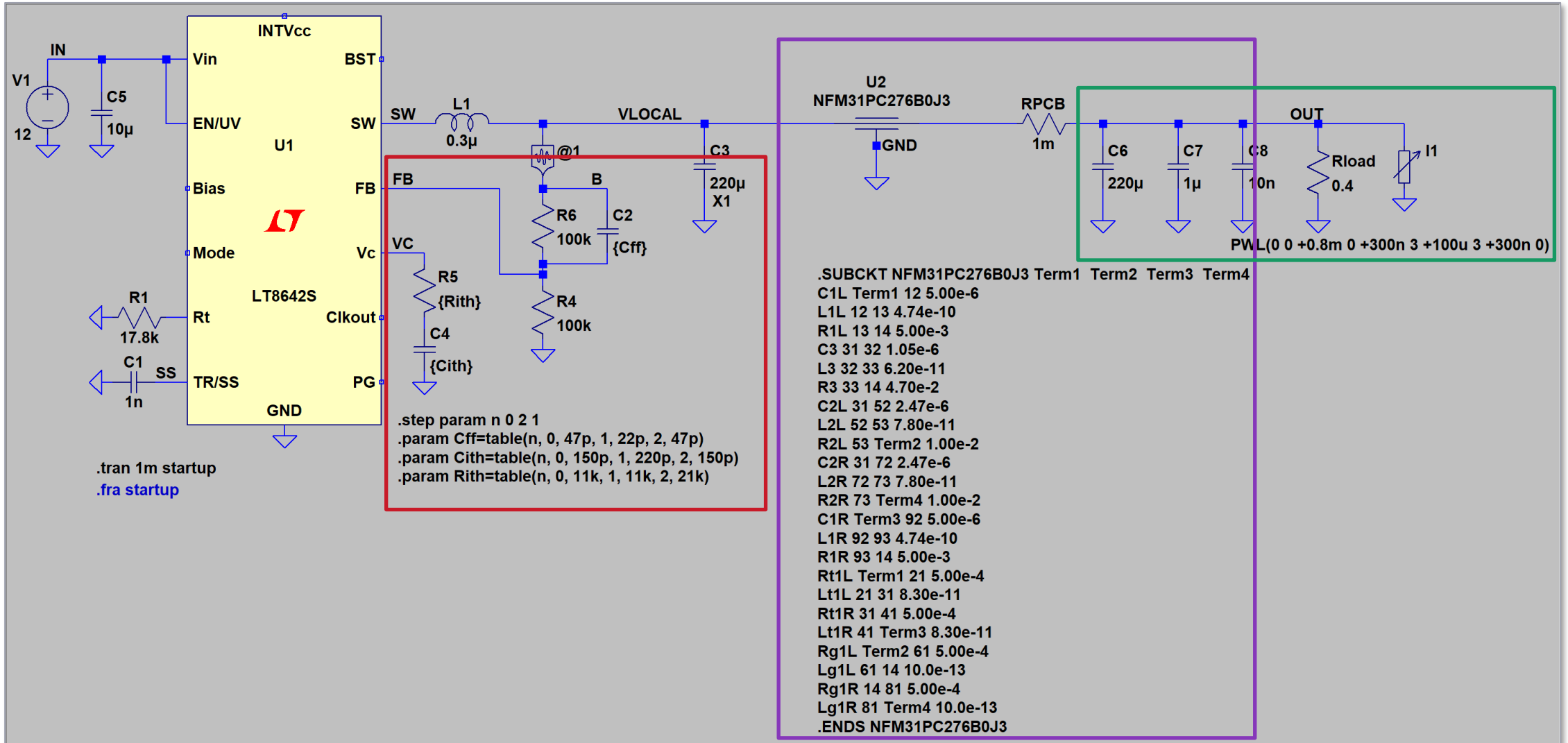
Why use LTspice?

To analyze more complexities in our circuits

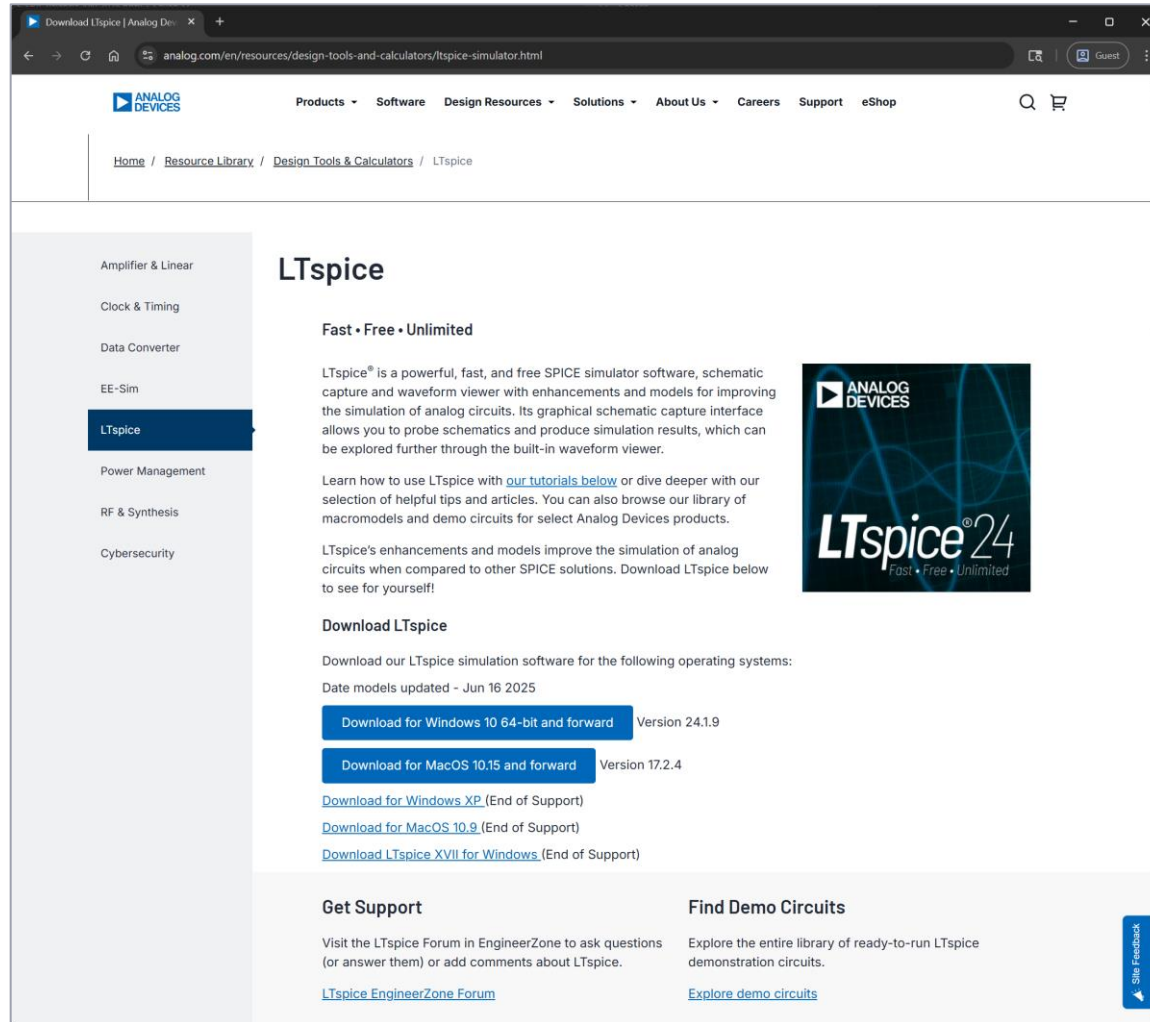
- Add complex input and output filtering
- Include complex component models



Why use LTspice?



Where to Find LTspice – analog.com/LTspice



Download LTspice | Analog Devices

analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html

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LTspice

Fast • Free • Unlimited

LTspice® is a powerful, fast, and free SPICE simulator software, schematic capture and waveform viewer with enhancements and models for improving the simulation of analog circuits. Its graphical schematic capture interface allows you to probe schematics and produce simulation results, which can be explored further through the built-in waveform viewer.

Learn how to use LTspice with [our tutorials below](#) or dive deeper with our selection of helpful tips and articles. You can also browse our library of macromodels and demo circuits for select Analog Devices products.

LTspice's enhancements and models improve the simulation of analog circuits when compared to other SPICE solutions. Download LTspice below to see for yourself!

Download LTspice

Download our LTspice simulation software for the following operating systems:
Date models updated - Jun 16 2025

[Download for Windows 10 64-bit and forward](#) Version 24.1.9

[Download for MacOS 10.15 and forward](#) Version 17.2.4

[Download for Windows XP](#) (End of Support)

[Download for MacOS 10.9](#) (End of Support)

[Download LTspice XVII for Windows](#) (End of Support)

Get Support

Visit the LTspice Forum in EngineerZone to ask questions (or answer them) or add comments about LTspice.

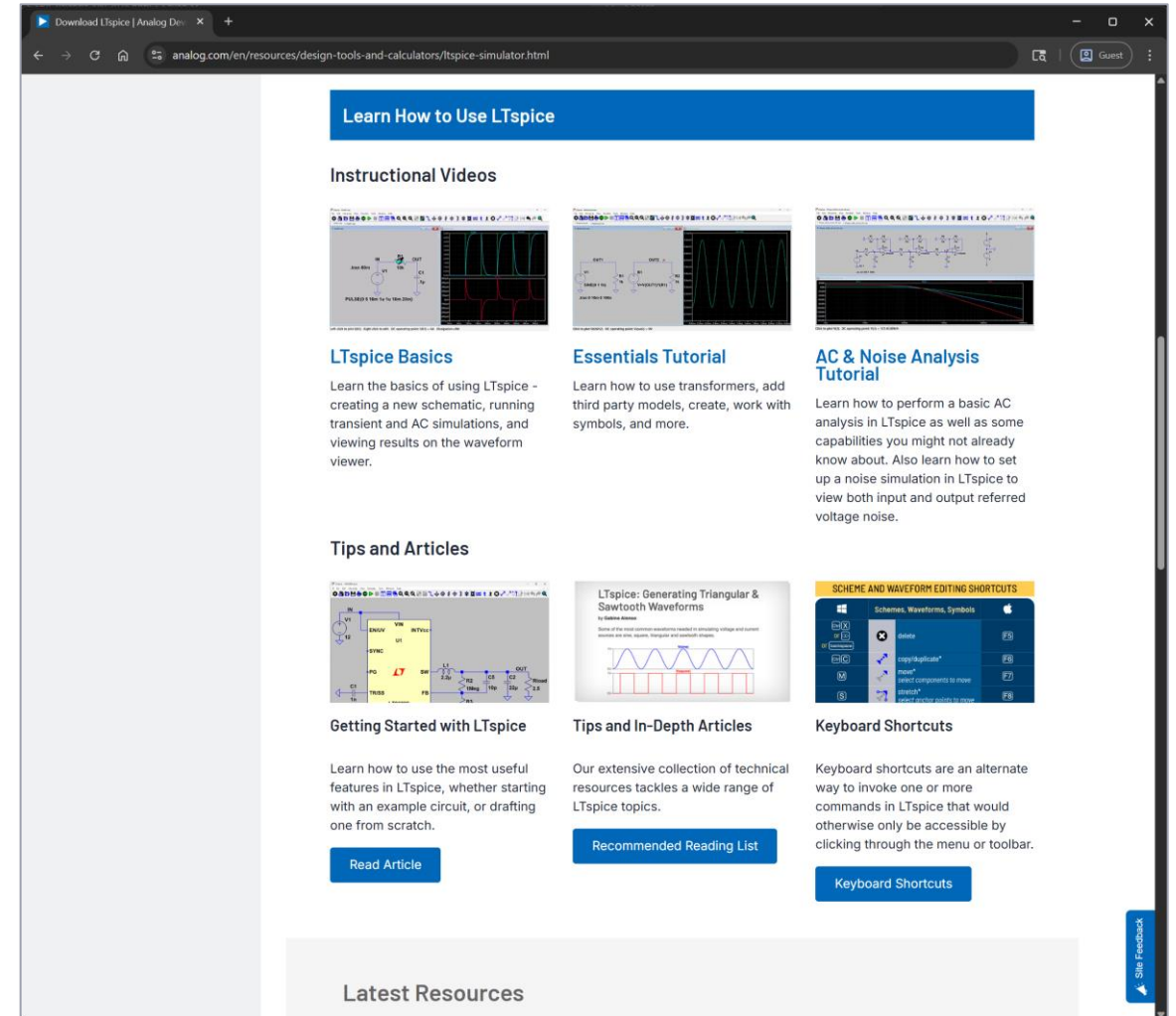
[LTspice EngineerZone Forum](#)

Find Demo Circuits

Explore the entire library of ready-to-run LTspice demonstration circuits.

[Explore demo circuits](#)

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Download LTspice | Analog Devices

analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html

Learn How to Use LTspice

Instructional Videos

LTspice Basics
Learn the basics of using LTspice – creating a new schematic, running transient and AC simulations, and viewing results on the waveform viewer.

Essentials Tutorial
Learn how to use transformers, add third party models, create, work with symbols, and more.

AC & Noise Analysis Tutorial
Learn how to perform a basic AC analysis in LTspice as well as some capabilities you might not already know about. Also learn how to set up a noise simulation in LTspice to view both input and output referred voltage noise.

Tips and Articles

Getting Started with LTspice
Learn how to use the most useful features in LTspice, whether starting with an example circuit, or drafting one from scratch.

[Read Article](#)

Tips and In-Depth Articles
Our extensive collection of technical resources tackles a wide range of LTspice topics.

[Recommended Reading List](#)

Keyboard Shortcuts
Keyboard shortcuts are an alternate way to invoke one or more commands in LTspice that would otherwise only be accessible by clicking through the menu or toolbar.

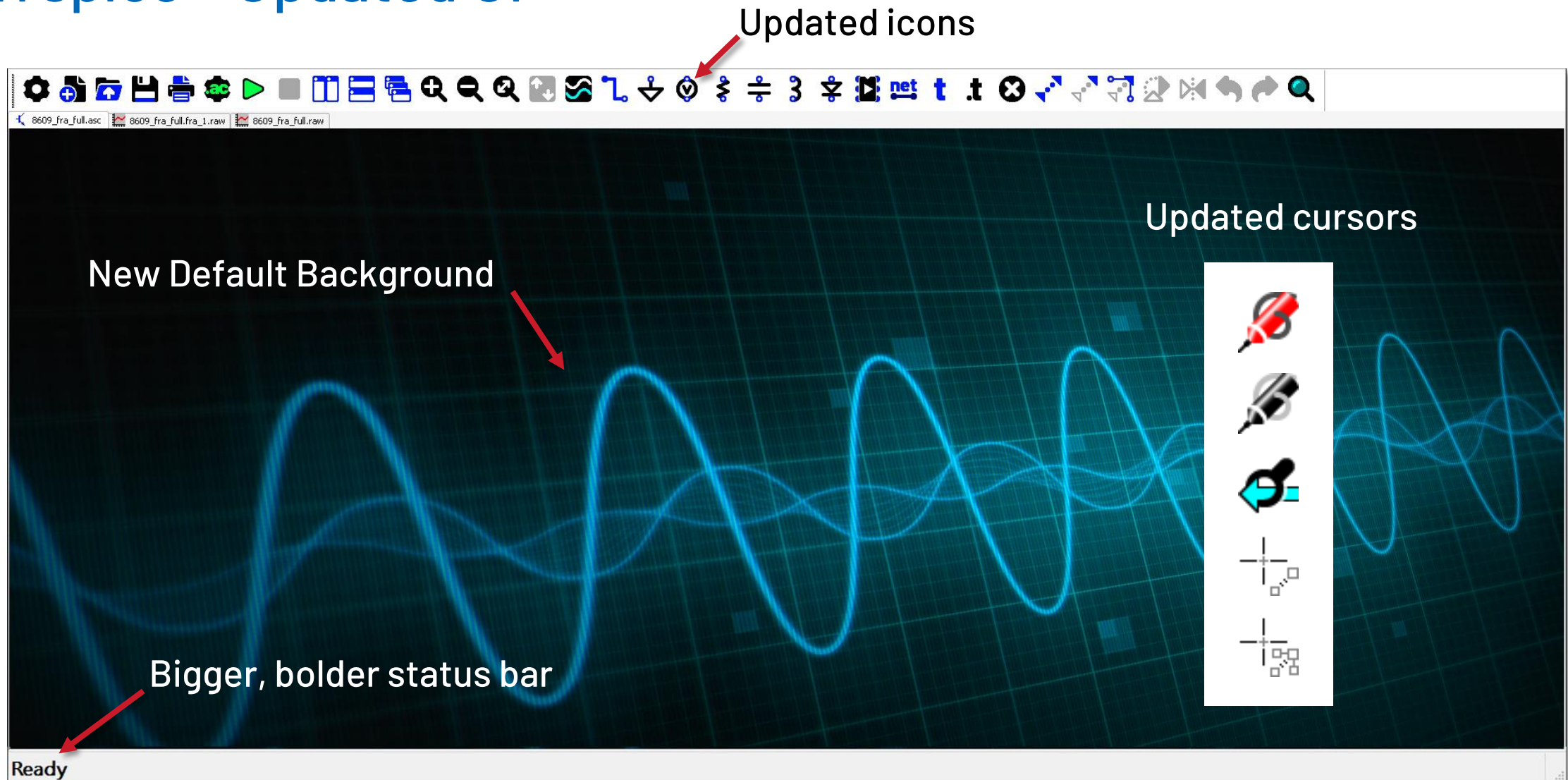
[Keyboard Shortcuts](#)

Latest Resources

[Site Feedback](#)

FRA Probe and FRA device are included in LTspice 24.1

LTspice – Updated UI



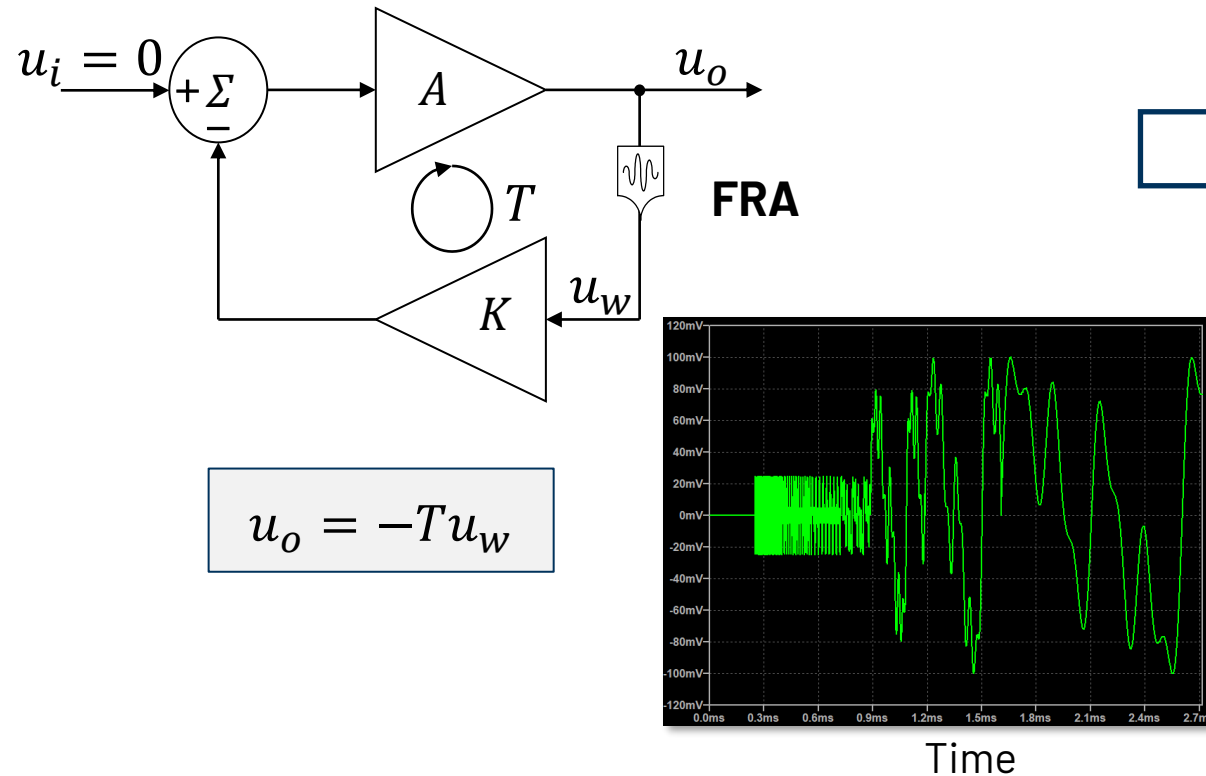
How to FRA



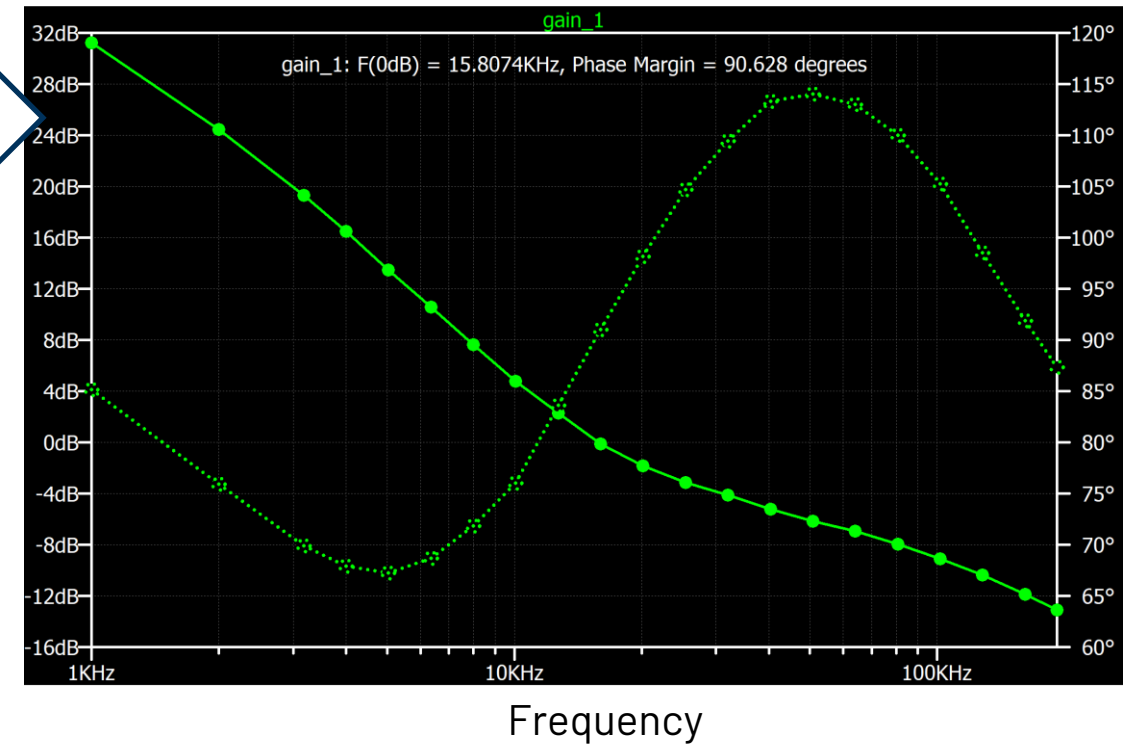
Simulate

What does the FRA Device do?

- Transient Simulation
- The FRA component injects sinusoidal stimulus



- Perturbation is measured at FRA nodes
- Bode plot is extracted by Fourier analysis



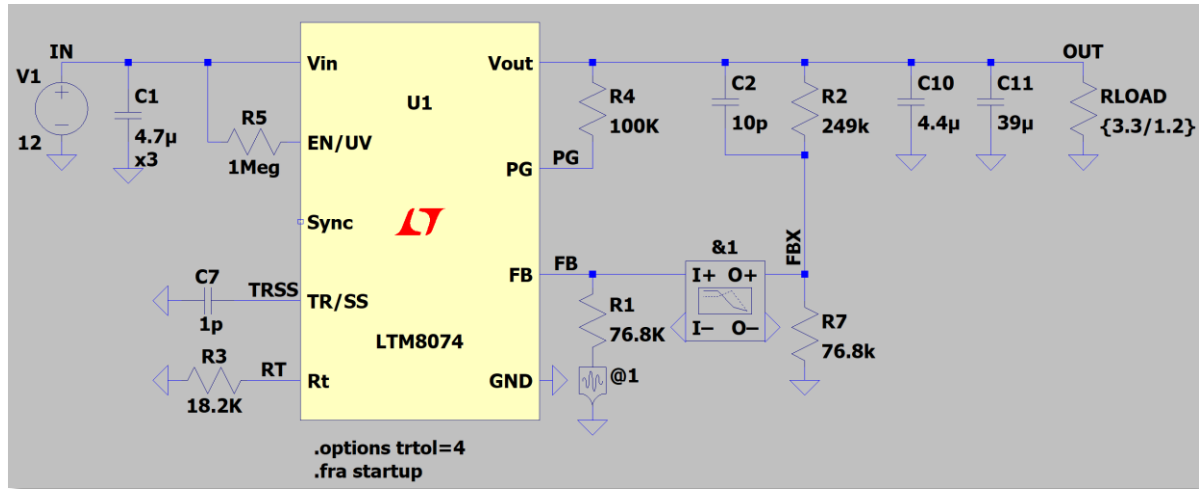


- Enables Bode Plots of any portion of the loop
- Enables analysis of uModules, Inverting Outputs, Current Feedback topologies

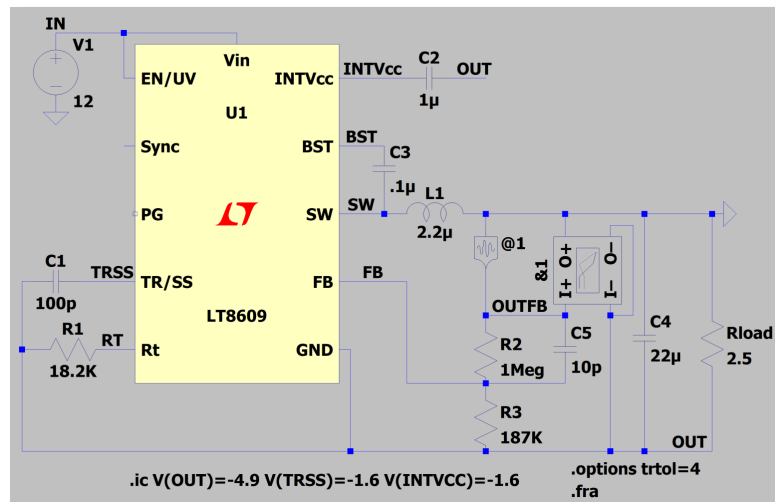
Inverting Output Example – requires FRA Probe

What does the FRA Probe do?

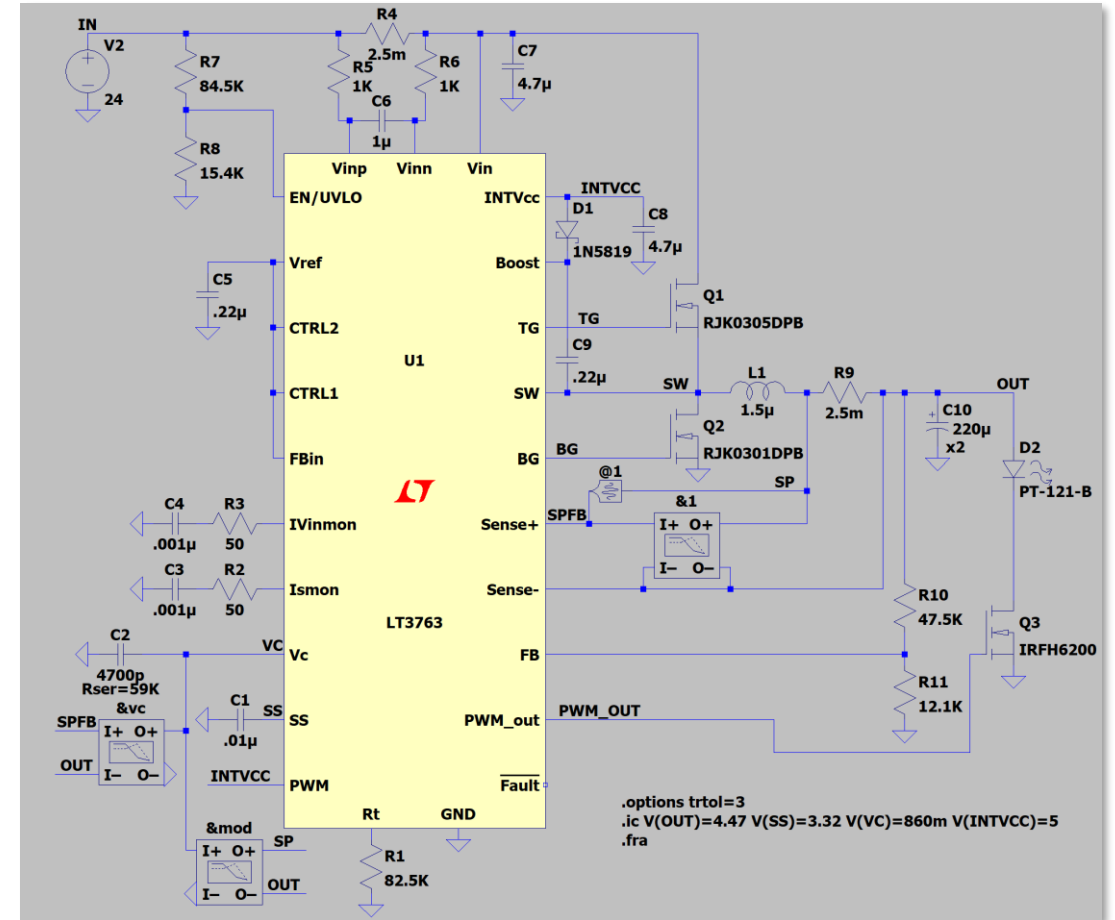
µModules with integrated top feedback resistors



Negative Outputs

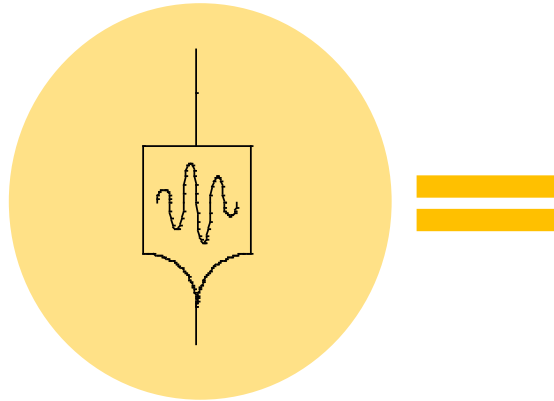


Current Feedback

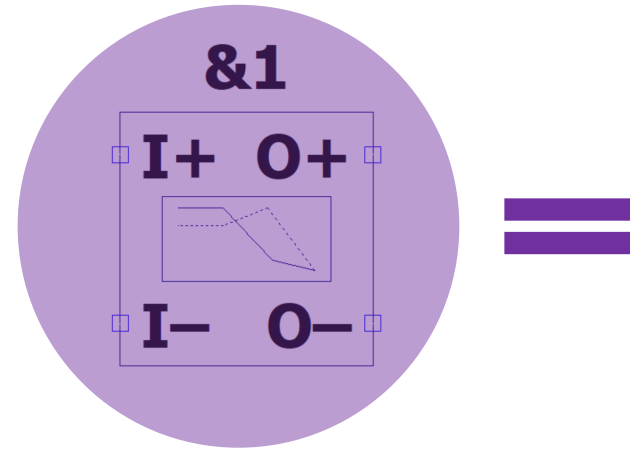


These examples are in
Open Examples -> Educational -> FRA

Functional FRA “Equivalencies”

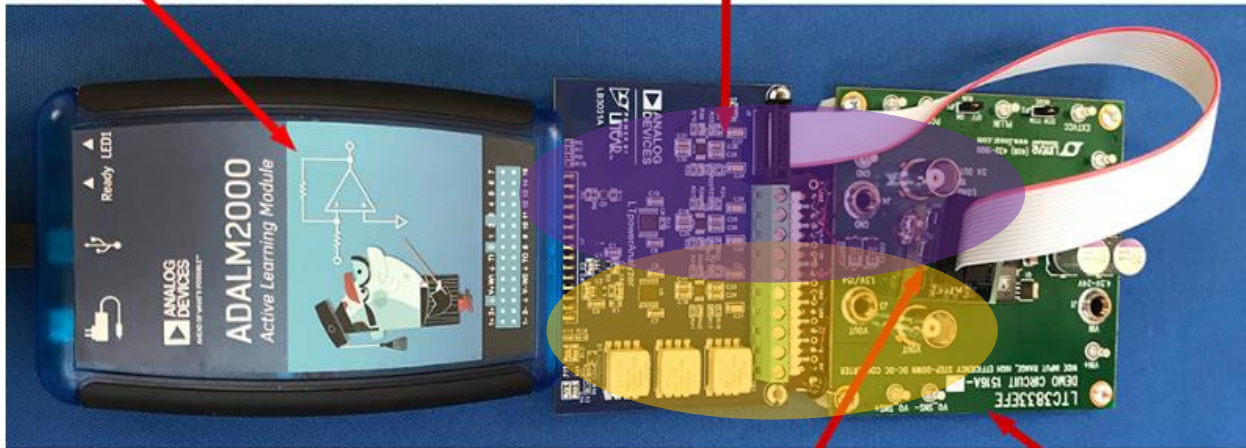


=



ADALM2000
(M2K)

LTpowerAnalyzer Board



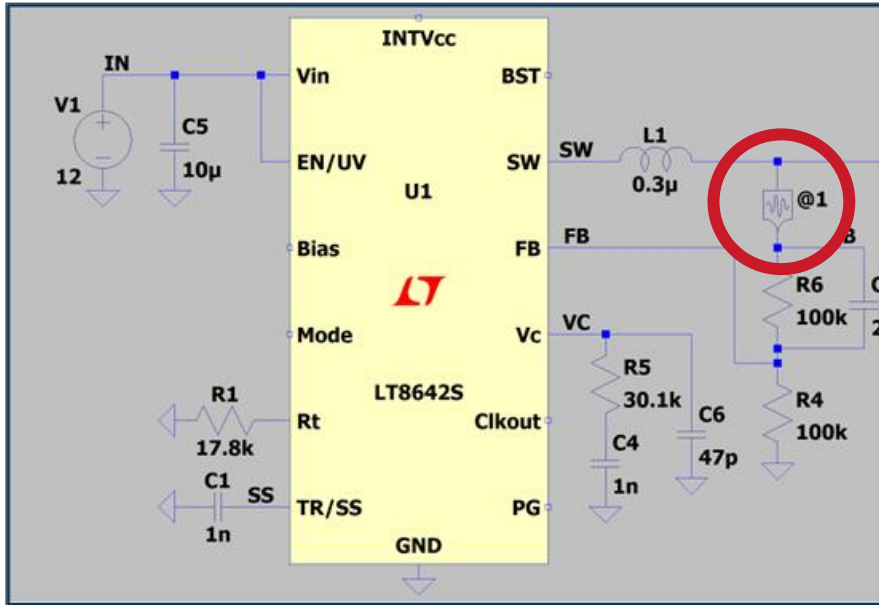
Current Probe
Module

DUT

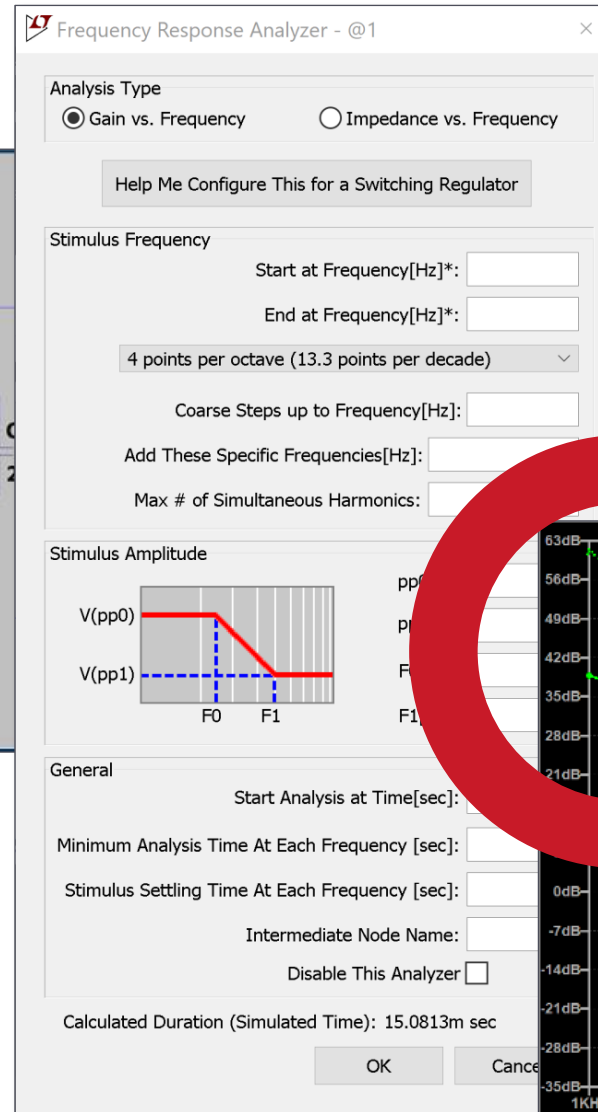


FRA Settings Overview

How To FRA

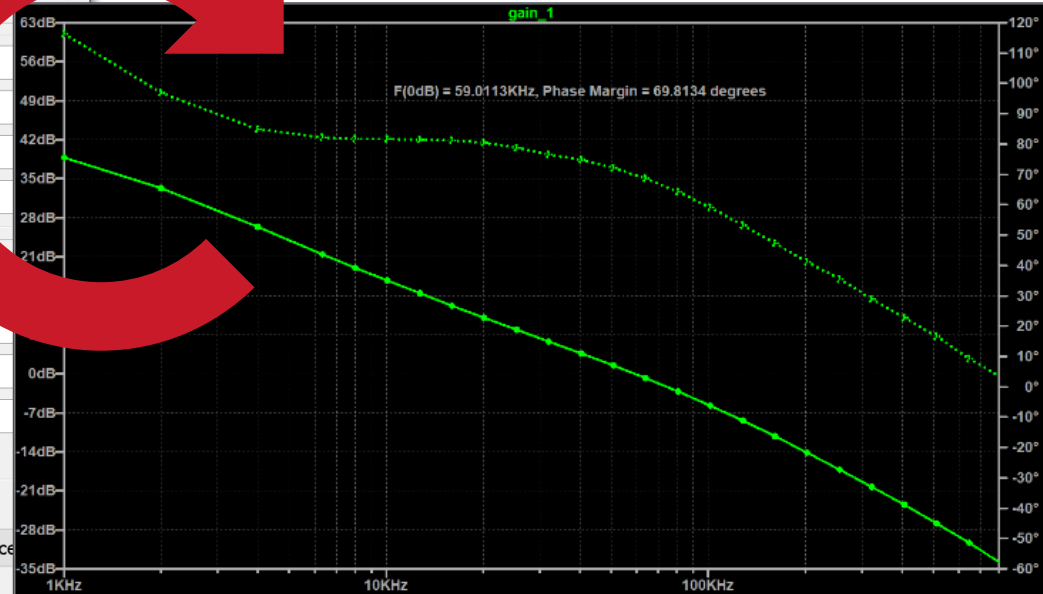


Place FRA Device



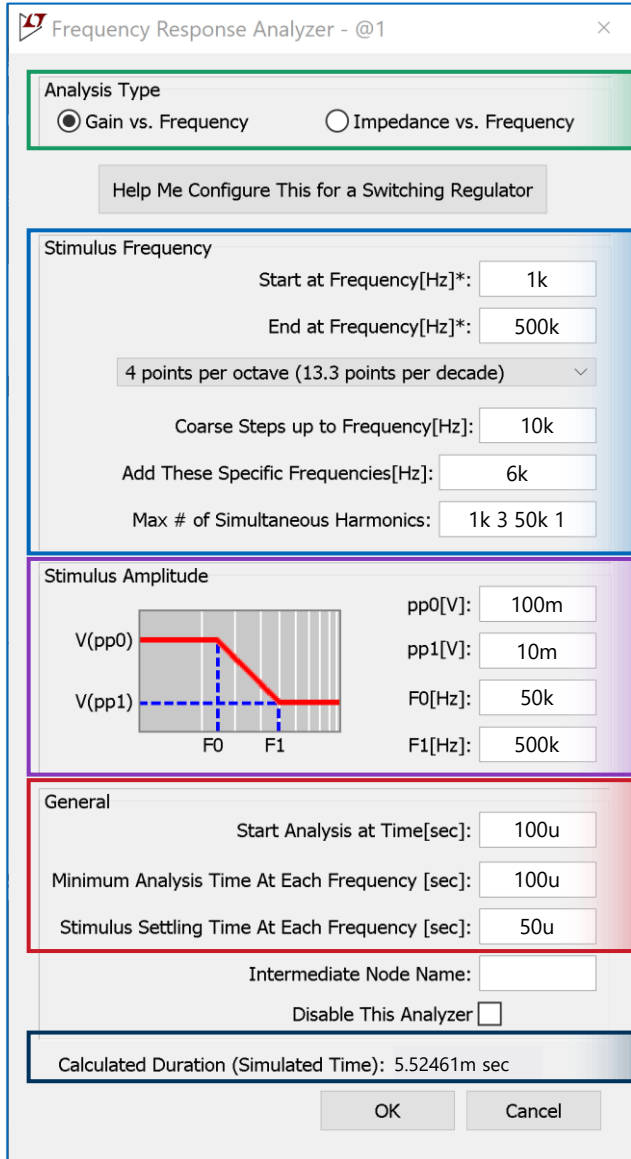
Configure FRA

Iterate



Simulate

FRA Device Settings: Overview



Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

Stimulus Frequency
Start at Frequency[Hz]*: 1k
End at Frequency[Hz]*: 500k
4 points per octave (13.3 points per decade)
Coarse Steps up to Frequency[Hz]: 10k
Add These Specific Frequencies[Hz]: 6k
Max # of Simultaneous Harmonics: 1k 3 50k 1

Stimulus Amplitude
V(pp0) V(pp1)
F0 F1
pp0[V]: 100m
pp1[V]: 10m
F0[Hz]: 50k
F1[Hz]: 500k

General
Start Analysis at Time[sec]: 100u
Minimum Analysis Time At Each Frequency [sec]: 100u
Stimulus Settling Time At Each Frequency [sec]: 50u
Intermediate Node Name:
Disable This Analyzer ☐
Calculated Duration (Simulated Time): 5.52461m sec
OK Cancel

Bode Plot of **Gain** or **Output Impedance** Sweep

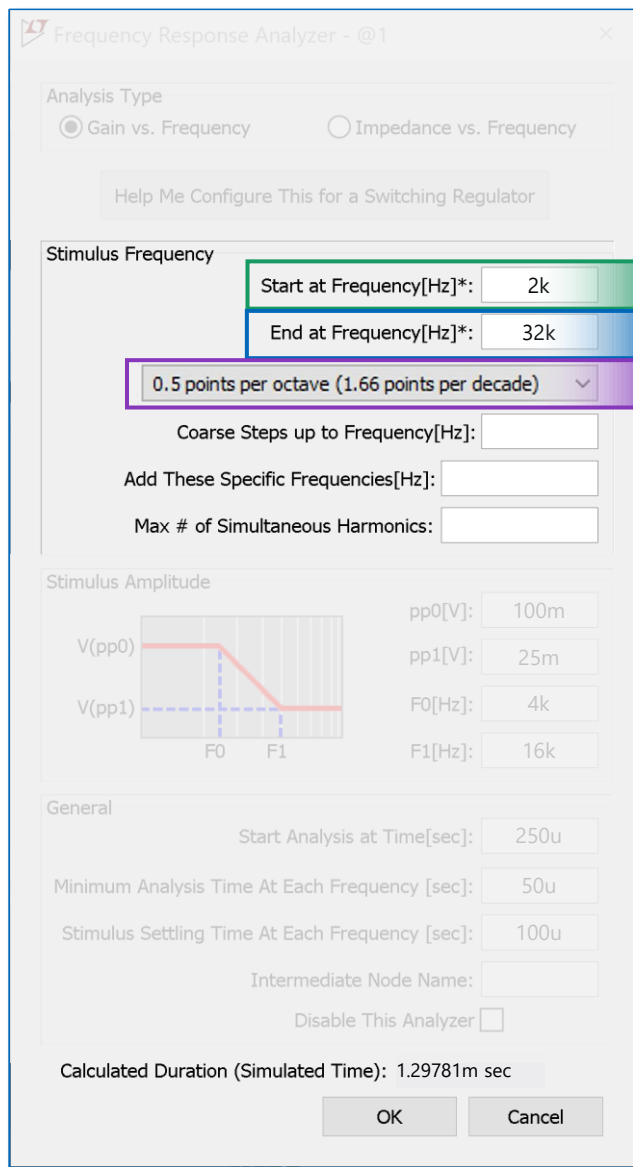
Set the **frequencies** applied in the analysis

Set stimulus **amplitude** as a function of frequency

Set the **timing** of analysis and stimulus transitions

Indicates simulated time based on settings above

FRA Device Settings: Required Frequency Settings



Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

Stimulus Frequency

Start at Frequency[Hz]*: 2k

End at Frequency[Hz]*: 32k

0.5 points per octave (1.66 points per decade)

Coarse Steps up to Frequency[Hz]:

Add These Specific Frequencies[Hz]:

Max # of Simultaneous Harmonics:

Stimulus Amplitude

V(pp0): 100m

V(pp1): 25m

F0[Hz]: 4k

F1[Hz]: 16k

General

Start Analysis at Time[sec]: 250u

Minimum Analysis Time At Each Frequency [sec]: 50u

Stimulus Settling Time At Each Frequency [sec]: 100u

Intermediate Node Name:

Disable This Analyzer ☐

Calculated Duration (Simulated Time): 1.29781m sec

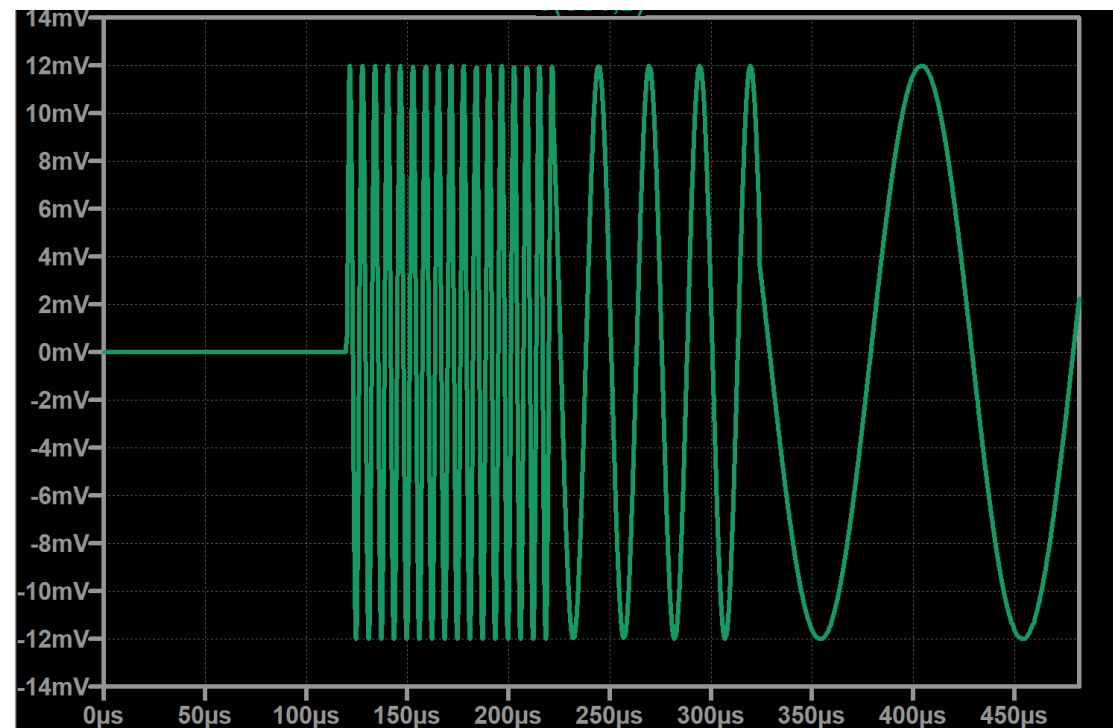
OK Cancel

Lowest Frequency

Highest Frequency

of Frequencies

Specified in octaves so
LTspice can apply
simultaneous harmonics to
speed up the simulation



Example with 3 frequency points

FRA Device Settings: Optional Frequency Settings

Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

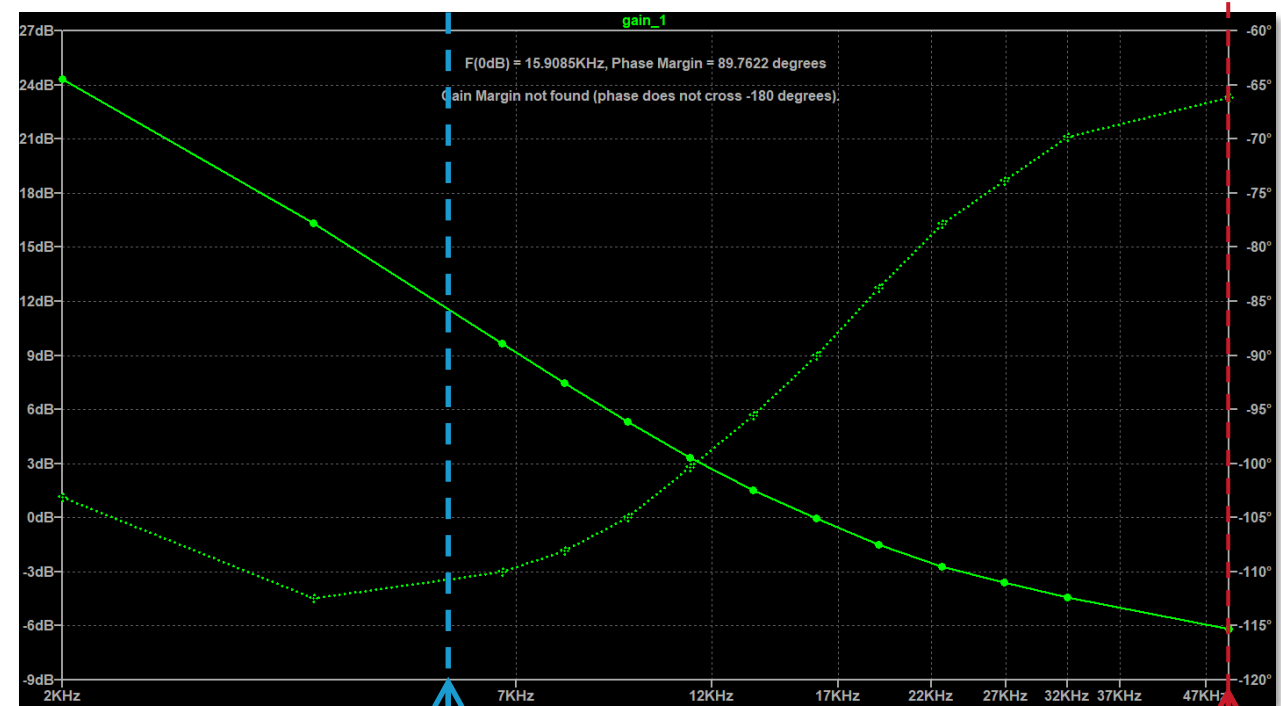
Help Me Configure This for a Switching Regulator

Stimulus Frequency
Start at Frequency[Hz]*: 2k
End at Frequency[Hz]*: 32k
4 points per octave (13.3 points per decade)
Coarse Steps up to Frequency[Hz]: 6k
Add These Specific Frequencies[Hz]: 50k
Max # of Simultaneous Harmonics: 2

Stimulus Amplitude
pp0[V]: 50m
pp1[V]:
F0[Hz]:
F1[Hz]:

General
Start Analysis at Time[sec]: 250u
Minimum Analysis Time At Each Frequency [sec]: 50u
Stimulus Settling Time At Each Frequency [sec]: 100u
Intermediate Node Name:
Disable This Analyzer ☐
Calculated Duration (Simulated Time): 1.29781m sec
OK Cancel

Increase Resolution
Reduce Low Freqs
Specific Freqs



Limit resolution at low frequency (Fcoarse)
LTspice applies just one point per octave below this frequency, to speed up simulations.
Recommend 2x~10x Fstart.

Specify individual frequencies
To be applied in addition to the sweep specified above

FRA Device Settings: Simultaneous Harmonics

Frequency Response Analyzer - @1

Analysis Type

☒ Gain vs. Frequency

☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

Stimulus Frequency

Start at Frequency[Hz]*:

2k

End at Frequency[Hz]*:

32k

4 points per octave (13.3 points per decade)

Coarse Steps up to Frequency[Hz]:

6k

Add These Specific Frequencies[Hz]:

50k

Max # of Simultaneous Harmonics:

1, 2, and 3

Stimulus Amplitude

V(pp0)

V(pp1)

F0

F1

pp0[V]:

50m

pp1[V]:

F0[Hz]:

F1[Hz]:

General

Start Analysis at Time[sec]:

250u

Minimum Analysis Time At Each Frequency [sec]:

50u

Stimulus Settling Time At Each Frequency [sec]:

100u

Intermediate Node Name:

Disable This Analyzer

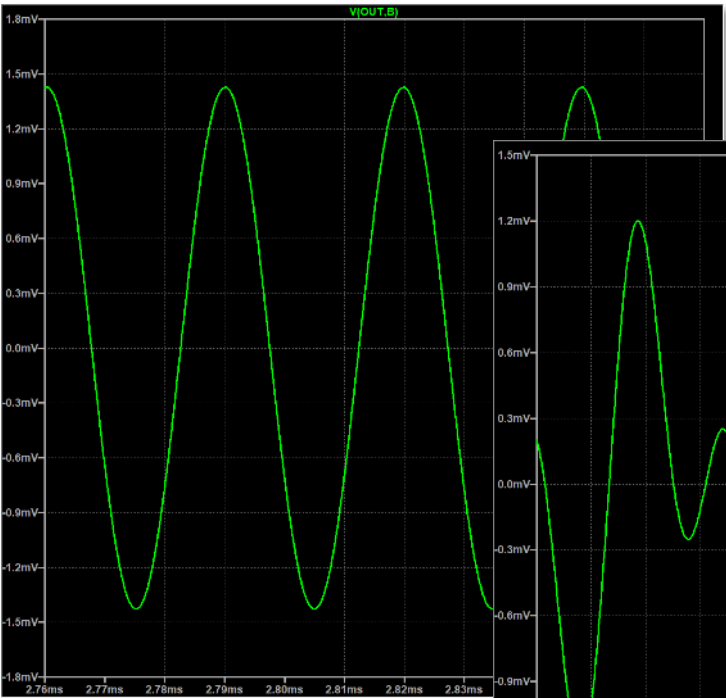
☐

Calculated Duration (Simulated Time):

1.29781m sec

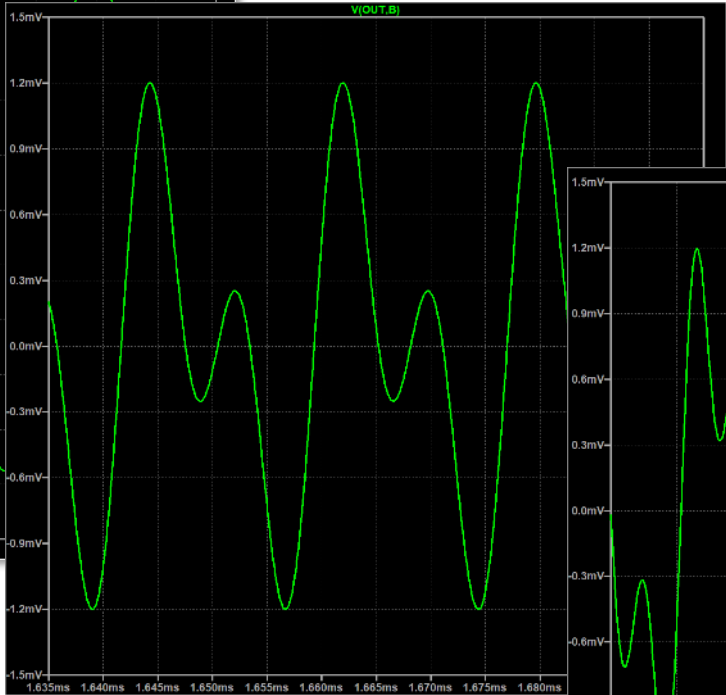
OK

Cancel



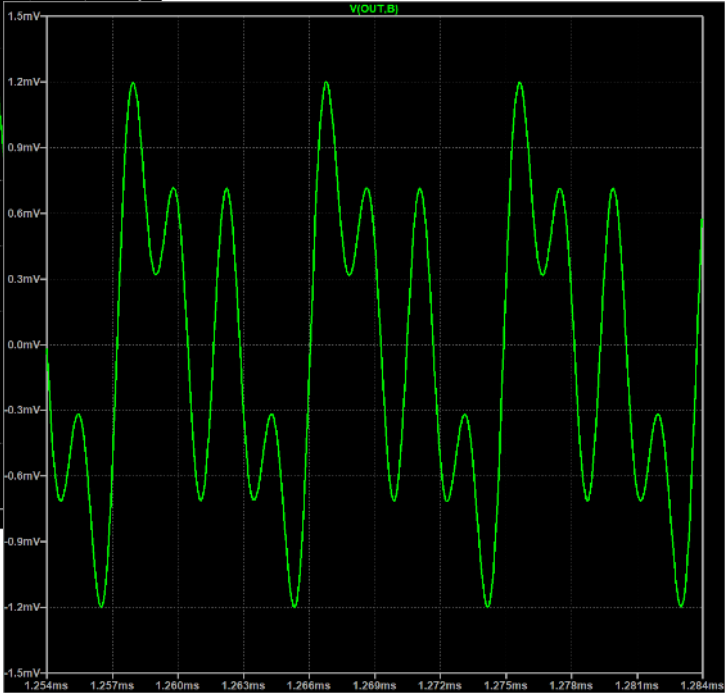
The waveform shows a single periodic sine wave. The y-axis is labeled from -1.8mV to 1.8mV in 0.3mV increments. The x-axis is labeled from 2.76ms to 2.83ms in 0.01ms increments. The signal is a green sine wave with an amplitude of approximately 1.5mV.

1 harmonic



The waveform shows a periodic signal with two cycles of a sine wave within the same time interval as the first harmonic. The y-axis is labeled from -1.5mV to 1.5mV in 0.3mV increments. The x-axis is labeled from 1.635ms to 1.680ms in 0.005ms increments. The signal is a green sine wave with an amplitude of approximately 1.2mV.

2 harmonics



The waveform shows a periodic signal with three cycles of a sine wave within the same time interval as the first harmonic. The y-axis is labeled from -1.5mV to 1.5mV in 0.3mV increments. The x-axis is labeled from 1.254ms to 1.284ms in 0.002ms increments. The signal is a green sine wave with an amplitude of approximately 1.2mV.

3 harmonics

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FRA Device Settings: Stimulus Amplitude

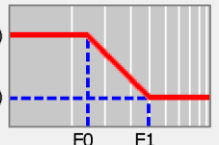
Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

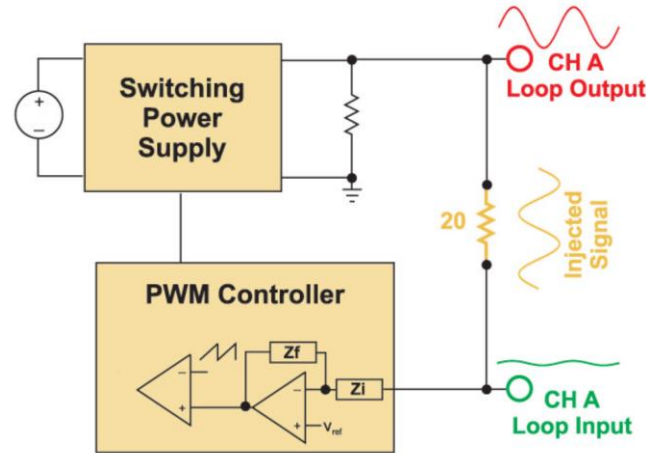
Stimulus Frequency
Start at Frequency[Hz]*: 2k
End at Frequency[Hz]*: 32k
0.5 points per octave (1.66 points per decade)
Coarse Steps up to Frequency[Hz]:
Add These Specific Frequencies[Hz]:
Max # of Simultaneous Harmonics:

Stimulus Amplitude

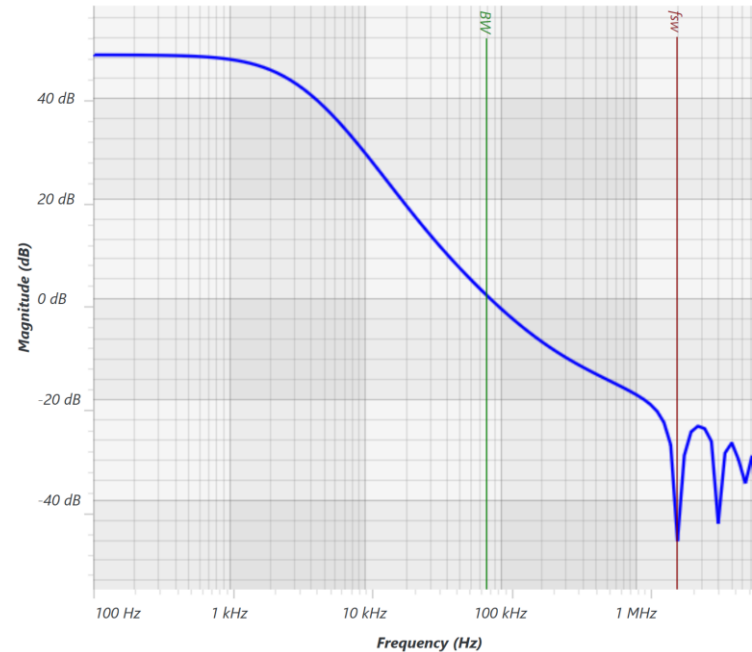
V(pp0)  F0 F1

pp0[V]: 100m
pp1[V]: 25m
F0[Hz]: 4k
F1[Hz]: 16k

General
Start Analysis at Time[sec]:
Minimum Analysis Time At Each Frequency [sec]:
Stimulus Settling Time At Each Frequency [sec]:
Intermediate Node Name:
Disable This Analyzer ☐
Calculated Duration (Simulated Time): 1.29781m sec
OK Cancel



Rail #1 (1.194V) Loop Gain



Signal needs to be large enough to resolve when divided by gain, yet small enough to avoid non-linear operation.

FRA Device Settings: Stimulus Amplitude

Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

Stimulus Frequency
Start at Frequency[Hz]*: 2k
End at Frequency[Hz]*: 32k
0.5 points per octave (1.66 points per decade)
Coarse Steps up to Frequency[Hz]:
Add These Specific Frequencies[Hz]:
Max # of Simultaneous Harmonics:

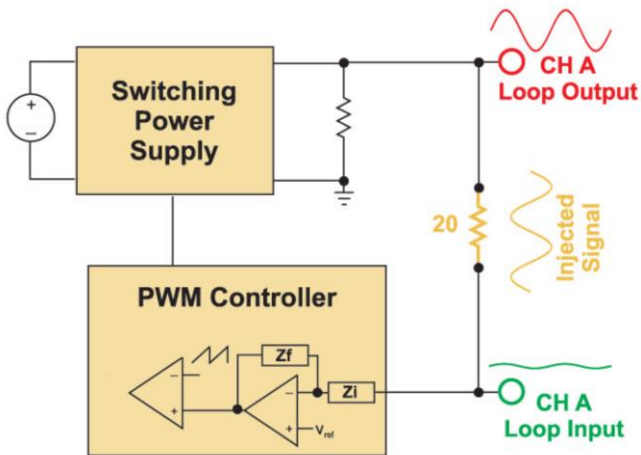
Stimulus Amplitude

V(pp0)
V(pp1)

F0 F1

pp0[V]: 100m
pp1[V]: 25m
F0[Hz]: 4k
F1[Hz]: 16k

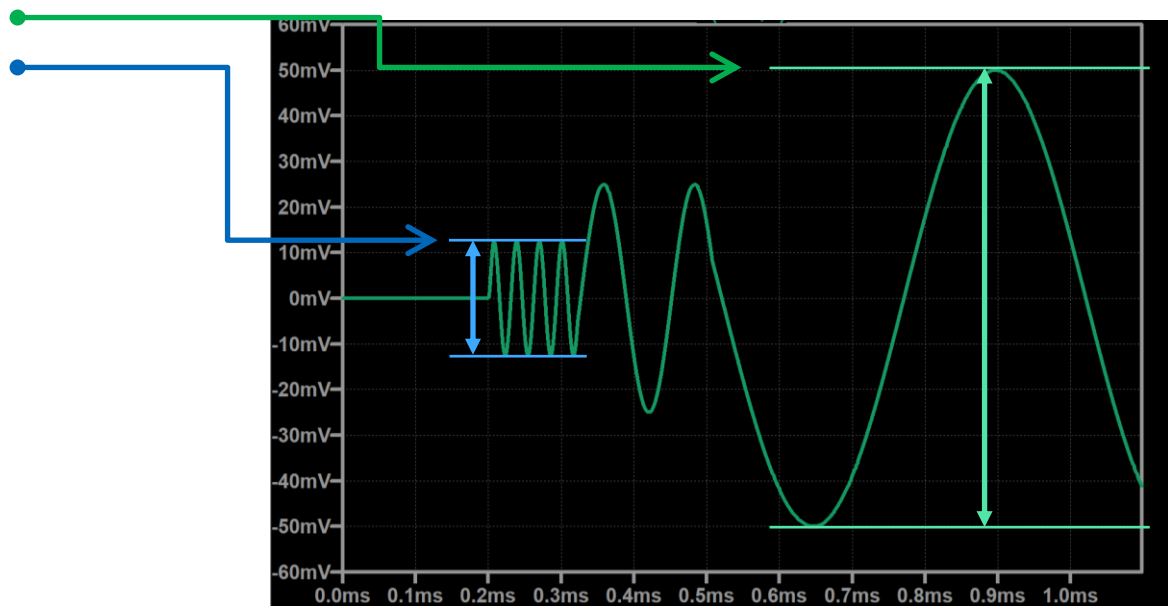
General
Start Analysis at Time[sec]:
Minimum Analysis Time At Each Frequency [sec]:
Stimulus Settling Time At Each Frequency [sec]:
Intermediate Node Name:
Disable This Analyzer ☐
Calculated Duration (Simulated Time): 1.29781m sec
OK Cancel



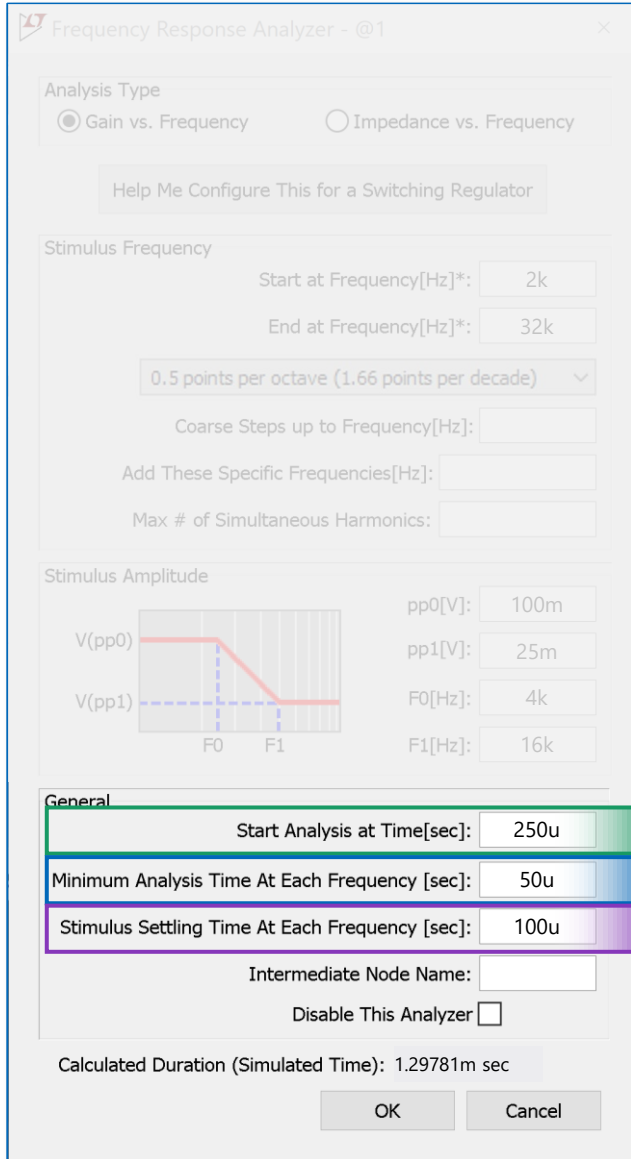
Amplitude at Low Freqs

Amplitude at High Freqs

Frequency Corners



FRA Device Settings: General



Averaging (T_{avgmin})

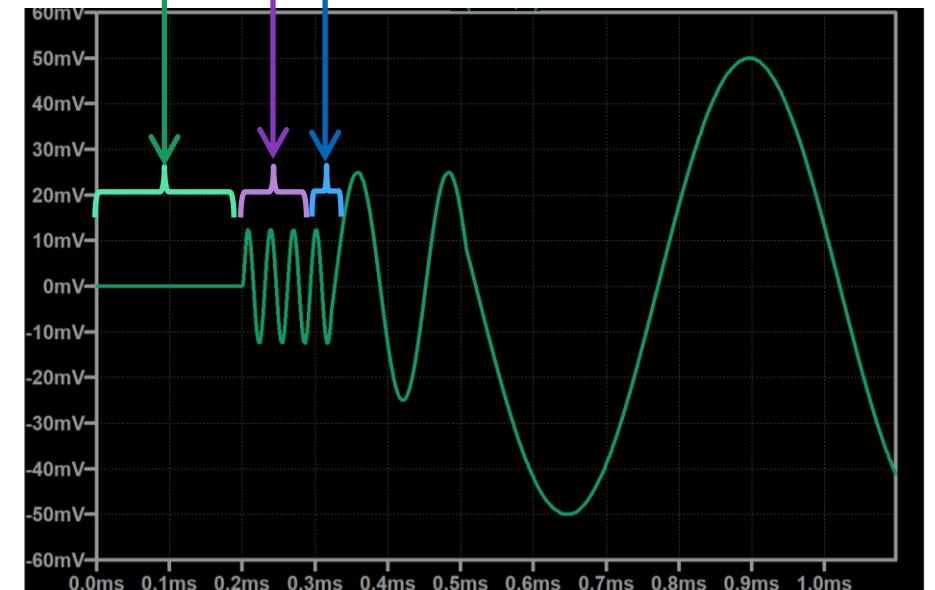
Must be sufficient to average out switching noise
($100 \div f_{sw}$ is a good starting point)

Settling (T_{settle})

Must be sufficient to allow system to adjust to new stimulus pattern, generally at least $2 \div f_{0dB}$

Start Analysis at Time Delay

Must be long enough to allow circuit to reach steady-state



FRA Device Settings: General

Frequency Response Analyzer - @1

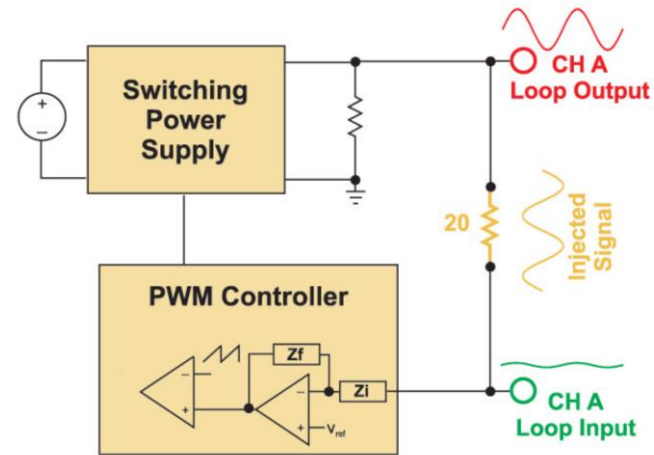
Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

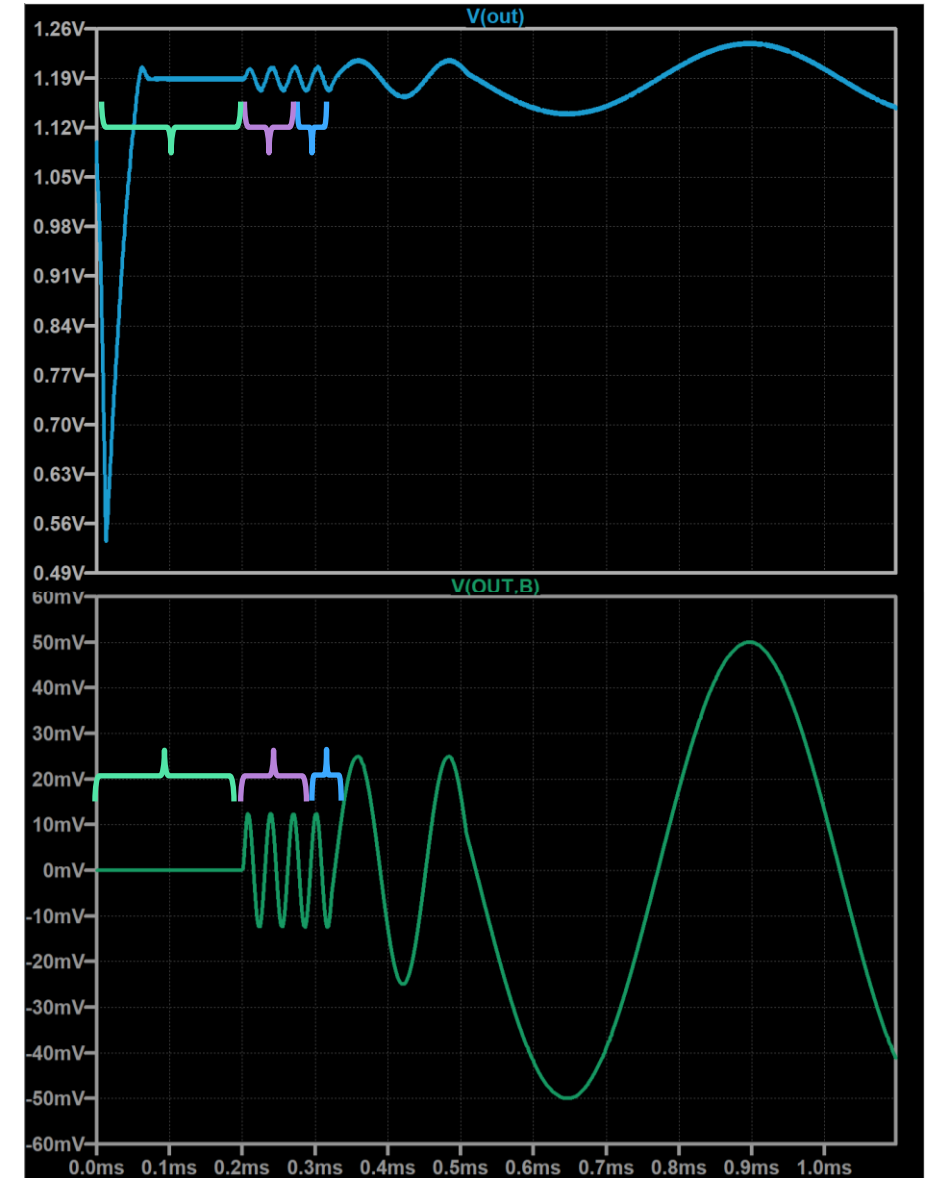
Stimulus Frequency
Start at Frequency[Hz]*: 2k
End at Frequency[Hz]*: 32k
0.5 points per octave (1.66 points per decade)
Coarse Steps up to Frequency[Hz]:
Add These Specific Frequencies[Hz]:
Max # of Simultaneous Harmonics:

Stimulus Amplitude
V(pp0): 100m
V(pp1): 25m
F0[Hz]: 4k
F1[Hz]: 16k

General
Start Analysis at Time[sec]: 250u
Minimum Analysis Time At Each Frequency [sec]: 50u
Stimulus Settling Time At Each Frequency [sec]: 100u
Intermediate Node Name:
Disable This Analyzer ☐
Calculated Duration (Simulated Time): 1.29781m sec
OK Cancel



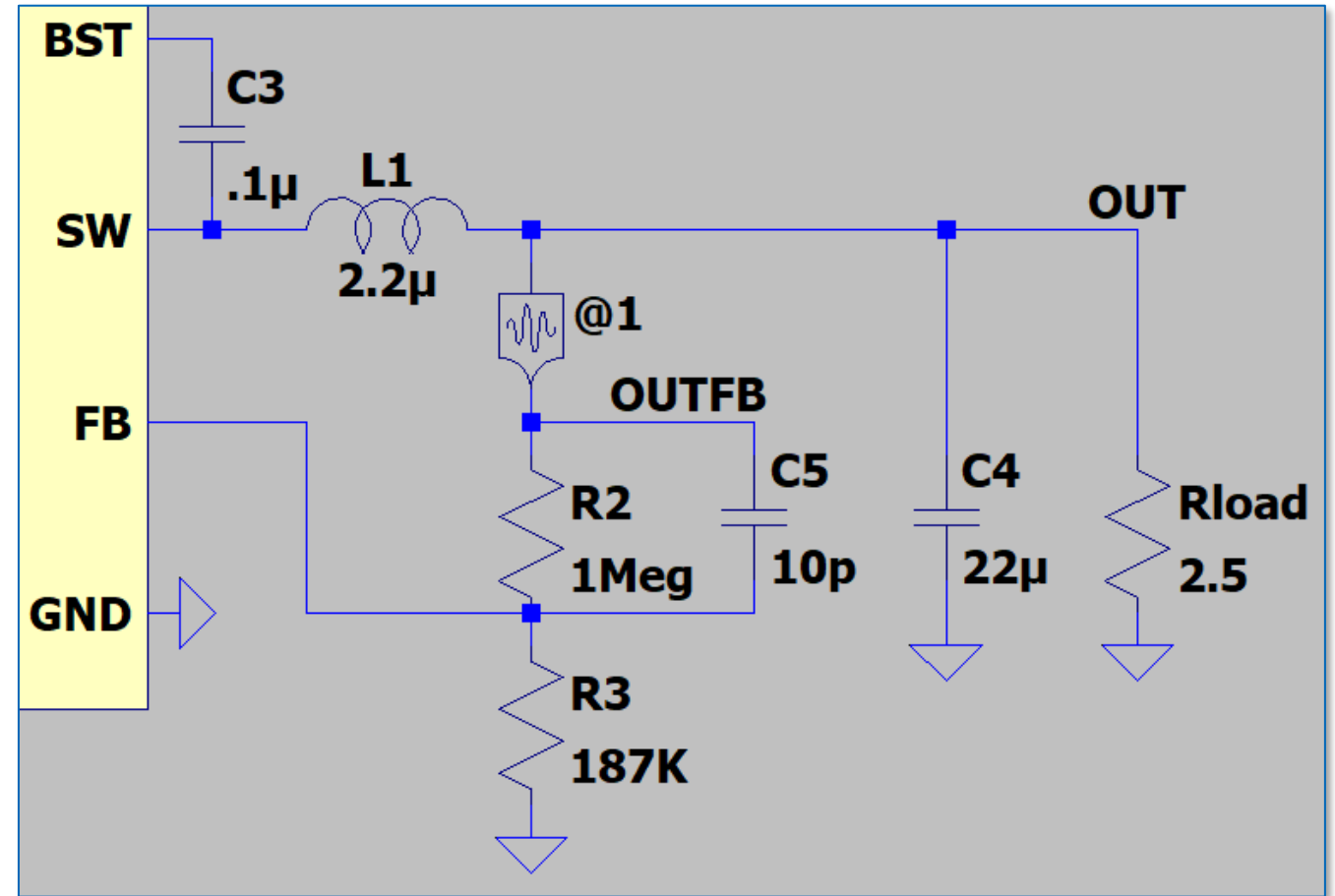
Inspect the resulting waveform in the transient simulation, and adjust these values accordingly



FRA Step by Step Setup Procedure

LTspice FRA: Step-by-Step How-To

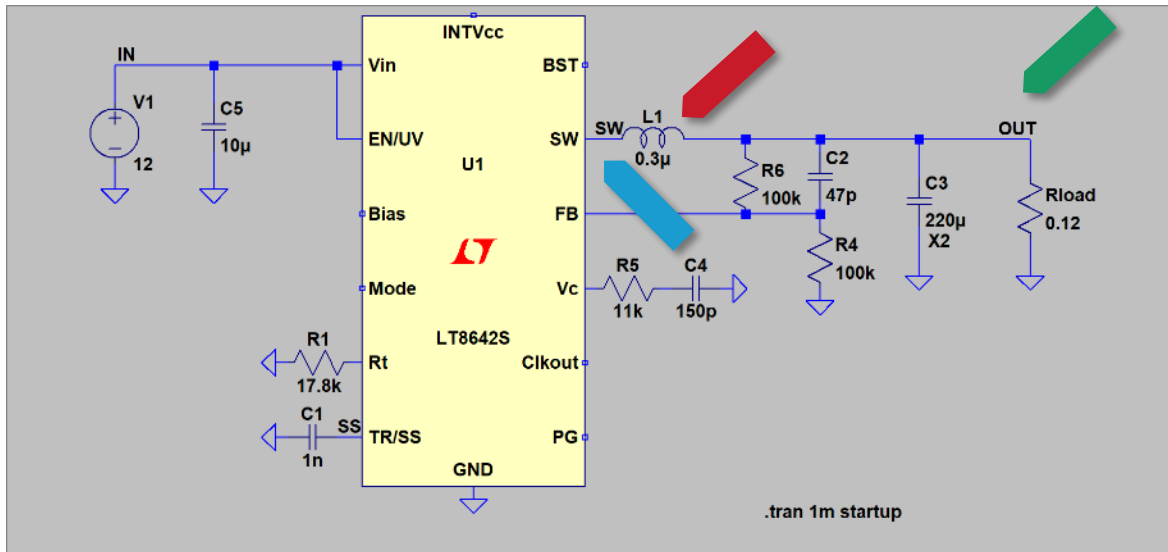
- 1) Investigate and prepare the circuit
- 2) Insert the FRA component
- 3) Initial FRA setup
- 4) Inspect the FRA transient waveforms
- 5) Initial Bode Analysis
- 6) Optimize the stimulus
- 7) Extend to more frequencies
- 8) Speed up the analysis



Live Demo

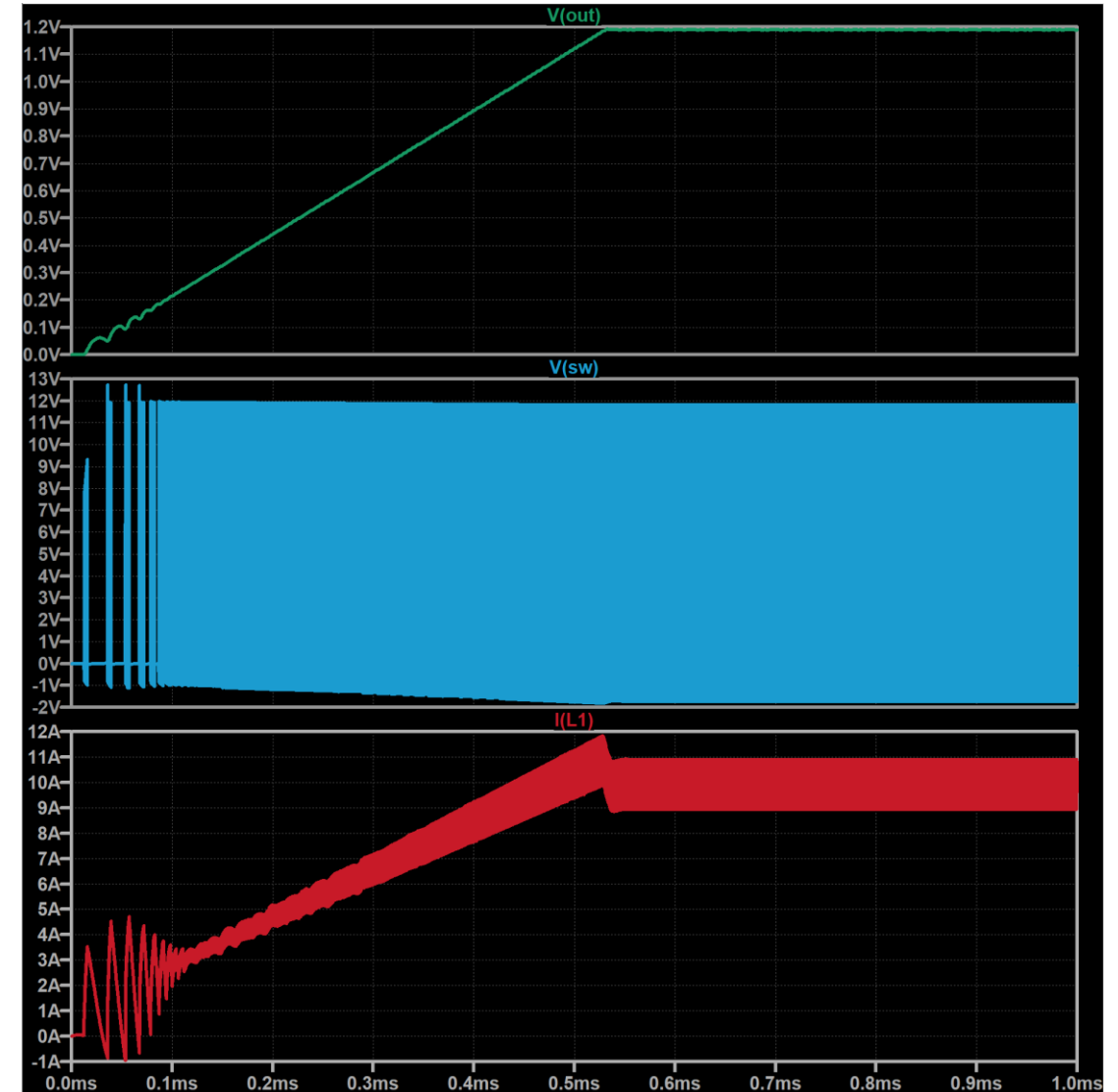
1a. Investigate and Prepare the Circuit–Basic

Explore Basic .tran Startup



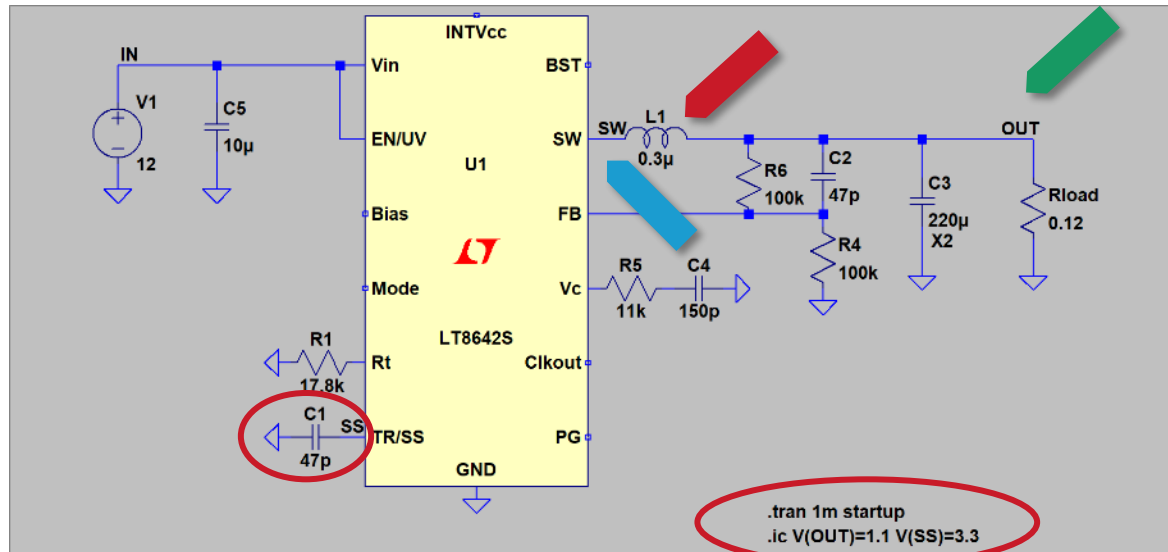
Plot and examine

- Switch node
- Inductor current
- Output voltage



1b. Investigate and Prepare the Circuit–Improved

Explore .tran with *Faster Startup*

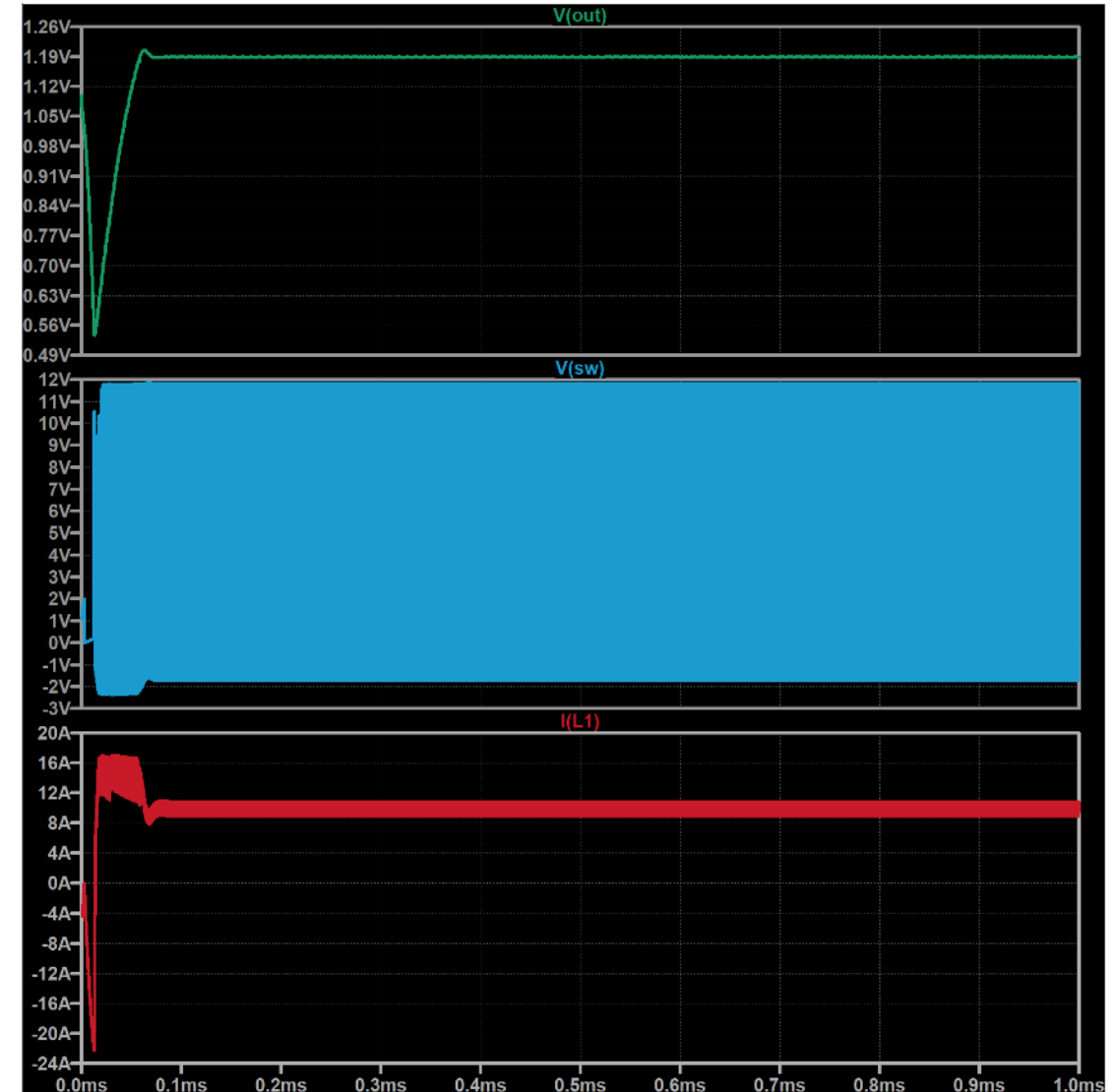


Consider speeding up the startup

- Add initial conditions, **reduce soft-start cap**
- Try higher trtol

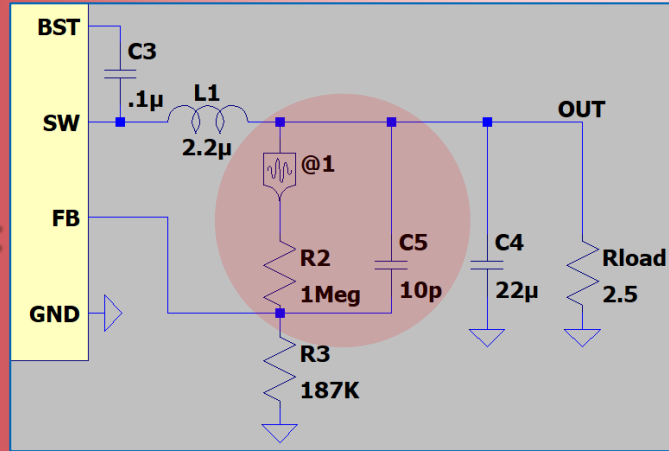
Make note of

- Time required to settle to steady-state (~80µs in this example compared to 550µs previously)
- Final output voltage (~1.2V)
- Switching frequency (~2MHz, not shown)

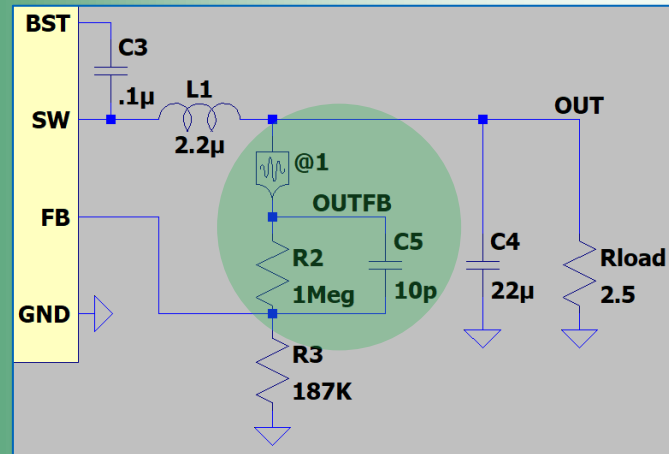


2. Break the Loop

Incorrect



Correct



Criteria

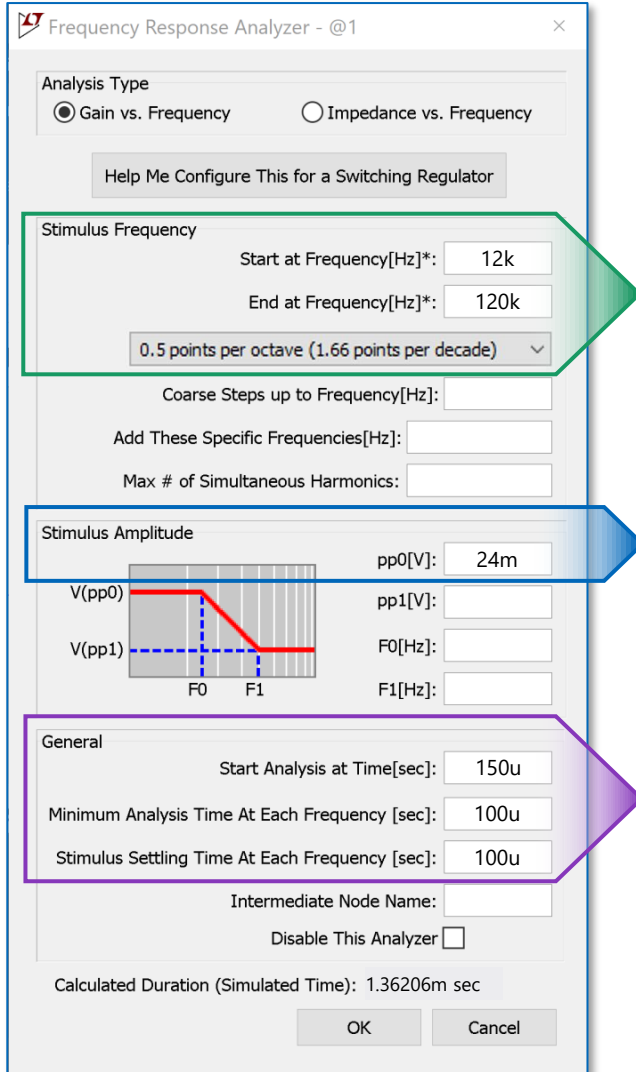
- Interrupt **all** feedback paths
- FRA component must point from **lower impedance (flat side)** to **higher impedance (pointy side)**

This requires engineering

- LTspice does not know the correct placement
- Many circuits have multiple places where the loop can be broken—if in doubt, try two places and compare the results (adjust the stimulus amplitude appropriately)

More examples near the end of this presentation

3. Initial FRA Device Setup—Start Small



Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency
☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

Stimulus Frequency

Start at Frequency[Hz]*: 12k

End at Frequency[Hz]*: 120k

0.5 points per octave (1.66 points per decade) ▼

Coarse Steps up to Frequency[Hz]:

Add These Specific Frequencies[Hz]:

Max # of Simultaneous Harmonics:

Stimulus Amplitude

pp0[V]: 24m

V(pp0)

V(pp1)

F0 F1

pp1[V]:

F0[Hz]:

F1[Hz]:

General

Start Analysis at Time[sec]: 150u

Minimum Analysis Time At Each Frequency [sec]: 100u

Stimulus Settling Time At Each Frequency [sec]: 100u

Intermediate Node Name:

Disable This Analyzer ☐

Calculated Duration (Simulated Time): 1.36206m sec

OK Cancel

Start with just two or three frequencies

- If you have an idea of the loop bandwidth, try $F_{end} = 2 \cdot f_{0dB}$ and $F_{start} = F_{end} \cdot 10$
- If you don't have a clue about the bandwidth, try $F_{start} = f_{SW} \div 160$ and $F_{end} = f_{SW} \div 10$
- Set resolution to 0.5 points per octave

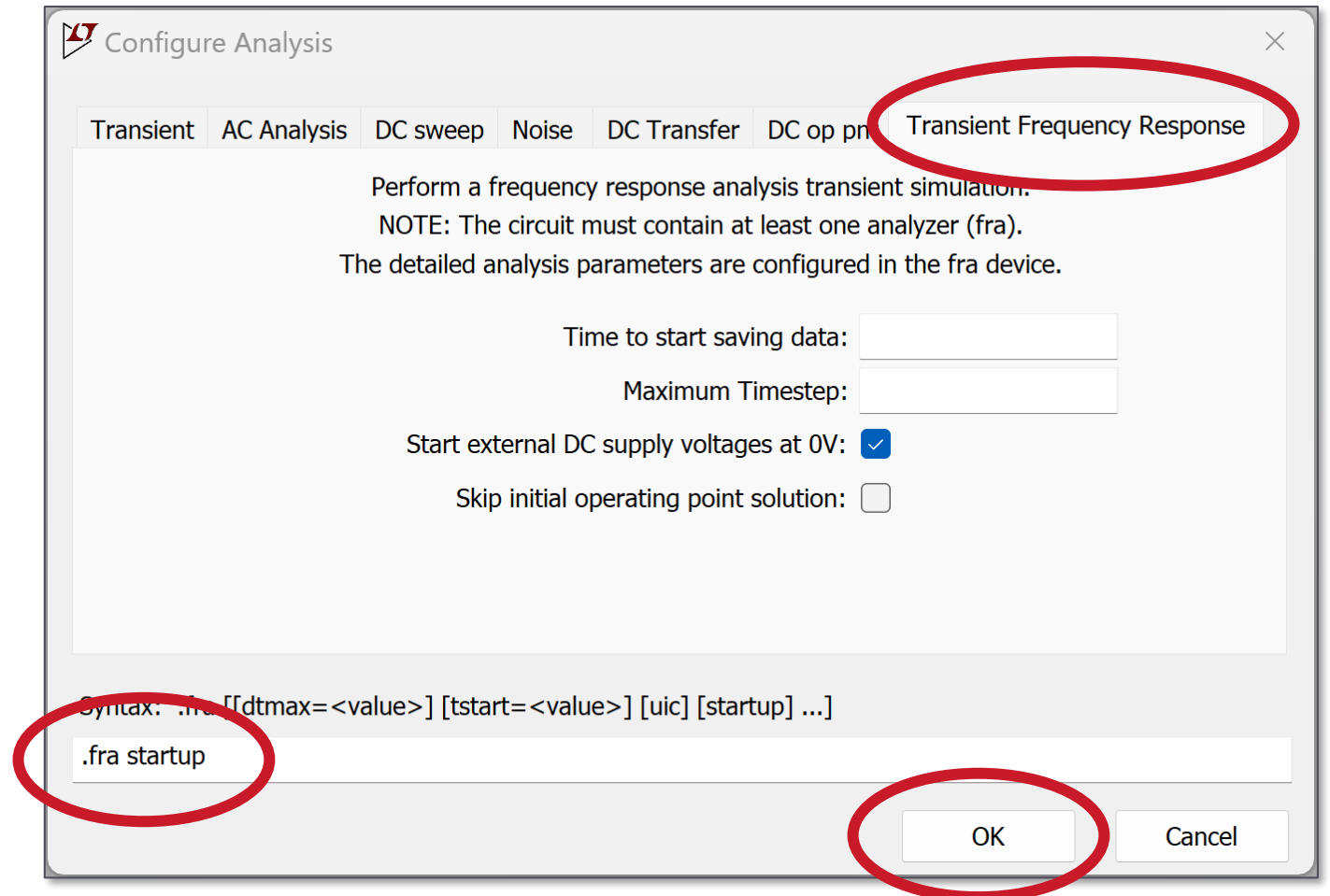
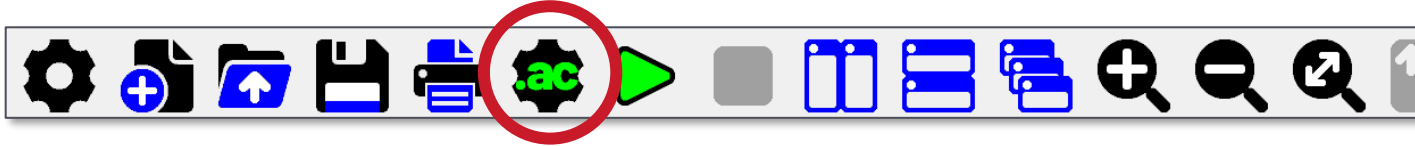
Start with pp0 = 1%~2% of common mode voltage

- Constant frequency parts tend to tolerate more stimulus than COT, boundary mode, etc.

Allow plenty of settling and averaging time

- 100~200 switching cycles is a good starting point

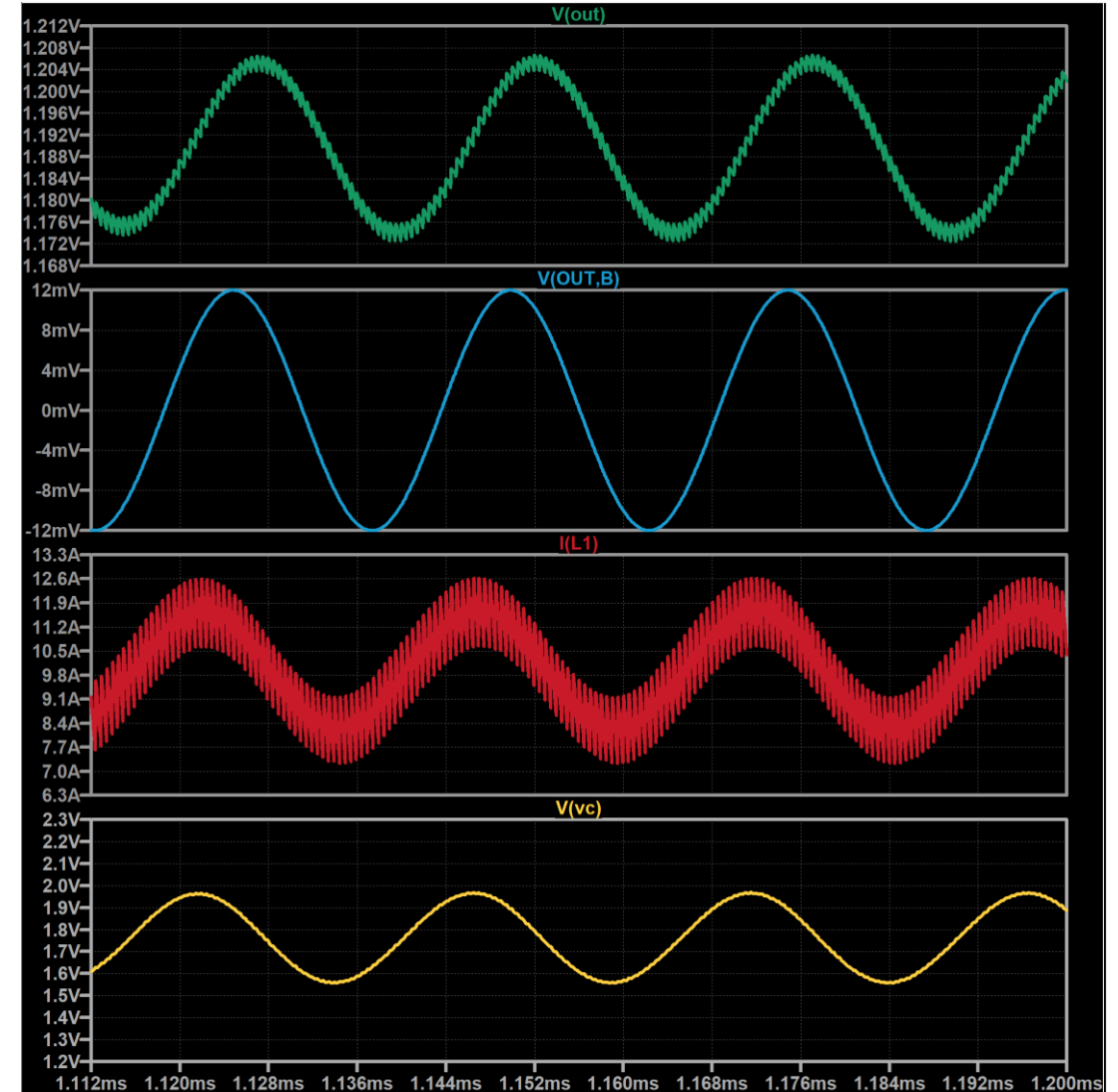
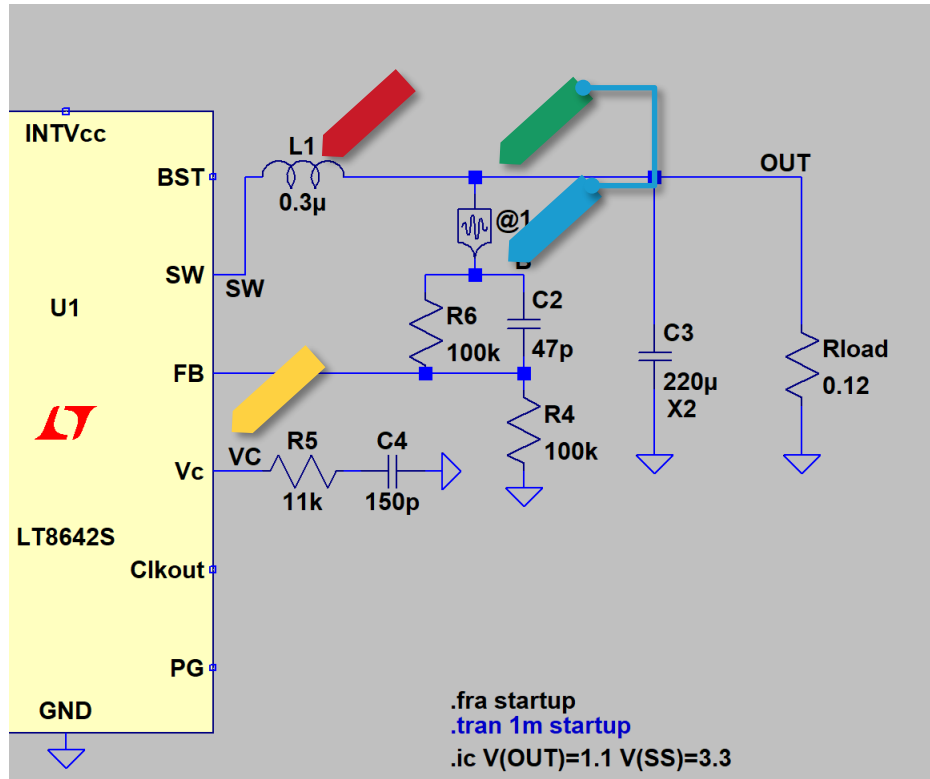
3. Add FRA SPICE directive to schematic



4a. Inspect the FRA transient waveforms

Plot and Examine

- Voltage at both FRA terminals, and the difference
- Inductor current
- Control voltage (if external)

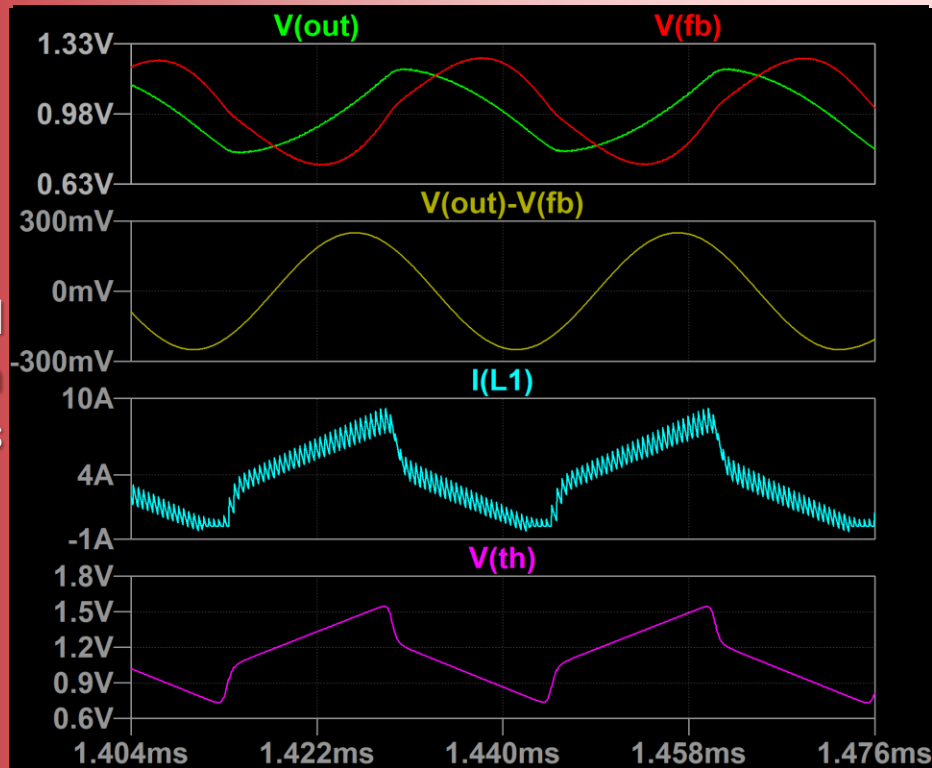


4b. Inspect the FRA transient waveforms

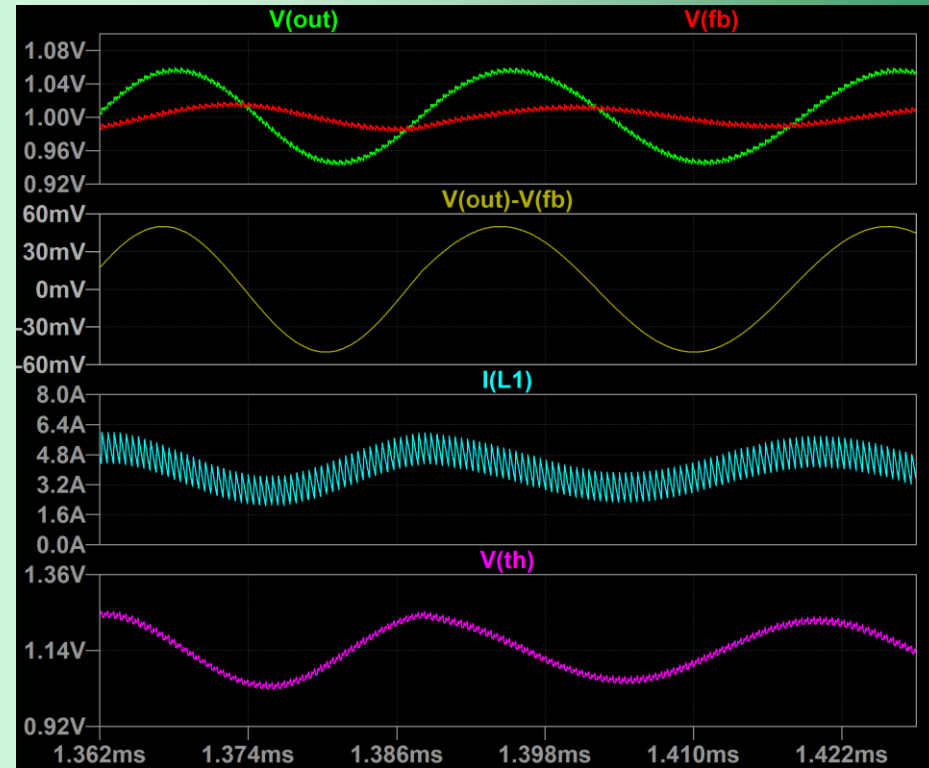
Ideally, sinusoidal pattern should be evident and symmetric

- Look for signs of non-linearity, which would indicate stimulus amplitude too large
- Note that there are discontinuities when the frequency changes – these are expected

Bad
too much
stimulus



Good



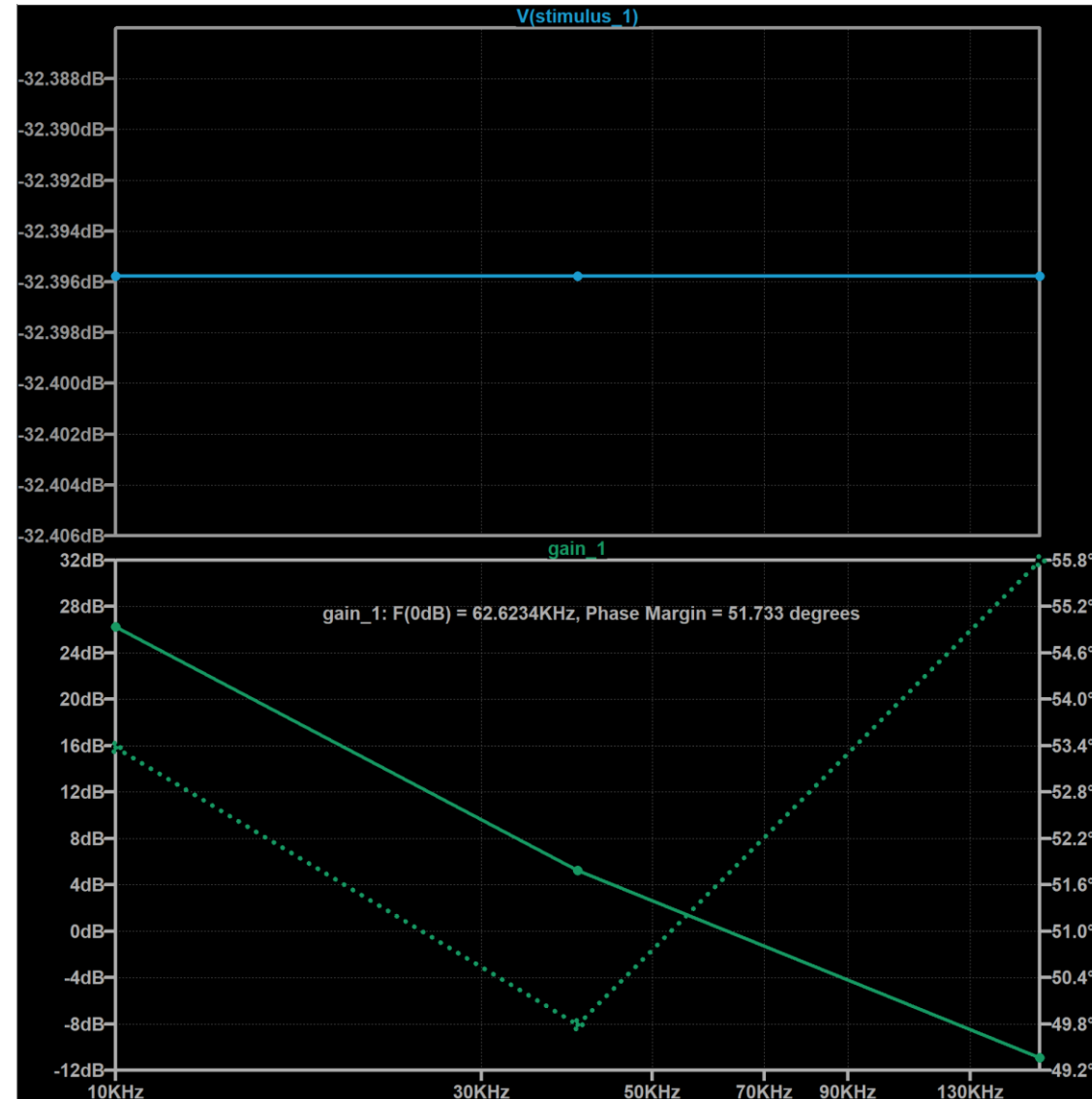
5. Initial Bode analysis

Find f_{0dB}

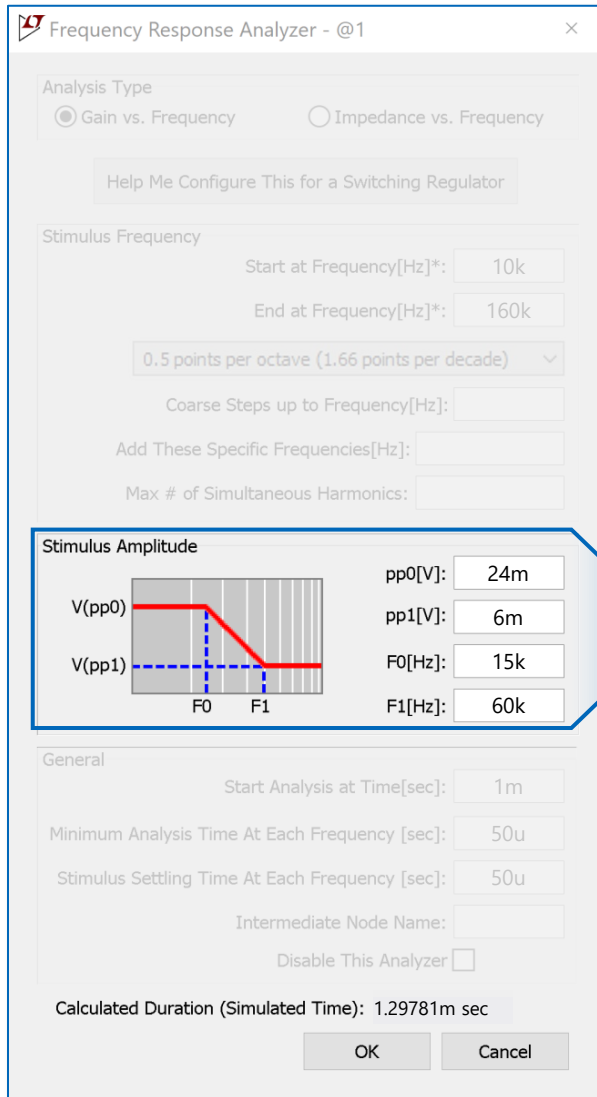
Adjust Fstart and Fend

- $F_{start} = f_{0dB} \div 8$
- $F_{end} = 2 \cdot f_{0dB}$

Re-run FRA



6a. Stimulus amplitude (Rule of Thumb)



Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency
☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

Stimulus Frequency
 Start at Frequency[Hz]*: 10k
 End at Frequency[Hz]*: 160k
 0.5 points per octave (1.66 points per decade)
 Coarse Steps up to Frequency[Hz]:
 Add These Specific Frequencies[Hz]:
 Max # of Simultaneous Harmonics:

Stimulus Amplitude

V(pp0)
 V(pp1)
 F0
 F1

pp0[V]: 24m
 pp1[V]: 6m
 F0[Hz]: 15k
 F1[Hz]: 60k

General
 Start Analysis at Time[sec]: 1m
 Minimum Analysis Time At Each Frequency [sec]: 50u
 Stimulus Settling Time At Each Frequency [sec]: 50u
 Intermediate Node Name:
 Disable This Analyzer ☐

Calculated Duration (Simulated Time): 1.29781m sec

OK Cancel

Specify frequency dependence

Need more signal at lower frequency due to high gain

Need less signal at higher frequency to avoid non-linear effects

A good starting point is:

- $pp0 = V_{CM} \div 25 \sim 50$
- $pp1 = pp0 \div 4$
- $F0 = f_{0dB} \div 4$
- $F1 = f_{0dB}$

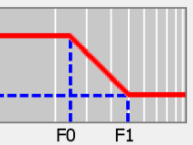
6a. Stimulus amplitude (Rule of Thumb)

Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

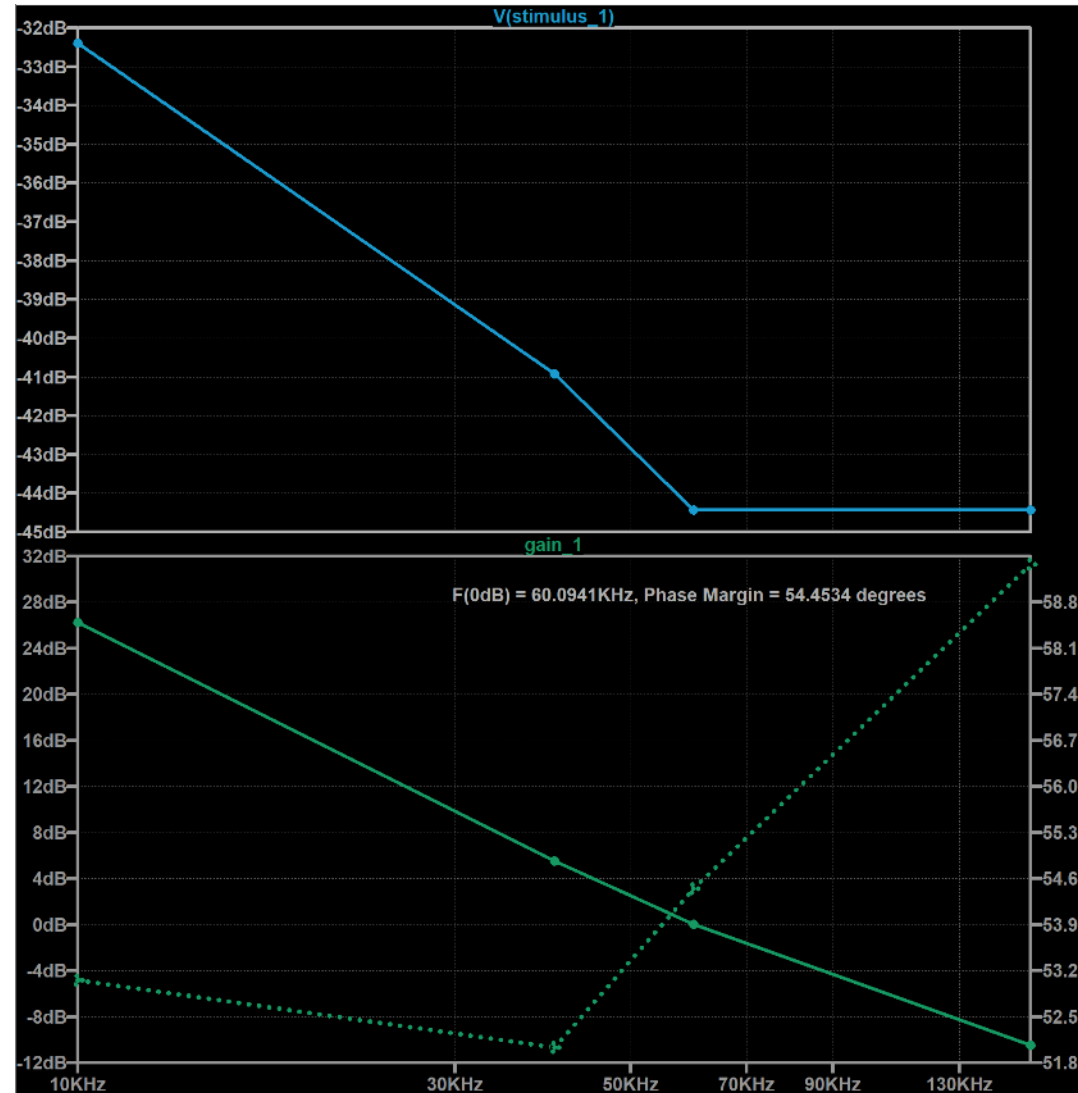
Stimulus Frequency
Start at Frequency[Hz]*: 10k
End at Frequency[Hz]*: 160k
0.5 points per octave (1.66 points per decade)
Coarse Steps up to Frequency[Hz]:
Add These Specific Frequencies[Hz]:
Max # of Simultaneous Harmonics:

Stimulus Amplitude


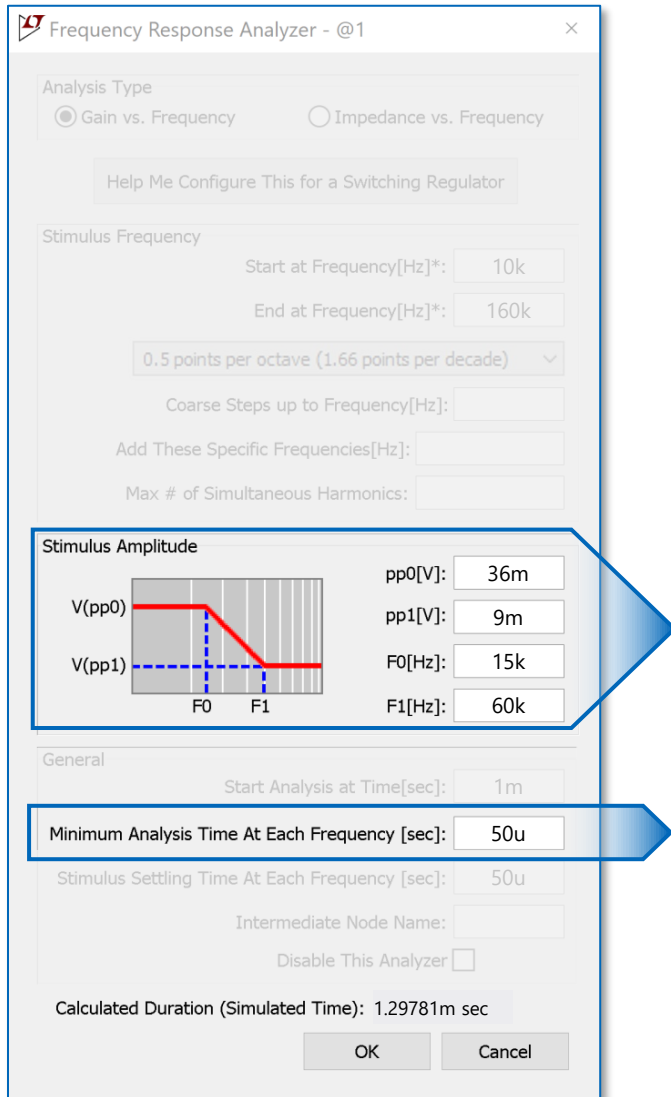
pp0[V]: 24m
pp1[V]: 6m
F0[Hz]: 15k
F1[Hz]: 60k

General
Start Analysis at Time[sec]: 1m
Minimum Analysis Time At Each Frequency [sec]: 50u
Stimulus Settling Time At Each Frequency [sec]: 50u
Intermediate Node Name:
Disable This Analyzer ☐
Calculated Duration (Simulated Time): 1.29781m sec

OK
Cancel

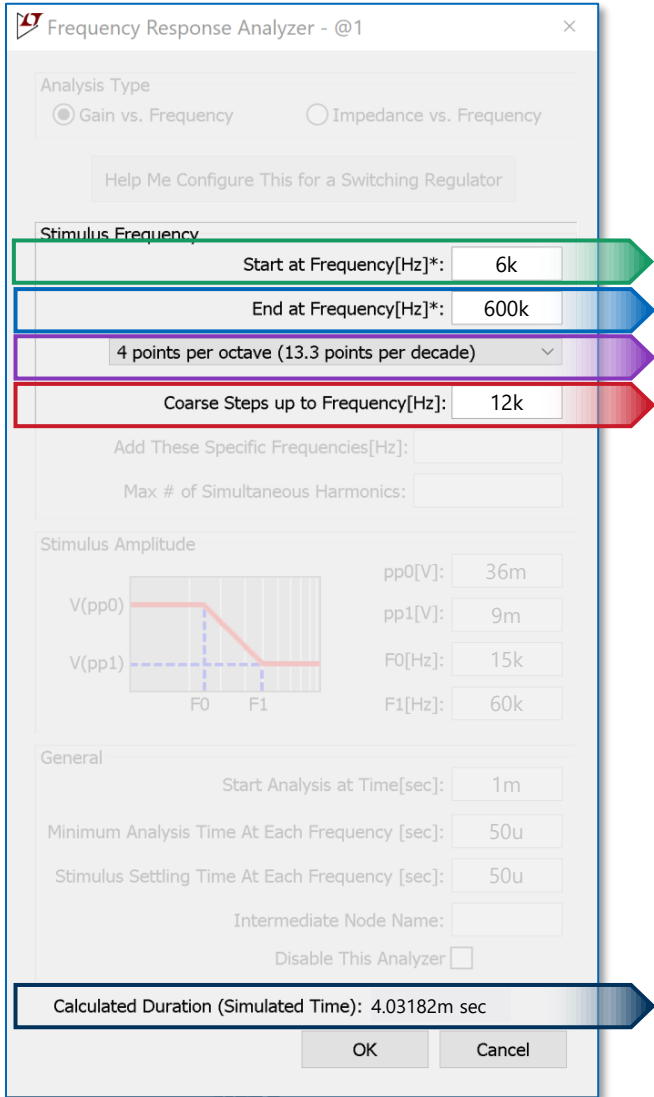


6b. Stimulus amplitude (fixes if needed)



- Increase amplitude up to 5~10% of V_{CM} , reducing if asymmetry or non-linearity becomes apparent
- If stimulus is still not apparent in waveforms, increase Tavgmin ($\sim 1000 \div f_{SW}$) to allow LTspice to resolve
- Note that phase tends to be more sensitive to errors than gain
- Adjust and re-run FRA until satisfied with the transient and Bode

7. Add more frequencies



The screenshot shows the 'Frequency Response Analyzer - @1' dialog box. It has several sections: 'Analysis Type' with 'Gain vs. Frequency' selected; a 'Stimulus Frequency' section with 'Start at Frequency[Hz]*:' set to 6k, 'End at Frequency[Hz]*:' set to 600k, a dropdown for '4 points per octave (13.3 points per decade)', and 'Coarse Steps up to Frequency[Hz]:' set to 12k; an 'Add These Specific Frequencies[Hz]:' field; a 'Max # of Simultaneous Harmonics:' field; a 'Stimulus Amplitude' section with a graph and fields for V(pp0), V(pp1), F0, and F1; a 'General' section with 'Start Analysis at Time[sec]:' set to 1m, 'Minimum Analysis Time At Each Frequency [sec]:' set to 50u, 'Stimulus Settling Time At Each Frequency [sec]:' set to 50u, and an 'Intermediate Node Name:' field; and a 'Disable This Analyzer' checkbox. At the bottom, it shows 'Calculated Duration (Simulated Time): 4.03182m sec' and 'OK' and 'Cancel' buttons. Colored arrows point to specific fields: a green arrow to 'Start at Frequency[Hz]*:', a blue arrow to 'End at Frequency[Hz]*:', a purple arrow to the points per octave dropdown, a red arrow to 'Coarse Steps up to Frequency[Hz]:', and a dark blue arrow to the 'Calculated Duration' field.

Starting frequency $\sim f_{\text{0dB}} \div 10$

- Lower Fstart leads to longer simulation time.
- Usually Fstart = $f_{\text{0dB}} \div 10 \sim 20$ is sufficient.

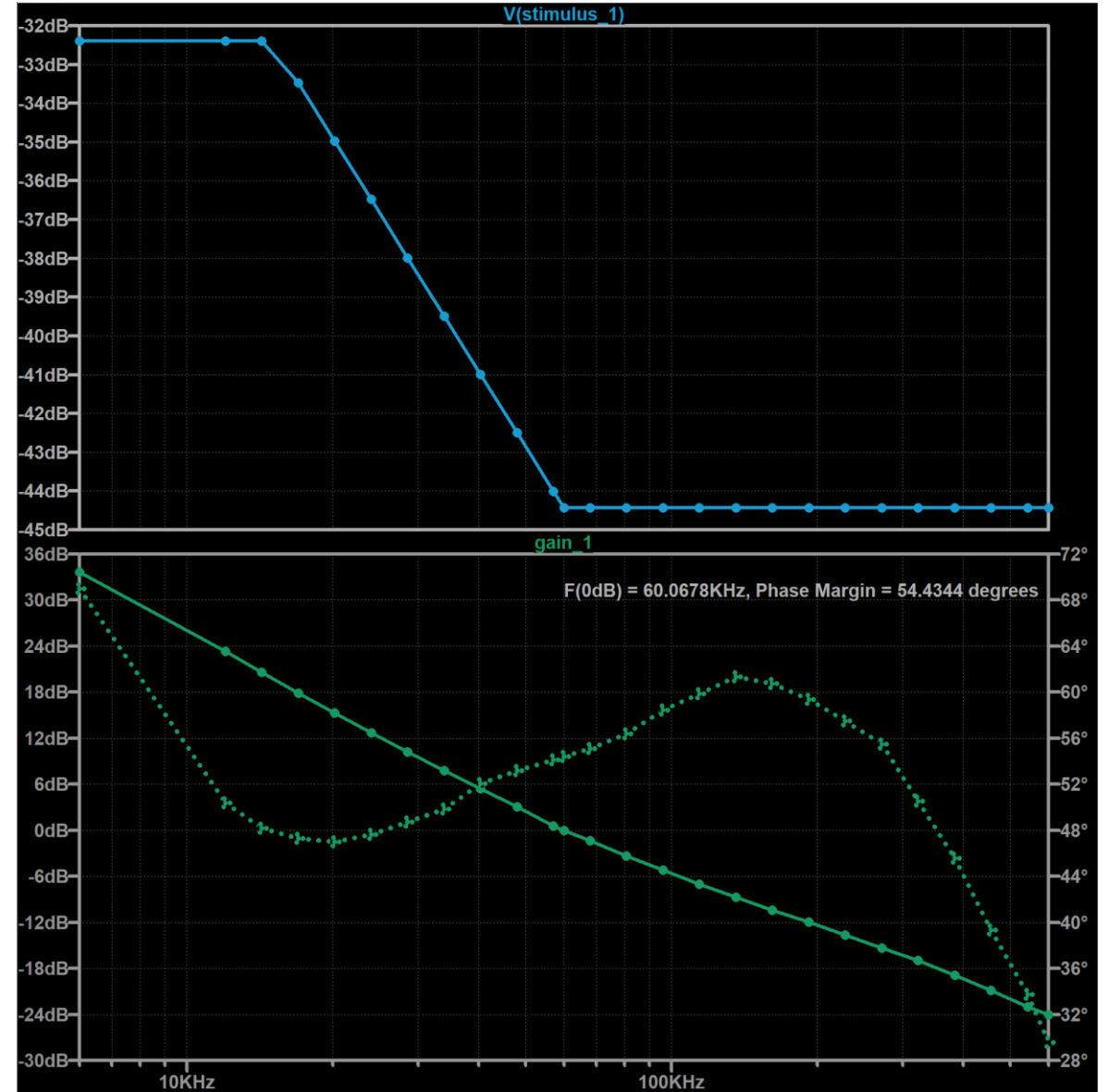
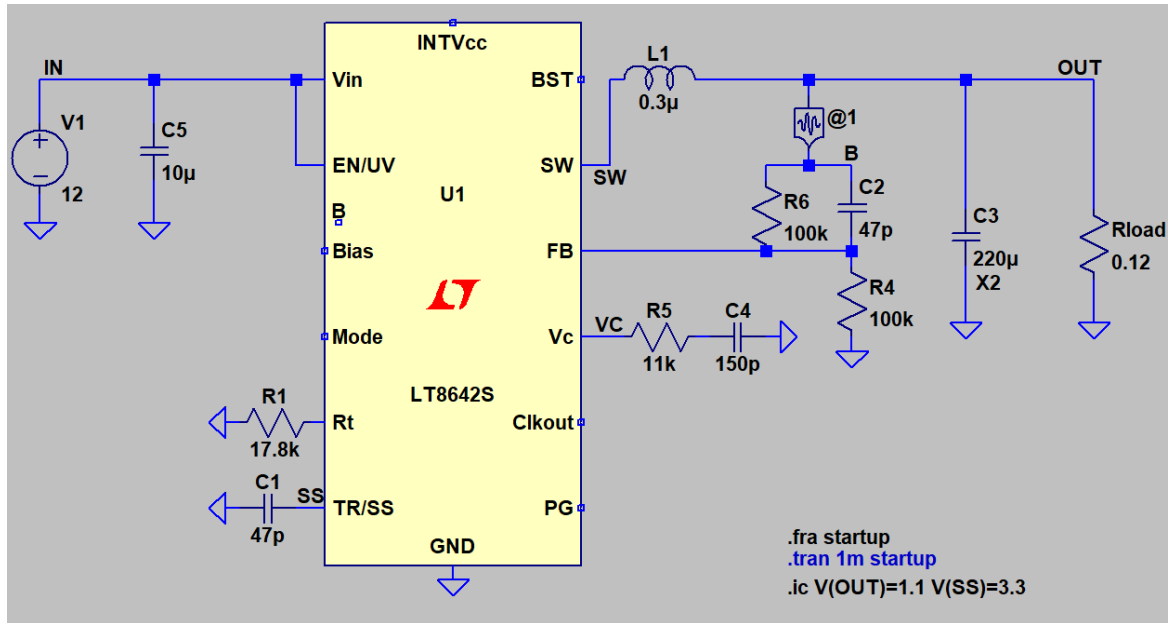
End frequency $\sim 10 \cdot f_{\text{0dB}}$

Increase frequency resolution to 2~3 points per Octave

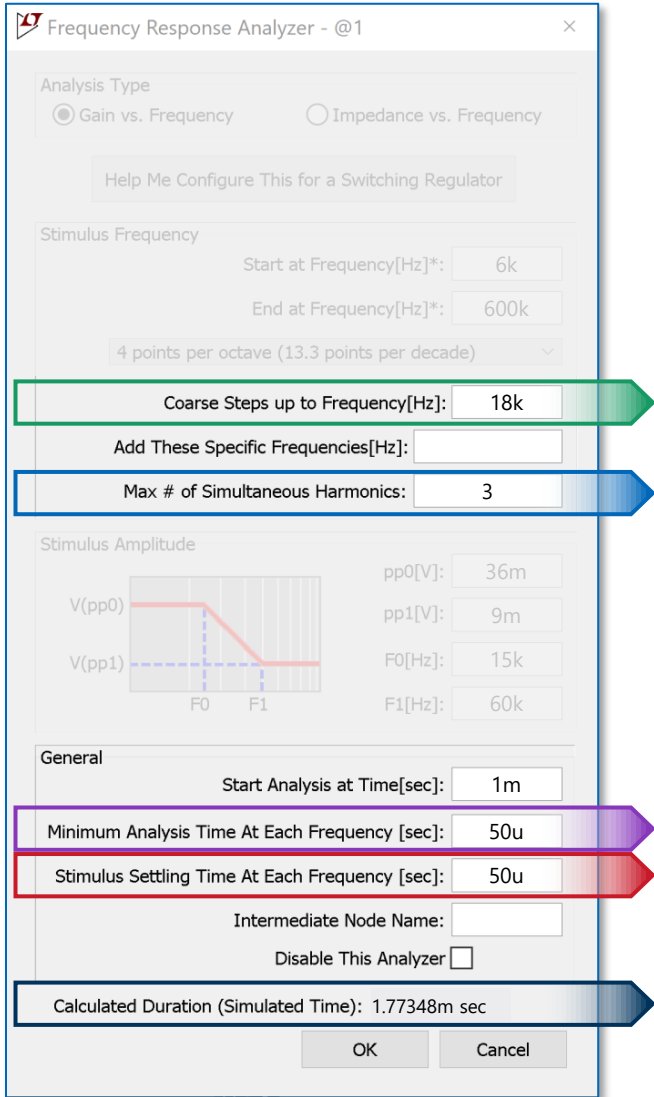
Specify Fcoarse = $2 \cdot \text{Fstart}$

Note: The Simulation time will increase as we added more data points

Voila!



8. Speed up the FRA simulation



Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency
☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

Stimulus Frequency
 Start at Frequency[Hz]*: 6k
 End at Frequency[Hz]*: 600k
 4 points per octave (13.3 points per decade)

Coarse Steps up to Frequency[Hz]: 18k

Add These Specific Frequencies[Hz]:

Max # of Simultaneous Harmonics: 3

Stimulus Amplitude
 V(pp0)
 V(pp1)
 F0
 F1

pp0[V]: 36m
 pp1[V]: 9m
 F0[Hz]: 15k
 F1[Hz]: 60k

General
 Start Analysis at Time[sec]: 1m

Minimum Analysis Time At Each Frequency [sec]: 50u

Stimulus Settling Time At Each Frequency [sec]: 50u

Intermediate Node Name:

Disable This Analyzer ☐

Calculated Duration (Simulated Time): 1.77348m sec

OK Cancel

Increase Fcoarse (usually at least $2 \sim 3 \cdot F_{start}$)

Increase max # of simultaneous harmonics (Nmax)

- Default is 1. Value of 2~4 can be quite beneficial.
- Note that this increases sensitivity to distortion → if increasing Nmax leads to phase/gain anomalies, reducing the amplitude may help
- The SPICE log indicates which harmonics are applied simultaneously

Decrease minimum analysis time (Tavgmin)

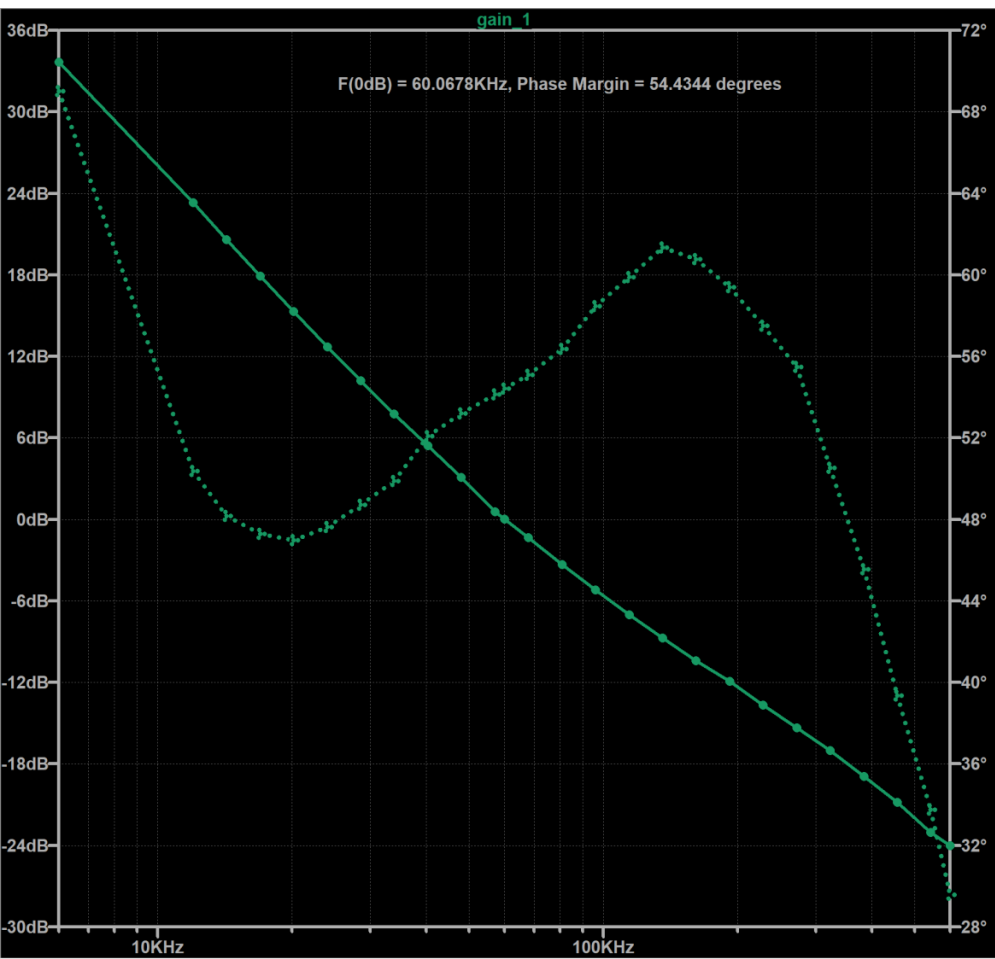
- If switching ripple is small relative to stimulus, less averaging is required (perhaps $50 \sim 100 \div f_{sw}$)
- Note that this only affects higher frequencies, so it's not worth making this too small

Reduce settling time (Tsettle) to $1 \sim 2 \div f_{0dB}$

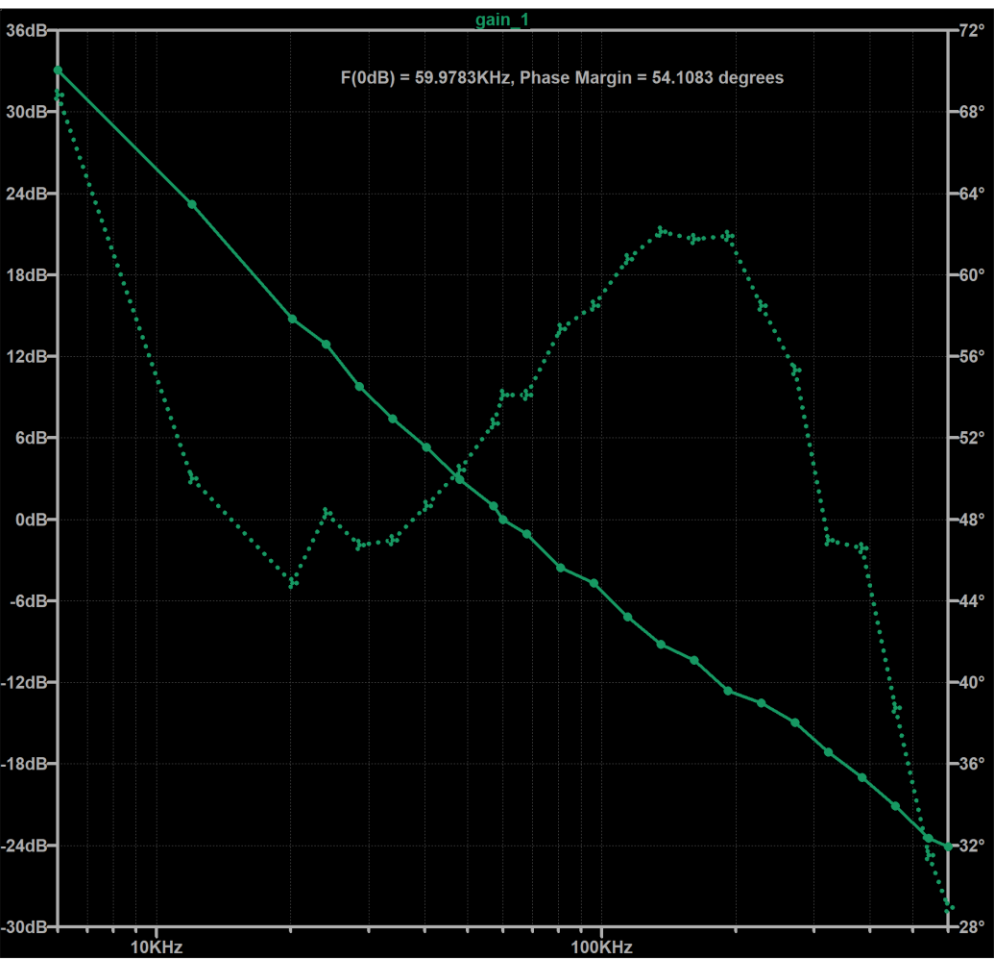
Pay attention to the sim time as you make changes

Presto!

Completed in 110 seconds

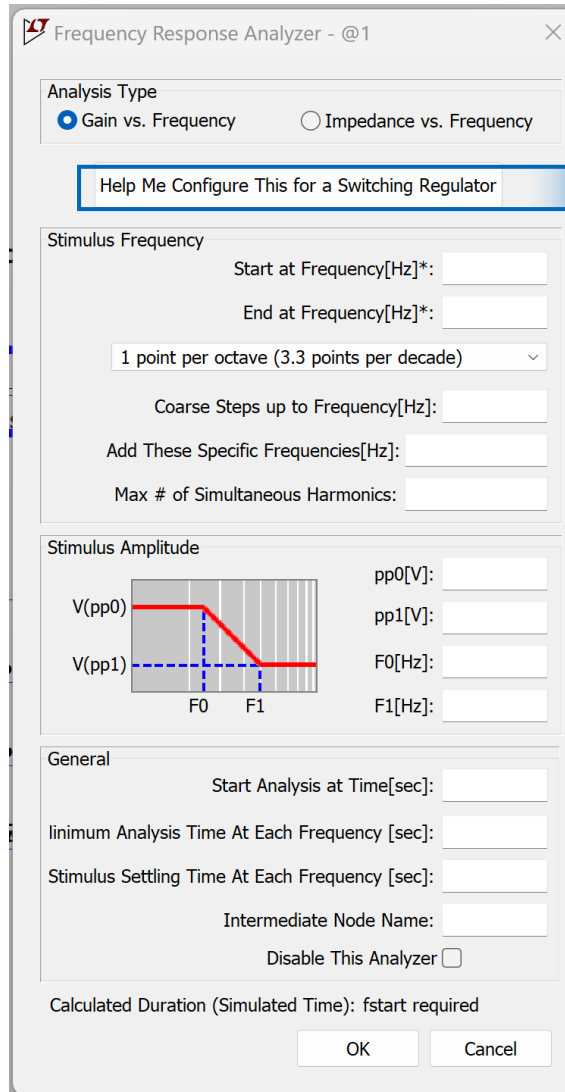


Completed in 45 seconds

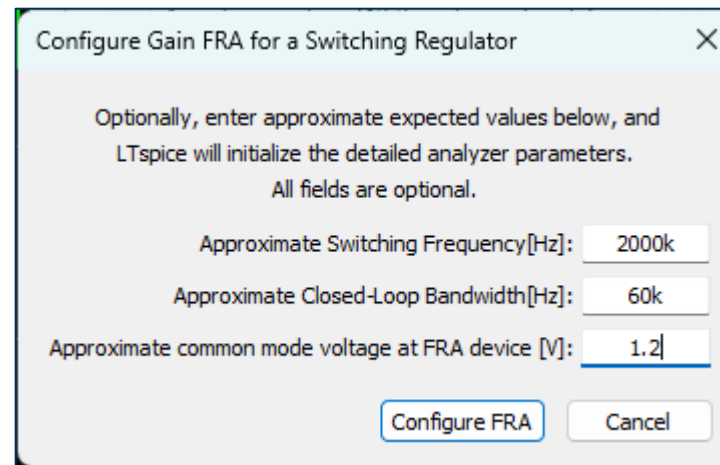


Config Wizard

FRA Device Settings: Configuration Wizard

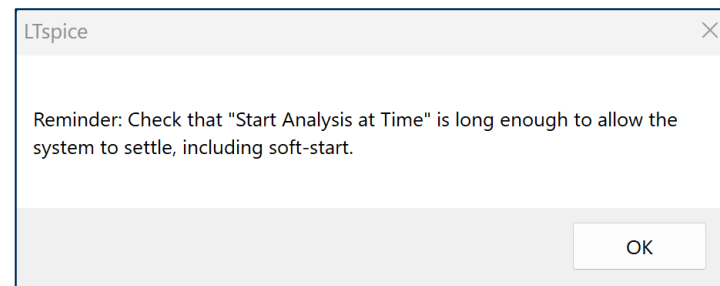


The configuration wizard is used to quickly setup a FRA simulation



Only three things to enter

- Switching frequency
- Approximate Bandwidth
- Output voltage



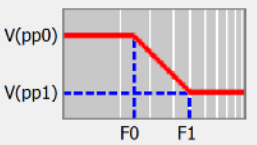
FRA Device Settings: Configuration Wizard Results

Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

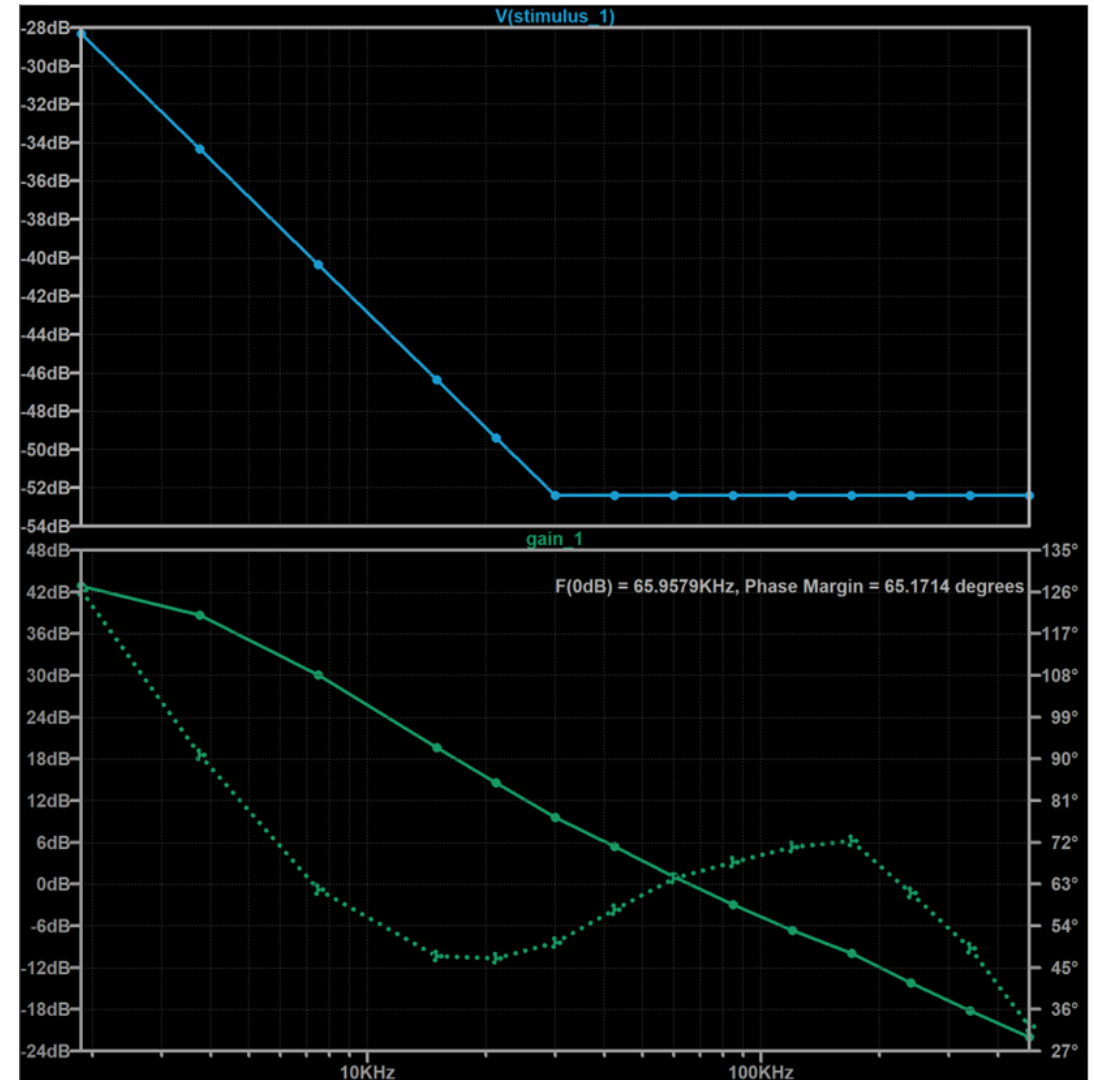
Help Me Configure This for a Switching Regulator

Stimulus Frequency
 Start at Frequency[Hz]*: 1.875K
 End at Frequency[Hz]*: 480K
 2 points per octave (6.6 points per decade)
 Coarse Steps up to Frequency[Hz]: 15K
 Add These Specific Frequencies[Hz]:
 Max # of Simultaneous Harmonics:

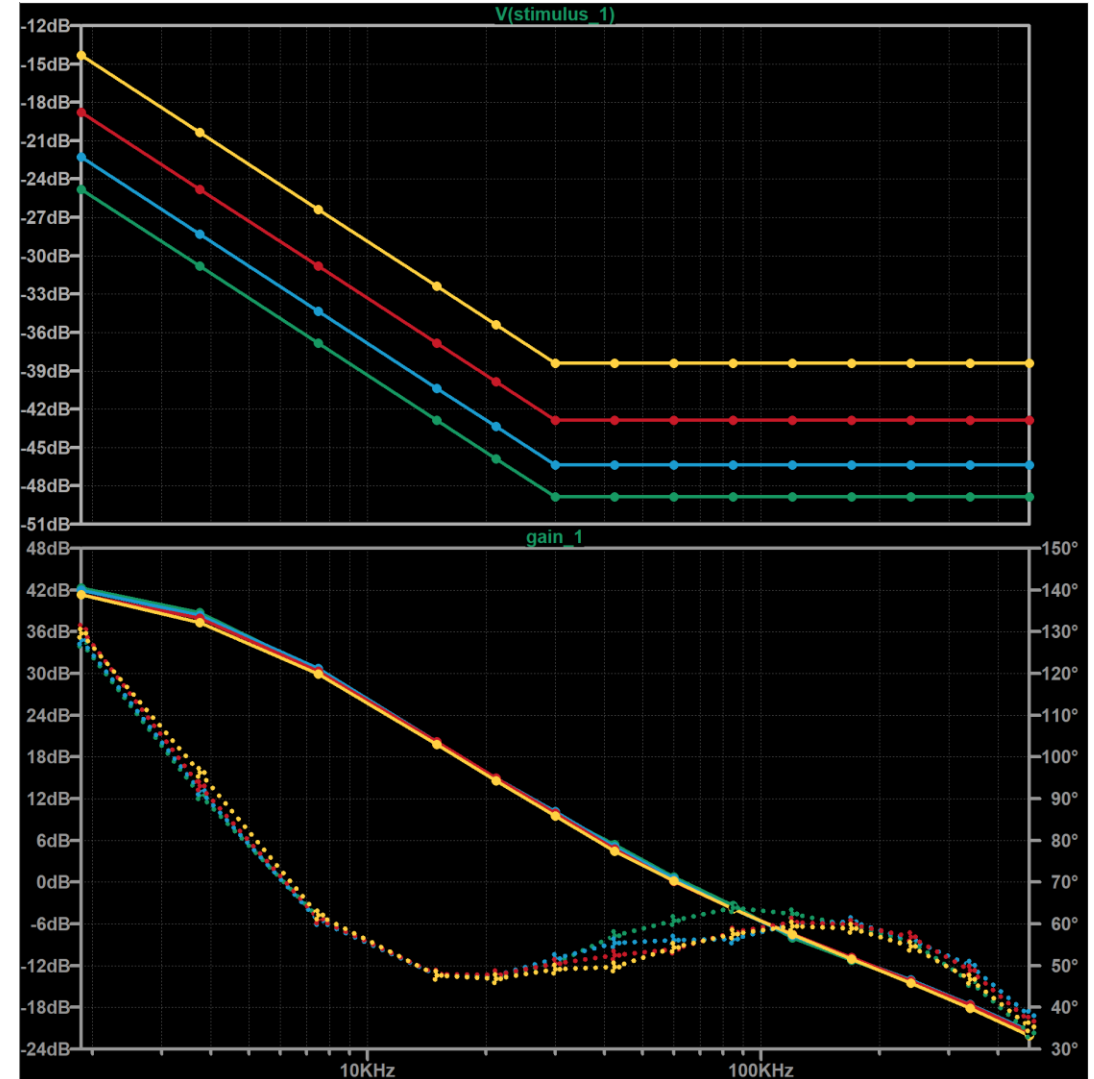
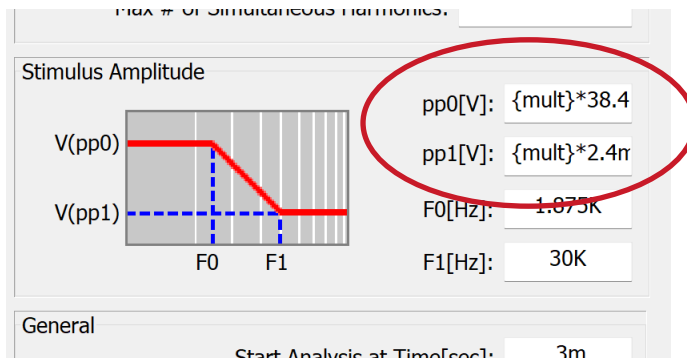
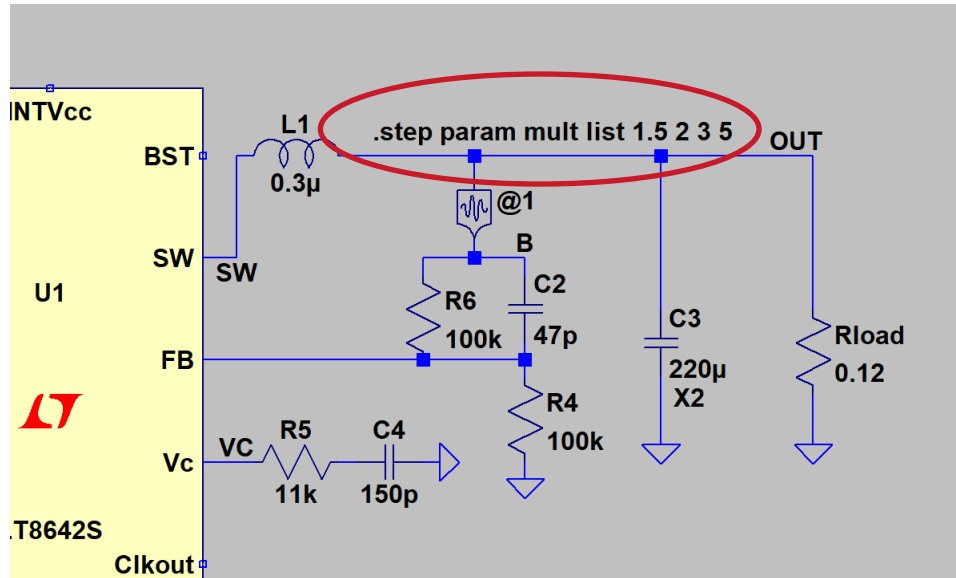
Stimulus Amplitude

 V(pp0): 38.4m
 V(pp1): 2.4m
 F0[Hz]: 1.875K
 F1[Hz]: 30K

General
 Start Analysis at Time[sec]: 1m
 Minimum Analysis Time At Each Frequency [sec]: 50μ
 Stimulus Settling Time At Each Frequency [sec]: 33.3333μ
 Intermediate Node Name:
 Disable This Analyzer ☐
 Calculated Duration (Simulated Time): 3.15336m sec

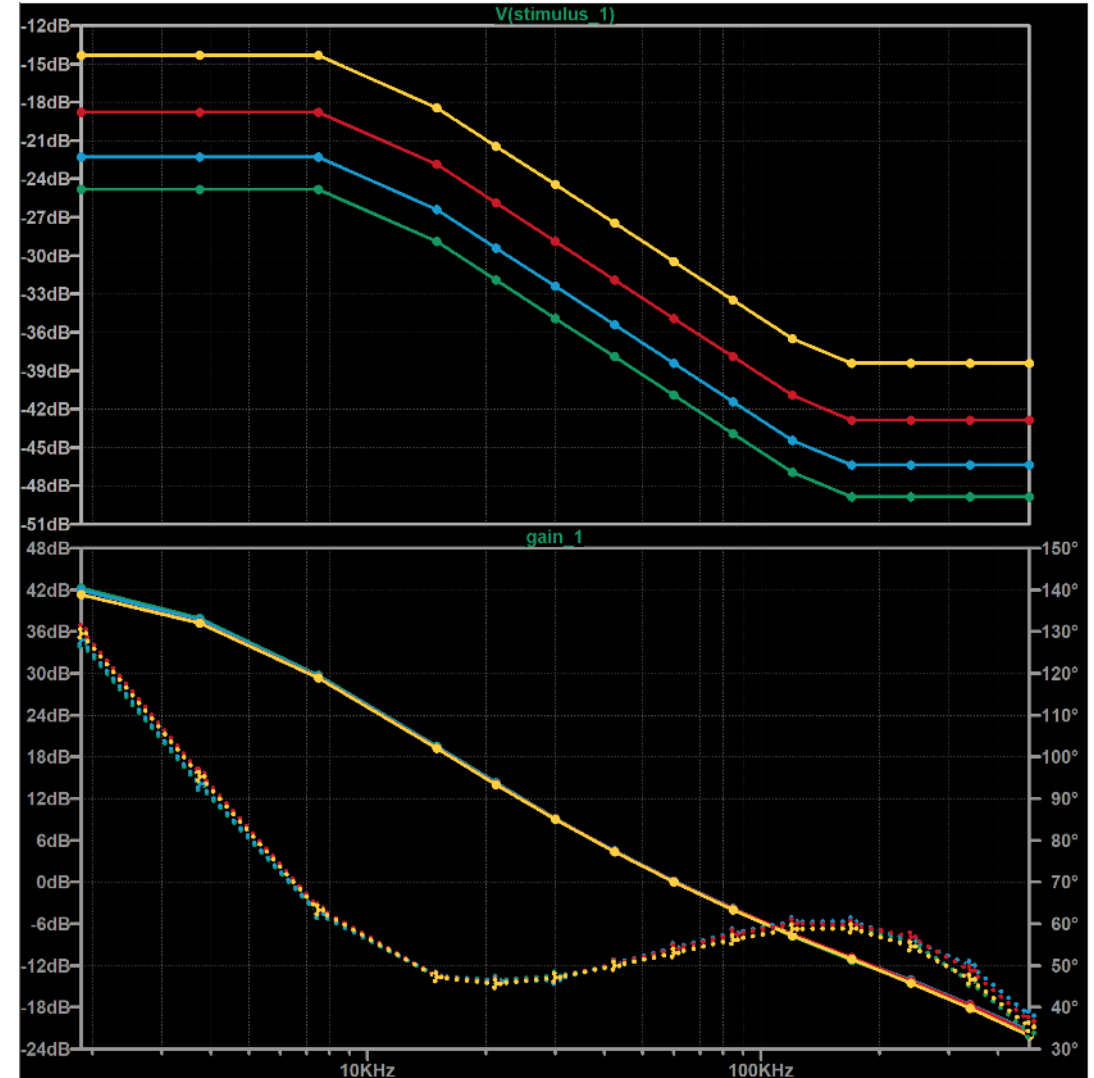
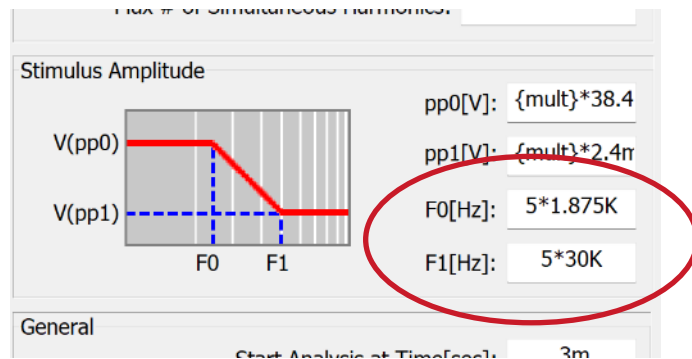
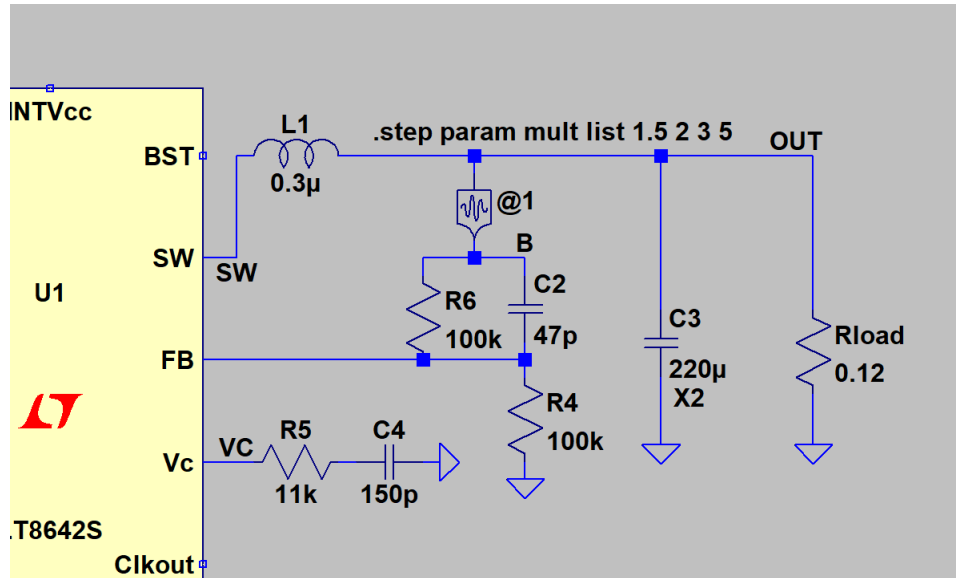
OK Cancel



FRA Device Settings: Stepping multiple amplitudes



FRA Device Settings: Shifting the corner frequencies

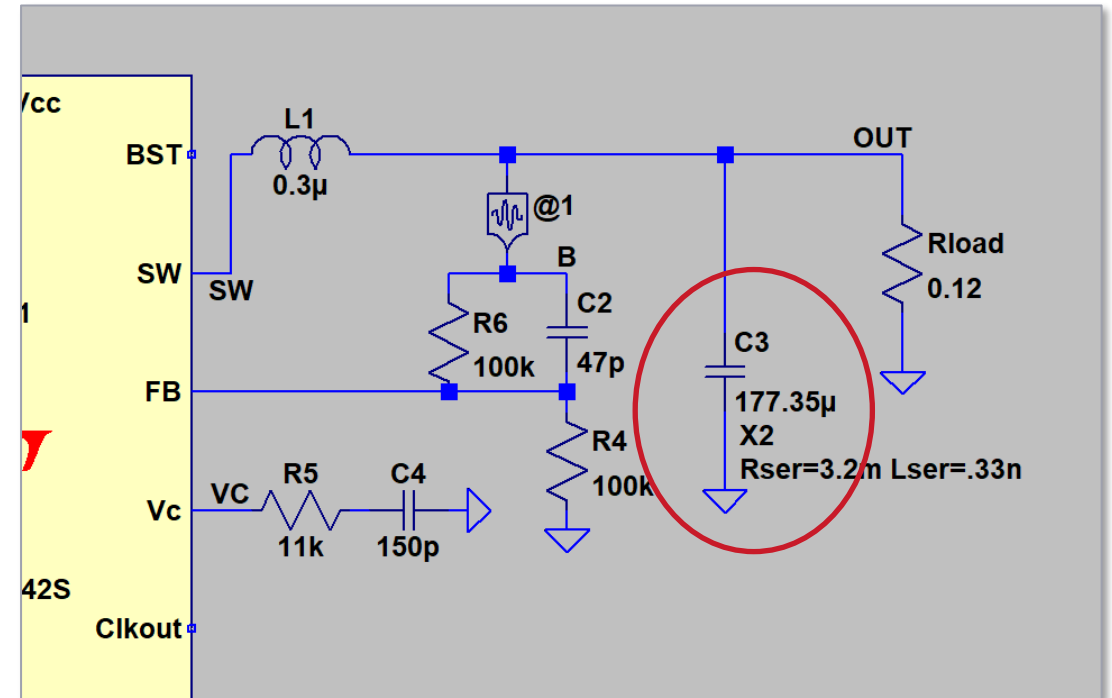
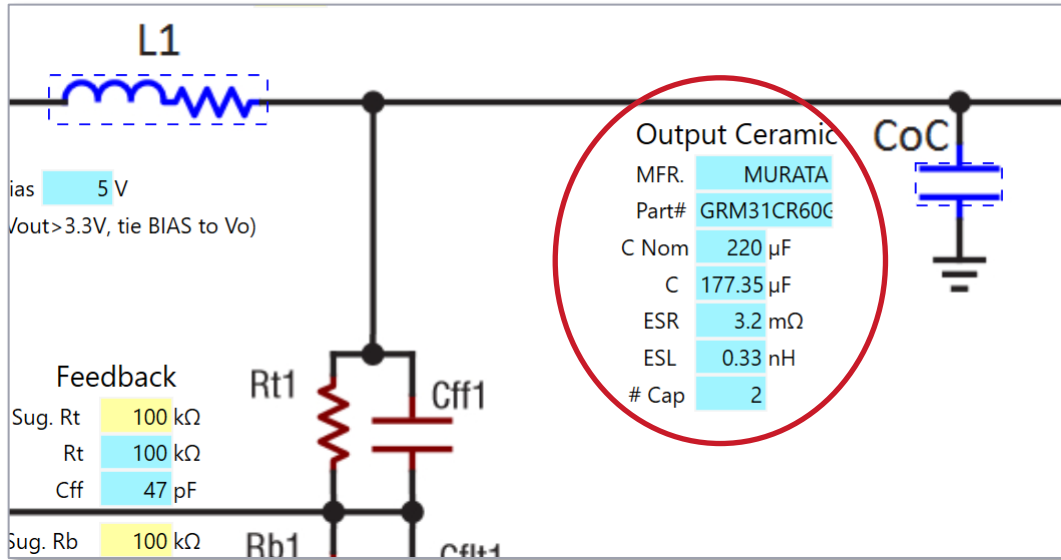




Note on Tool Correlation:

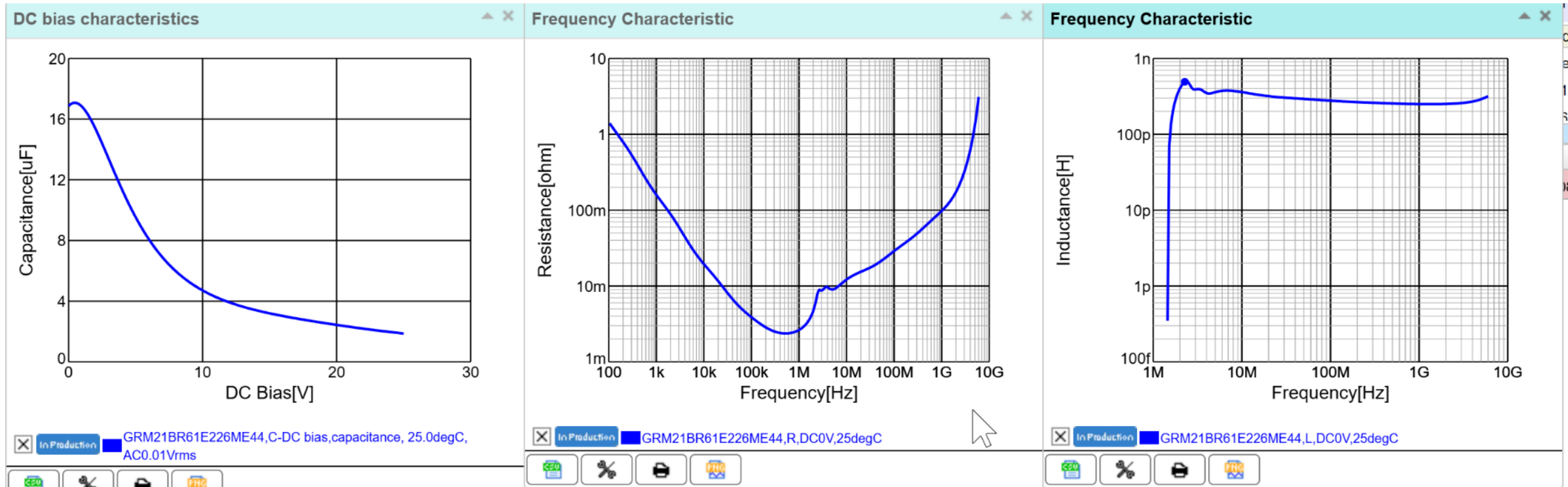
LTspice vs. LTpowerCAD

Simulation results depend on model details



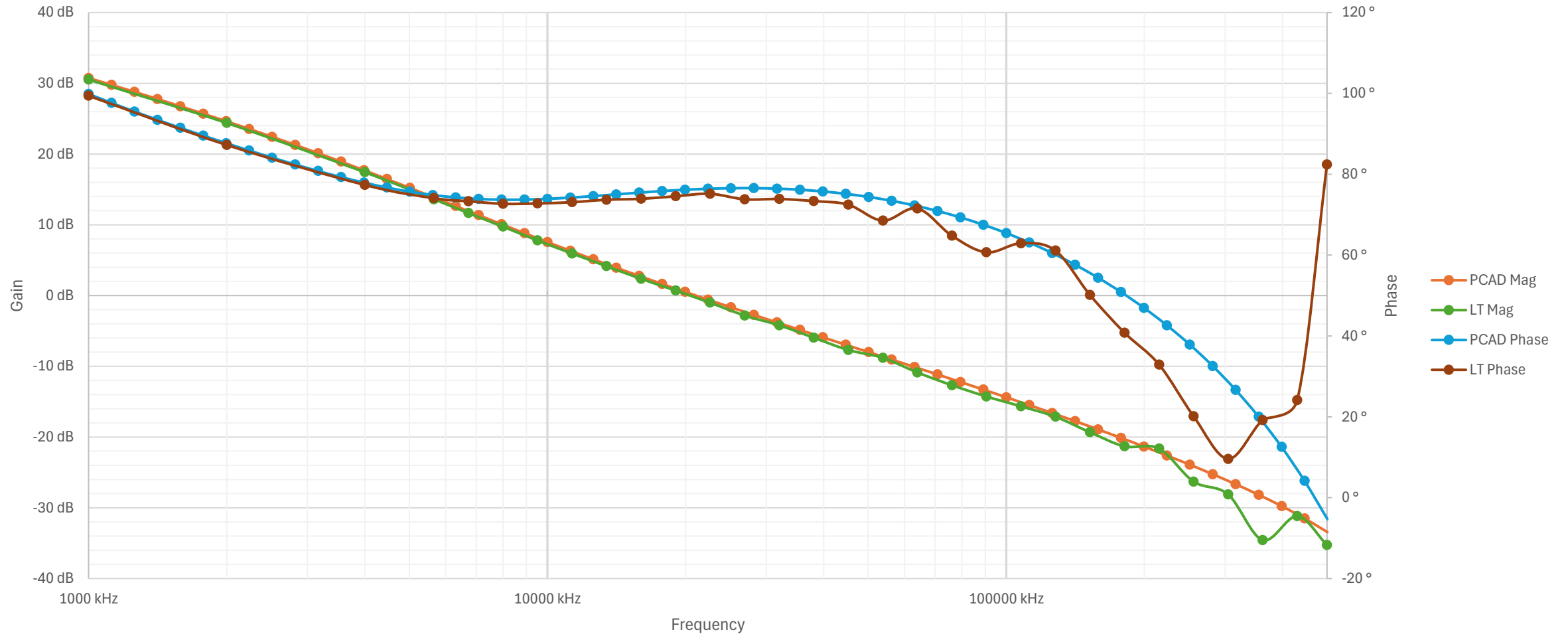
Derating caps, adding parasitics

Using Murata SimSurfing to understand capacitor parasitic values



Comparing FRA to LTpowerCAD – LT3763 example

Spice vs CAD Comparison

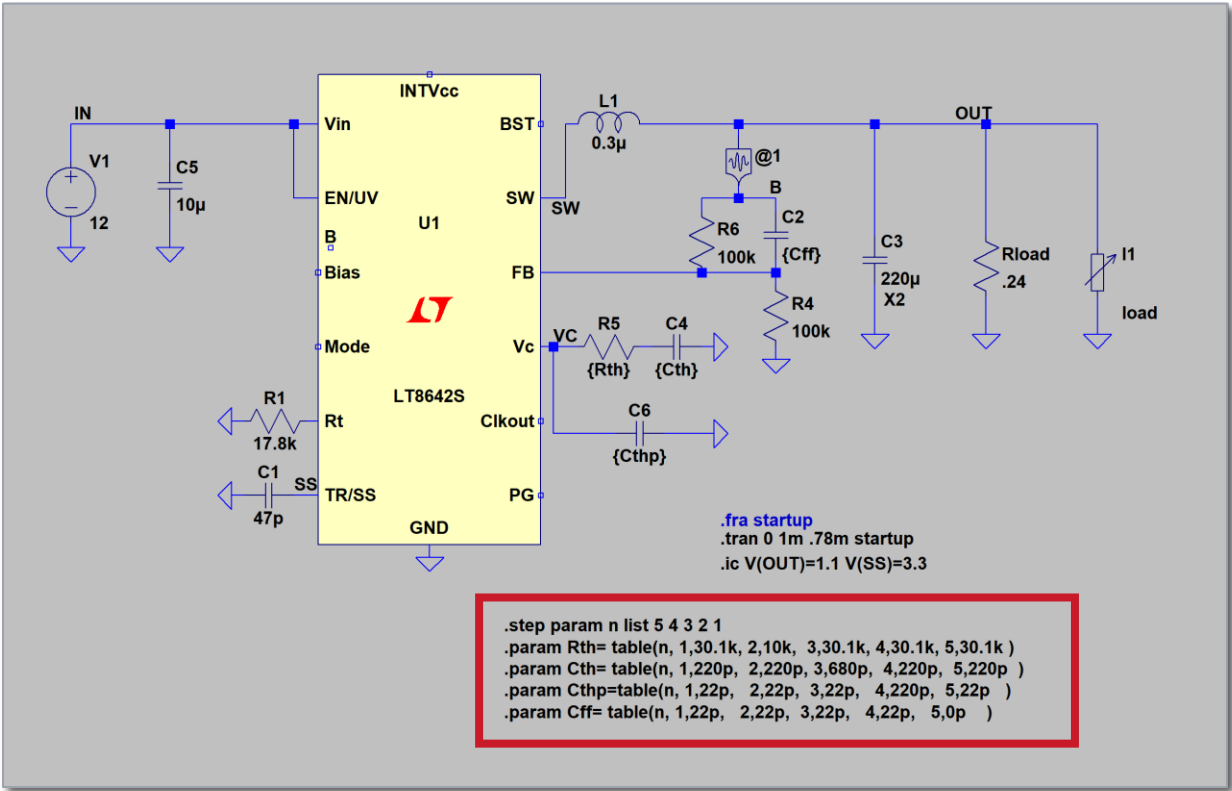


In LTspice Examples -> Educational -> FRA

Comparing Bode to Transient Response

Transient Response Comparisons

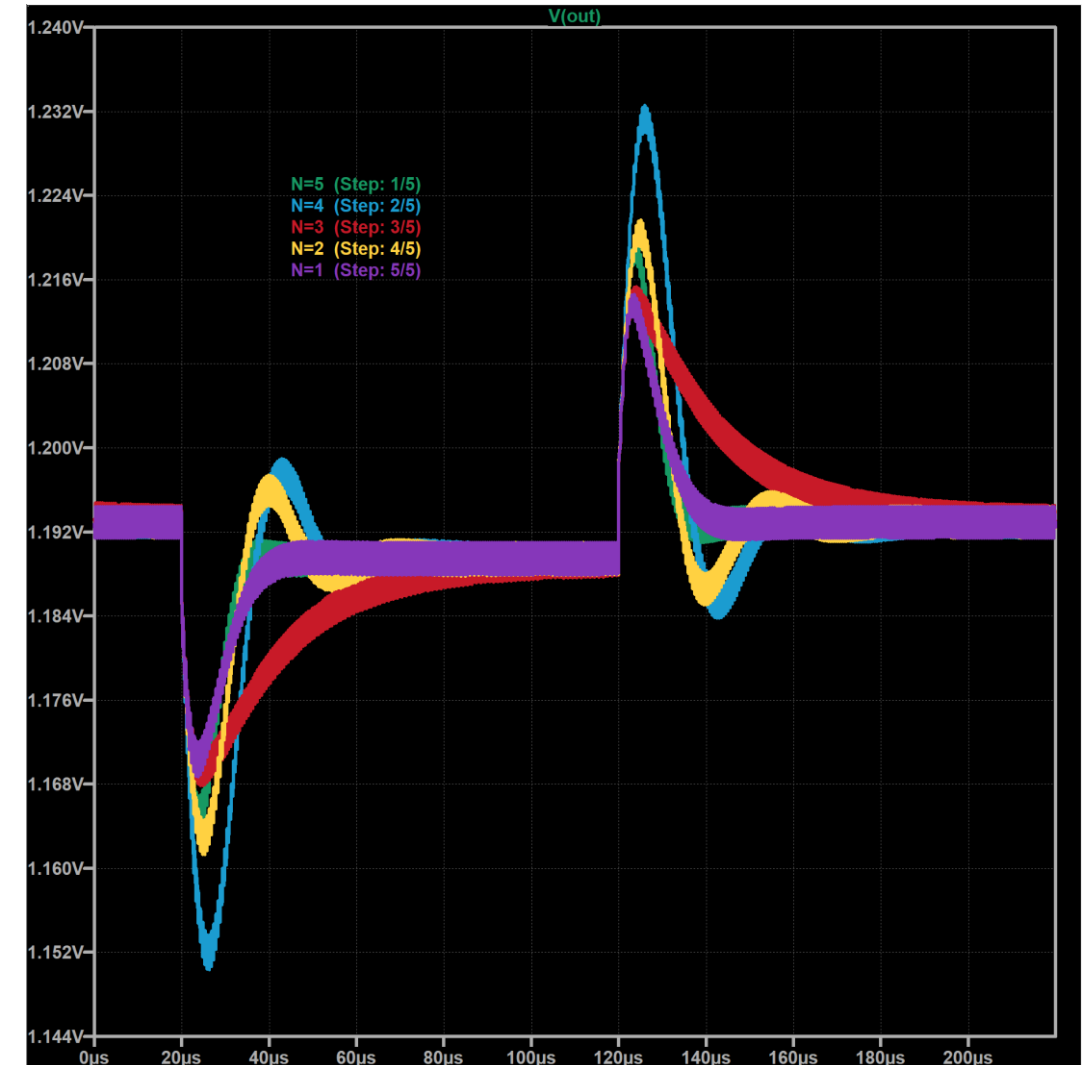
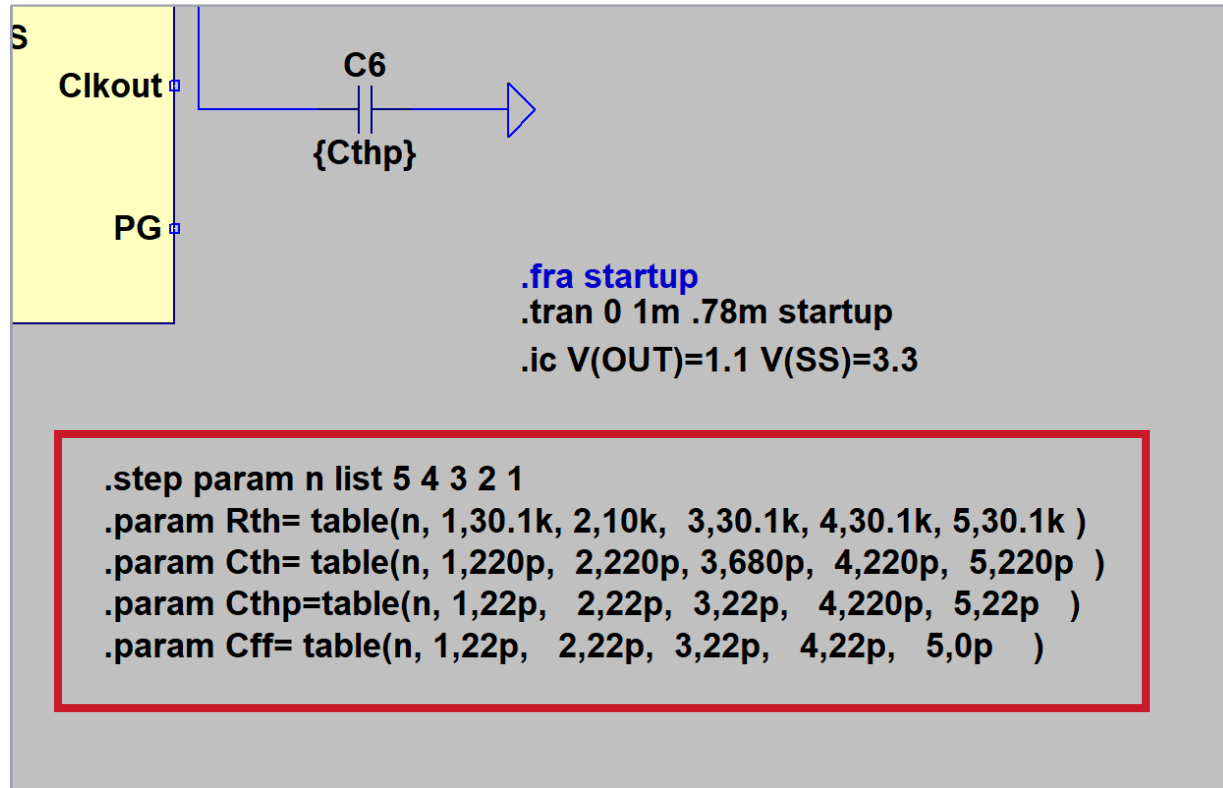
Step through different Rth, Cth, Cthp, and Cff values



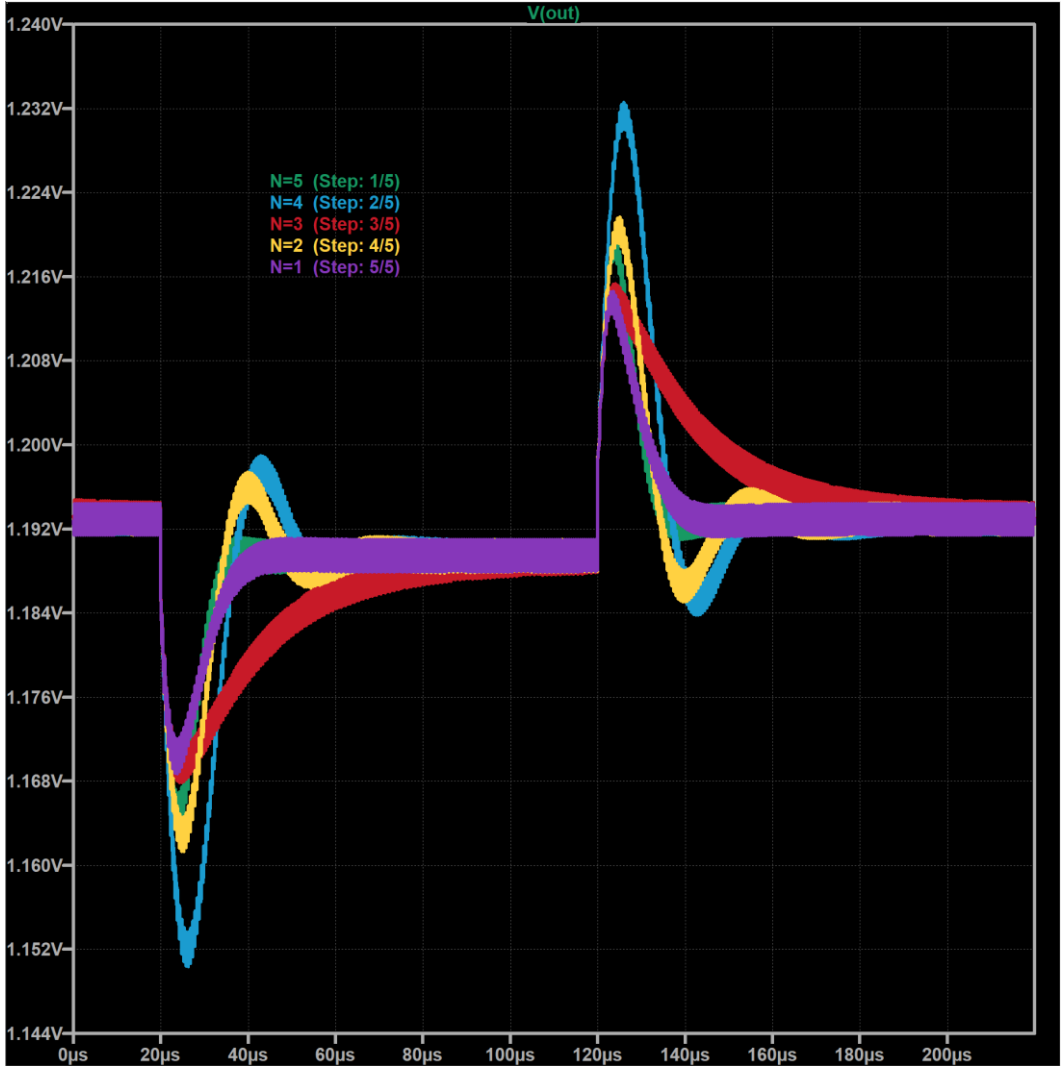
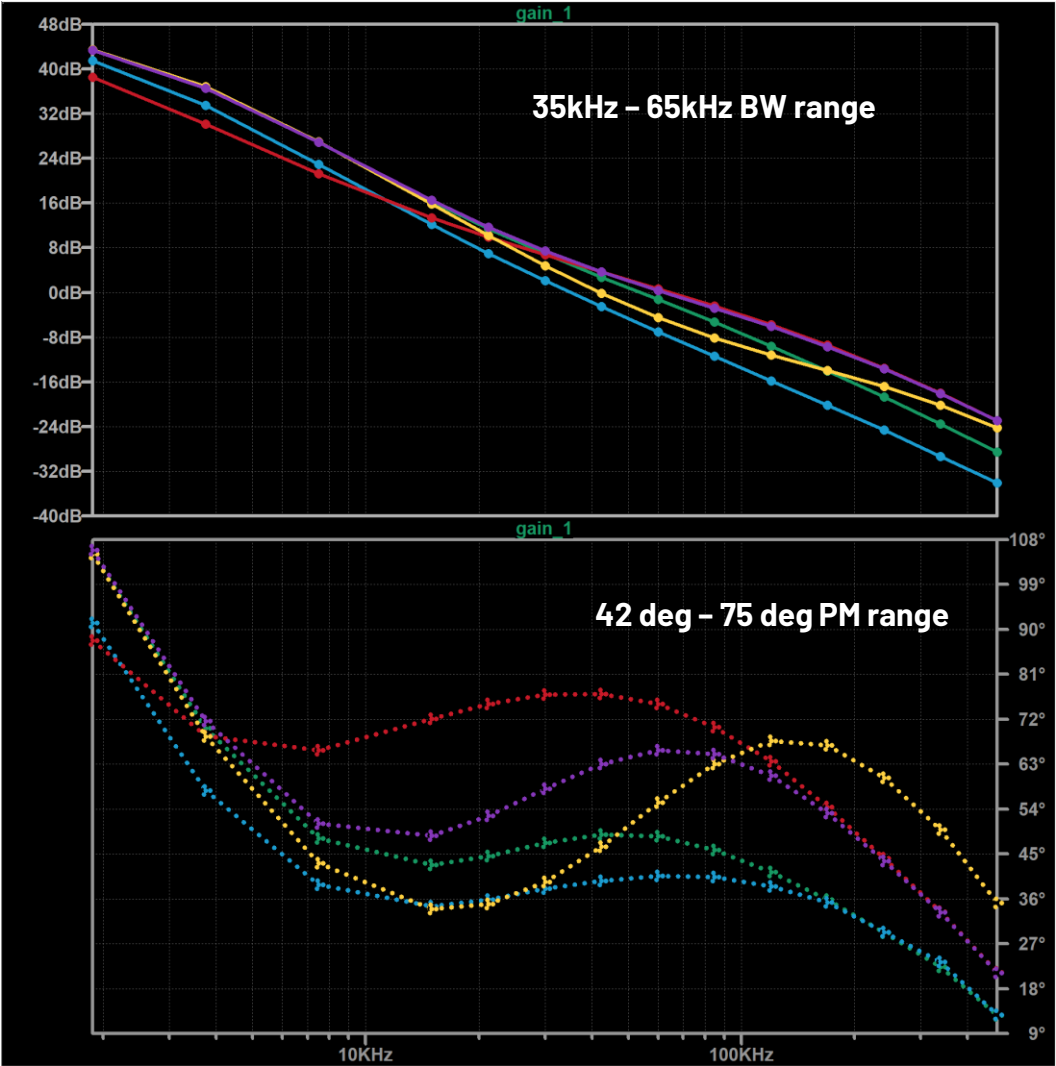
	Run 1	Run 2	Run 3	Run 4	Run 5
Rth	30.1k	10k	30.1k	30.1k	30.1k
Cth	220p	220p	680p	220p	220p
Cthp	22p	22p	22p	220p	22p
Cff	22p	22p	22p	22p	5p

Transient Response Comparisons

Step through different R_{th} , C_{th} , C_{thp} , and C_{ff} values



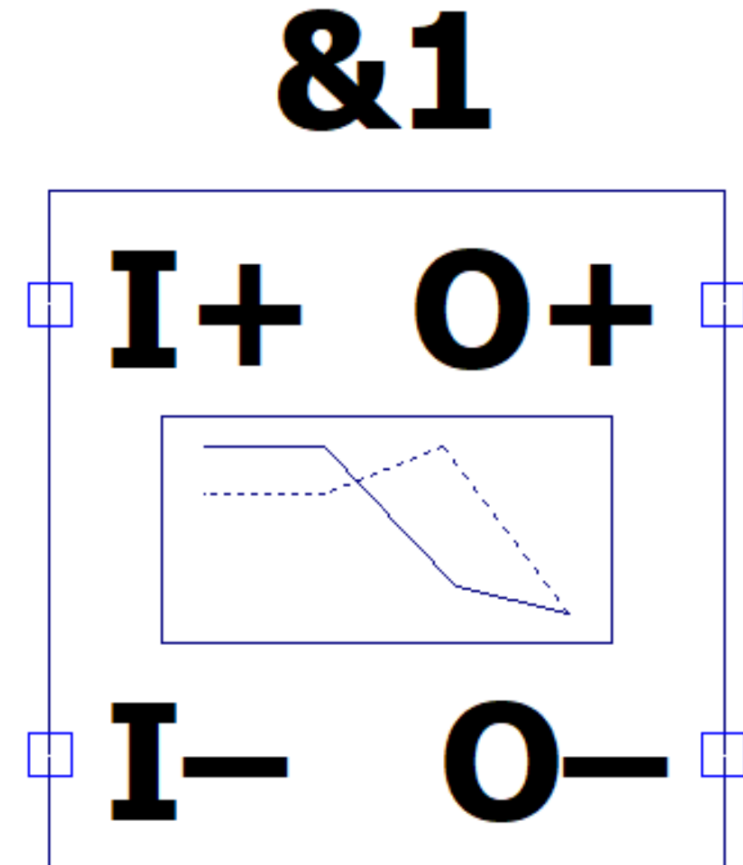
Transient Response Comparisons



Using the FRA Probe

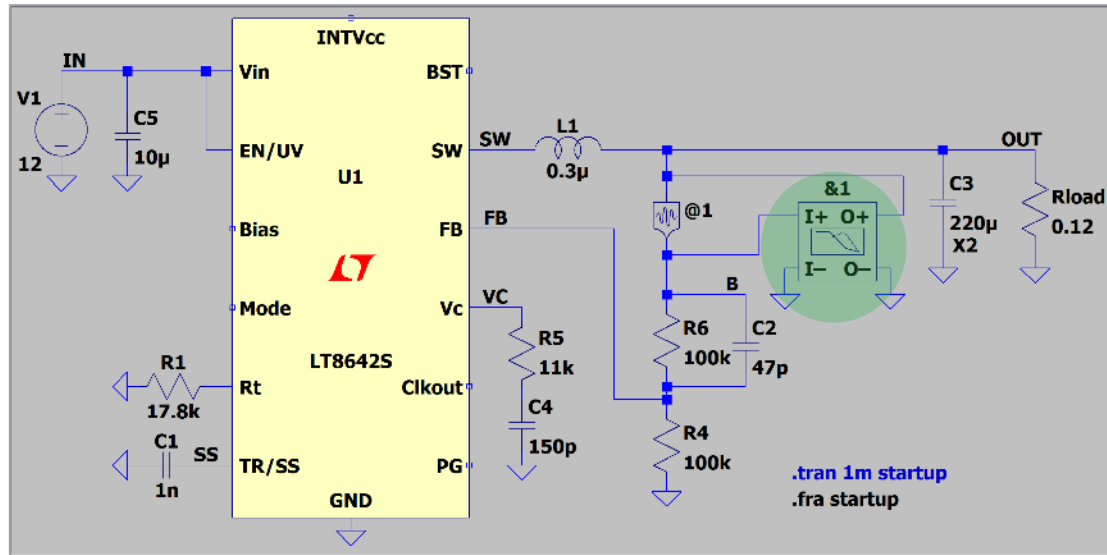
FRA Probe – New in LTspice 24

- The **FRA probe** identifies nodes for **gain analysis** in **an FRA simulation**
- **All terminals are inputs**, with zero capacitance and infinite impedance
- **LTspice** analyzes the **complex gain**, $V(0+, 0-) / V(1+, 1-)$
 - This is saved in the frequency-domain raw file and can be plotted in the Bode plot window
- **The FRA probe must be used in conjunction with an FRA device**
 - The stimulus is controlled by the FRA device
 - The probe has no configuration parameters



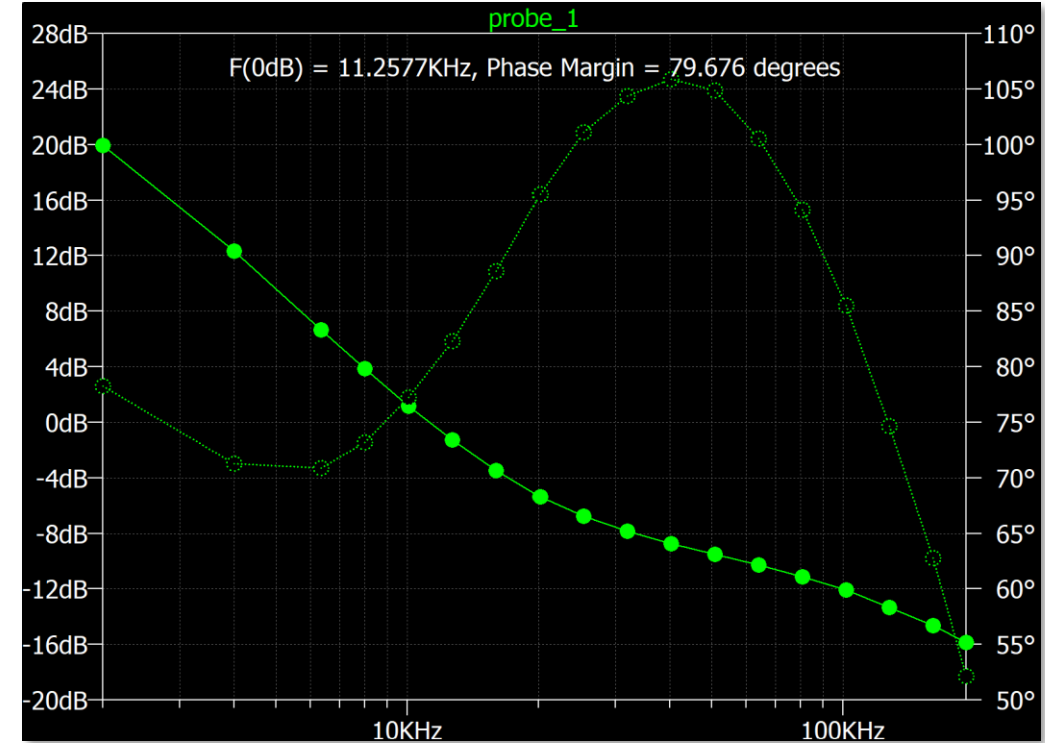
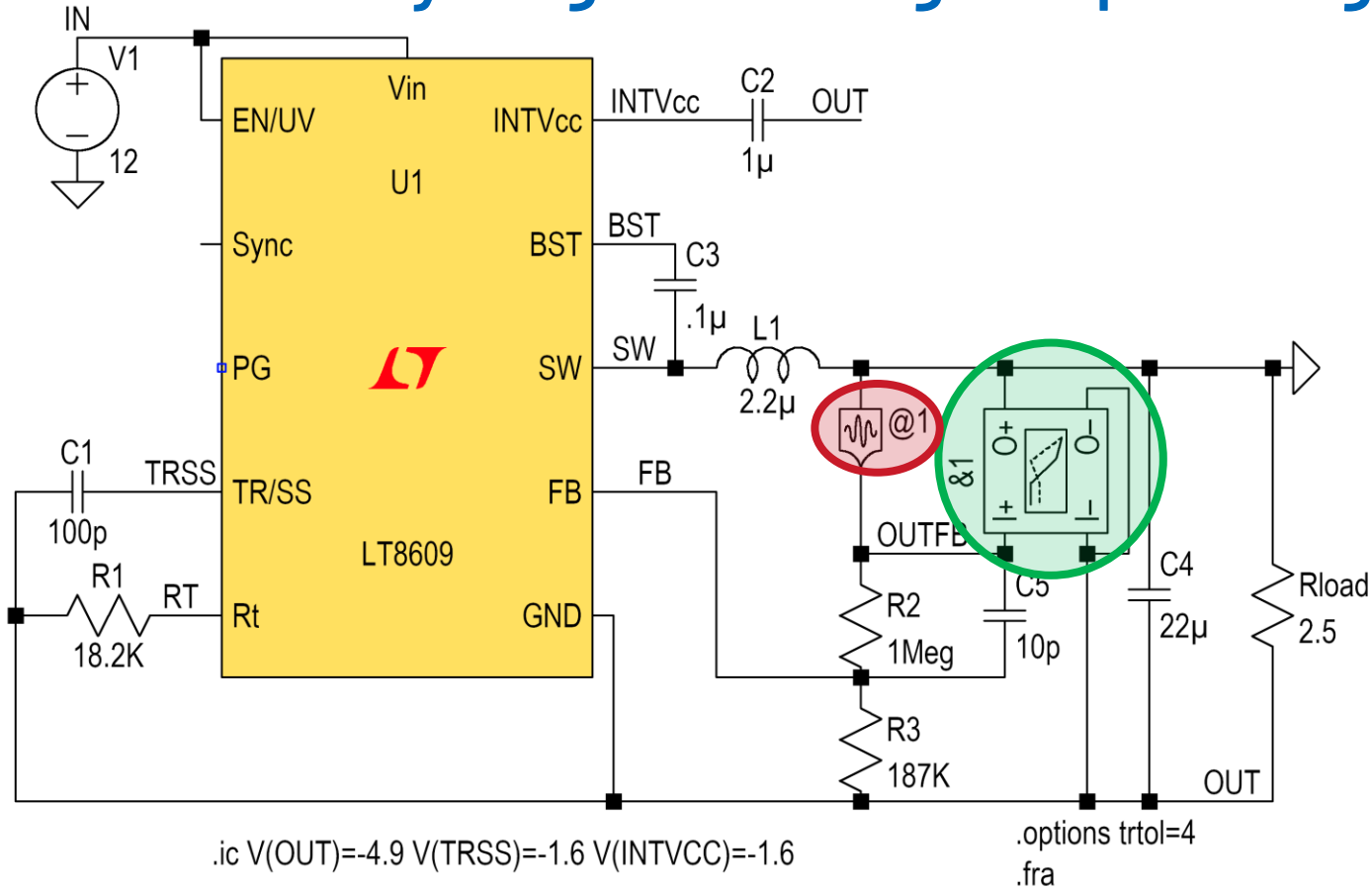
FRA Probe – Standard Buck Converter

Criteria



- Interrupt **all** feedback paths just as before
- FRA component must be point from **lower impedance (flat side)** to **higher impedance (pointy side)**
- FRA Probe used in conjunction with FRA component
 - Not required in the standard buck converter in the FB path
 - Good practice to place it as it is used in hardware
 - In this standard buck converter setup, we probe
 - **OUT node**
 - **B node**
- Assignment of FRA to FRA Probe **@1 and &1**
 - There can be more than one FRA **@** and FRA Probe **&** in a circuit

Ex. 1: Analyzing Inverting Output Regulators

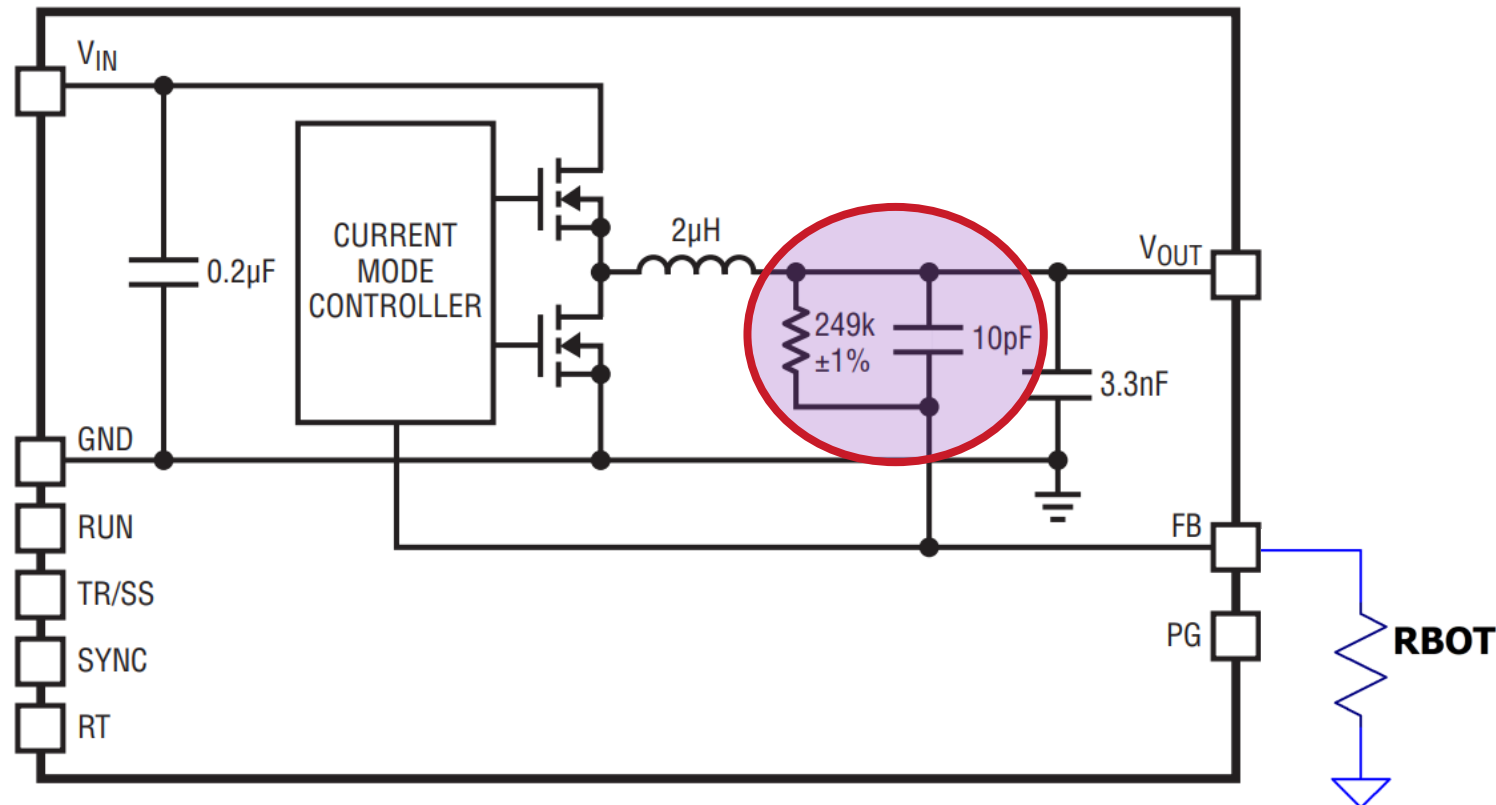


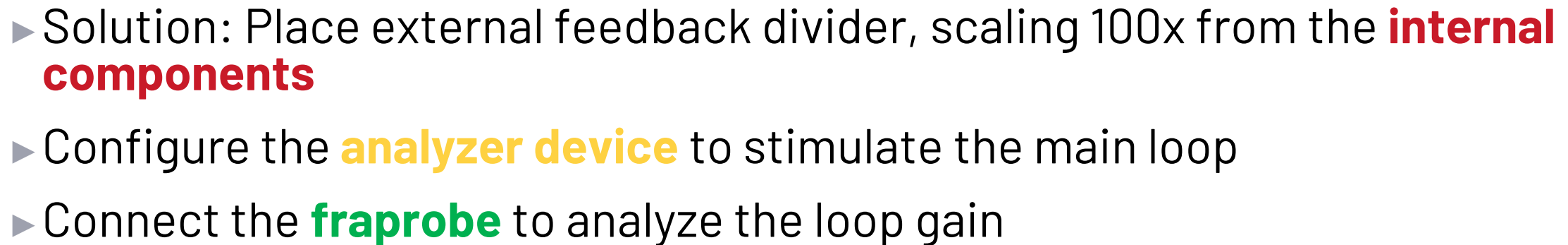
- Connect the positive (O+, I+) terminals of the **fraprobe** across the **fra stimulus device**, negative fraprobe terminals (O-, I-) to the negative output

Ex. 2: Analyzing μ Modules with Internal Feedback

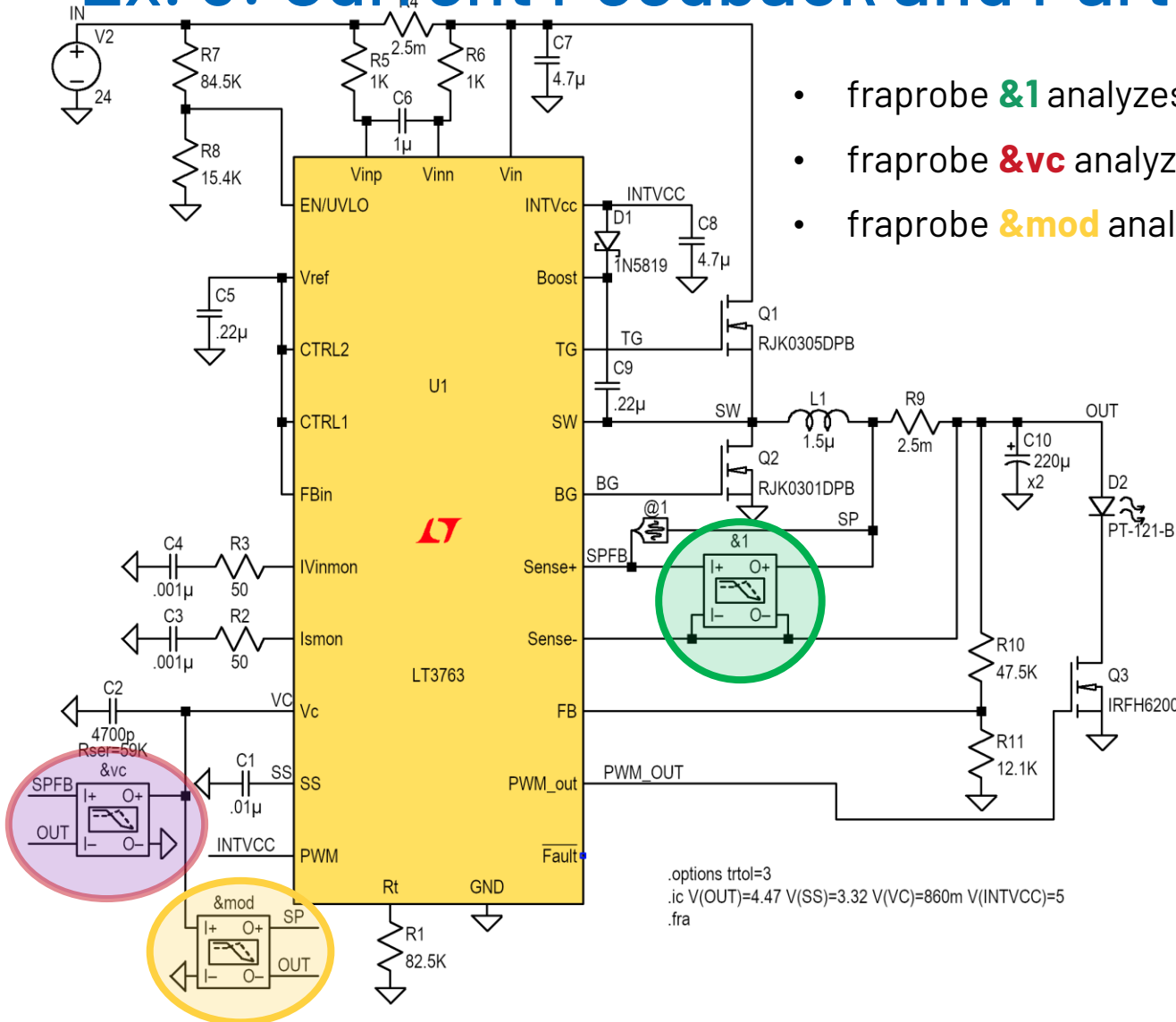
- Many μ Modules have **integrated feedback components**
 → There is no way to break the loop outside the module!

LTM8074 Block Diagram

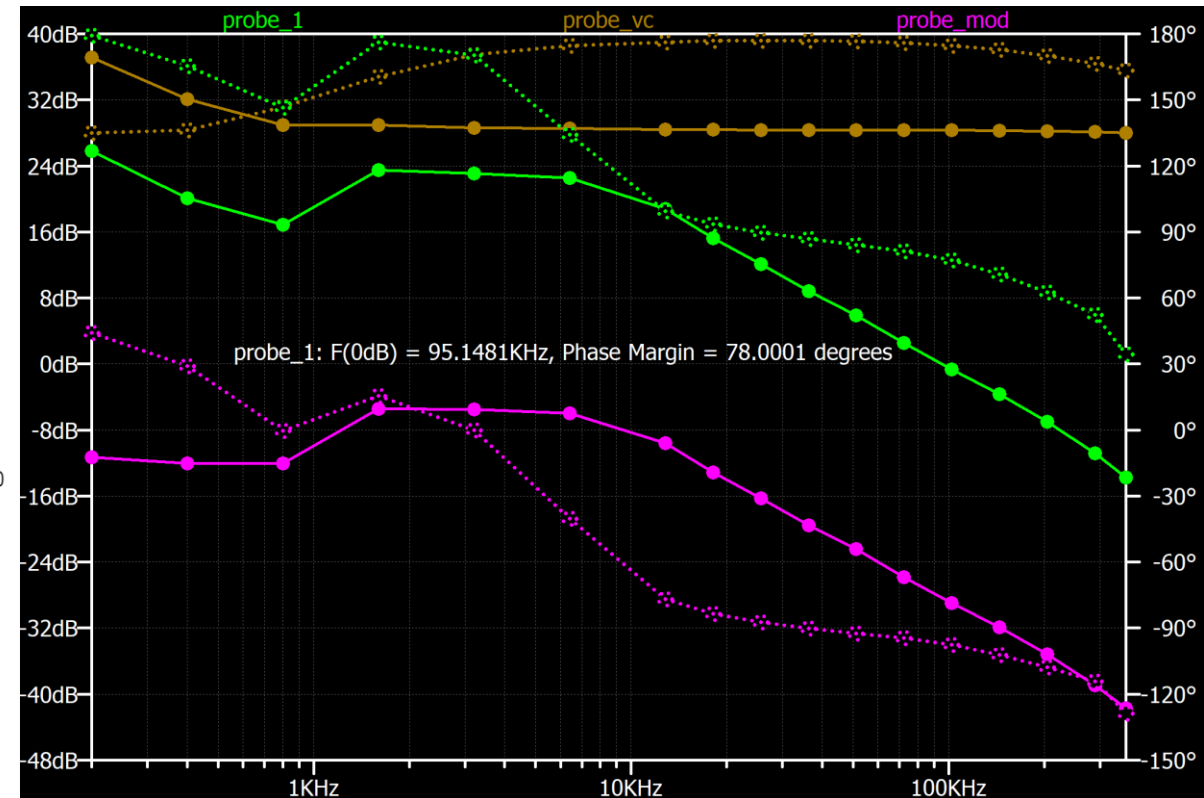




Ex. 3: Current Feedback and Partial Loop Analysis



- fraprobe **&1** analyzes the full loop differentially across the sense resistor R9
- fraprobe **&vc** analyzes from current sense resistor to compensation point
- fraprobe **&mod** analyzes from compensation point to current sense resistor



Questions?

